Near-Cubic Gate and Near-Quartic Wire Lower Bounds for SYM ∘ THR Circuits

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Abstract

We prove that a language in E^{NP} requires $n^3/\operatorname{polylog} n$ gates, and more strongly $n^4/\operatorname{polylog} n$ wires, for SYM ∘ THR circuits, even with a sufficiently small constant fraction of errors at every sufficiently large input length. These are standard SYM ∘ THR circuits: arbitrary real-weight threshold gates feed an arbitrary symmetric output gate. More generally, the gate bound allows the top gate to read an integer-weighted sum of the bottom gates, with each gate charged by the magnitude of its weight. For SYM ∘ THR circuits the best previous published bounds were $n^{5/2-\varepsilon}$, for gates and hence for wires (Chen–Tal–Wang). A single technique raises both exponents: the gate exponent to $3-o(1)$ and the wire exponent to $4-o(1)$.

Following Williams' algorithmic method, we give an algorithm that, given a witness, returns a certified estimate of the acceptance probability of the exclusive-or of two such circuits below this wire cap in $2^n/n^{\Omega(\log\log n)}$ time, and apply the almost-everywhere transfer of Chen–Lyu–Williams. The algorithm works on products of Hamming-code stars, on which a circuit's score splits into a left part, a right part, and a mixed part to which, on average, only $\widetilde{O}(W/n^{3/2})$ of the bottom gates contribute, for W wires. A cycle-sum identity shows that, summed over Fourier frequencies, the error potential of a derandomized sampler is a weighted count of closed walks whose signed sums of mixed parts vanish. Our main new tool is discrete tomography: by a quantitative form of a theorem of Hajdu and Tijdeman, counts of residues modulo many primes determine the exact joint distribution of these sums. To our knowledge, this is the first use of discrete tomography in circuit complexity. The exponent $4 = 3/2 + 1 + 3/2$ comes from the cells, from localizing the sums by nested subsampling, and from tomography on windows of side $n^{3/2}/\operatorname{polylog} n$. The tomography theorem may be of independent interest. Both lower bounds are formalized in Lean 4, assuming only six published results that the paper states (github.com/devnag/wire4_gate3).

Keywords. Circuit lower bounds; threshold circuits; symmetric functions; gate and wire complexity; certified counting; matrix concentration; discrete tomography; exact polynomial representations; formal verification; Lean.

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1 Introduction

How many threshold tests are needed to compute a Boolean function when their outputs may be combined by any symmetric rule? We prove a near-cubic gate lower bound and a stronger near-quartic wire lower bound for a language in E^{NP}, with constant error at every large input length. Both results improve the $n^{5/2-\varepsilon}$ bound of Chen, Tal, and Wang [CTW26] for SYM $\circ$ THR, which bounds gates and hence wires. One technique gives both: the gate theorem is a corollary of the wire theorem. The gate theorem first appeared in the author's unrefereed preprint [Nag26c]; this paper supersedes that preprint and contains complete proofs of both theorems, which are also checked in Lean 4 from six published results (Section 1.5).

A SYM $\circ$ THR circuit first computes threshold gates

$$T_i(x) = \mathbf{1}\Big[\sum_{l=1}^{n} w_{i,l}x_l \geq \theta_i\Big], \qquad w_{i,l}, \theta_i \in \mathbb{R},$$

and applies an arbitrary Boolean lookup to the number $\sum_i T_i(x)$ of gates that output one. The lookup may test a majority, a parity, an exact count, or any set of counts. The ordinary gate count is the number of bottom gates, with repeated gates counted with multiplicity. The wire count additionally charges each input read by a bottom gate. Integer top weights give a useful generalization, defined below; they are unnecessary for the standard gate theorem.

Main results, informally. *There are a language $F \in \mathrm{E}^{\mathrm{NP}}$ and a constant $\varepsilon > 0$ such that, at every sufficiently large length n, every* SYM $\circ$ THR *circuit with at most $n^3/\operatorname{polylog} n$ bottom gates disagrees with F on more than an ε fraction of inputs. The same language requires $n^4/\operatorname{polylog} n$ wires.* The precise wire theorem is Theorem 1.1; Corollary 1.2 derives the gate theorem. "Almost everywhere" means every sufficiently large input length, rather than infinitely many lengths. A constant error bound forces disagreement on a fixed positive fraction of the uniform inputs, a stronger guarantee than a worst-case lower bound, which only forces one error.

The language is fixed before the length or circuit is chosen. The standard class E^{NP}, a usual target of the algorithmic method, consists of languages decided in $2^{O(n)}$ time with a satisfiability oracle; the hard language is not claimed to be in P.

1.1 The precise theorem

Definition 1.1 (Circuits and wires)**.** A SYM $\circ$ THR circuit on n inputs has the form

$$C(x) = g\Big(\sum_{i=1}^{G} T_i(x)\Big),$$

where $T_1, \ldots, T_G$ are threshold gates with real parameters, called the *bottom gates* of the circuit (repetitions are allowed and counted), and $g : \{0, \ldots, G\} \to \{0, 1\}$ is arbitrary, called the *lookup*. Let f_i be the number of inputs l with $w_{i,l} \neq 0$, the *fan-in* of T_i. The *wire count* is

$$W(C) = G + \sum_{i=1}^{G} f_i,$$

the top fan-in plus the total bottom fan-in.

Integer-weighted extension. Corollary 1.2 also covers circuits

$$C(x) = g\Big(\sum_{i=1}^{G} c_i T_i(x)\Big), \qquad c_i \in \mathbb{Z} \setminus \{0\}, \quad g : \mathbb{Z} \to \{0,1\},$$

charged by their *total top weight* $\mathrm{wt}(C) = \sum_i |c_i|$, which equals G when all $c_i = 1$. Bottom fan-in and real bottom weights are unrestricted in both models.

The number of bottom gates and the total bottom fan-in are each at most $W(C)$. Fixing some inputs to constants does not increase W, and neither does complementing the output (replace g by $1 - g$). For $n \geq 4$ write

$$\lambda_n = \lceil \log_2 n \rceil, \qquad \mu_n = \lceil \log_2 \lambda_n \rceil, \qquad \mathcal{A}_4(n) = \frac{n^4}{\lambda_n^{17/4} \mu_n^{15/2}},$$

and for an n-input circuit C and a language F write $\mathrm{agr}(C, F) = \Pr_{x \in \{0,1\}^n}[C(x) = F(x)]$.

Theorem 1.1 (Almost-everywhere near-quartic wire lower bound)**.** *Let $0 < \gamma < 1$ be a fixed absolute constant: the rational rejection gap of the three-query PCP of proximity of Chen and Williams [CW19, Lemma 24] (supporting statement A.4). For every fixed $0 < d < \gamma/5$ there are a positive integer K_d, a language $F_d \in \mathrm{E}^{\mathrm{NP}}$, and an integer $n_0 \geq 4$ such that, for every $n \geq n_0$ and every* SYM ∘ THR *circuit C on n inputs,*

$$\boxed{W(C) \leq \Big\lfloor \frac{n^4}{K_d \lambda_n^{17/4} \mu_n^{15/2}} \Big\rfloor \quad \Longrightarrow \quad \mathrm{agr}(C, F_d) < 1 - d.}$$

Corollary 1.2 (Almost-everywhere near-cubic gate lower bound)**.** *With γ, d, K_d, F_d, and n_0 as in Theorem 1.1, for every $n \geq n_0$, every* SYM ∘ THR *circuit C on n inputs satisfies*

$$\boxed{G \leq \Big\lfloor \frac{n^3}{2K_d \lambda_n^{17/4} \mu_n^{15/2}} \Big\rfloor \quad \Longrightarrow \quad \mathrm{agr}(C, F_d) < 1 - d.}$$

More generally, the same conclusion holds for $C(x) = g(\sum_i c_i T_i(x))$, with integers $c_i \neq 0$, when $\sum_i |c_i|$ is at most the same cap.

Proof. Write $a_i = |c_i|$ and $h = \sum_{c_i < 0} a_i$. For $c_i > 0$, take a_i copies of T_i; for $c_i < 0$, take a_i copies of $1 - T_i$. A complement is again a threshold gate with the same support, including when the original threshold is attained. Indeed, for $T_i = \mathbf{1}[s_i \geq \theta_i]$, choose $\eta_i > 0$ no larger than every positive gap $\theta_i - s_i(x)$ on the finite Boolean cube (choose $\eta_i = 1$ if there is no such gap). Then

$$1 - T_i(x) = \mathbf{1}[-s_i(x) \geq -\theta_i + \eta_i].$$

The unweighted count of the resulting bottom gates is

$$t(x) = \sum_{c_i > 0} a_i T_i(x) + \sum_{c_i < 0} a_i (1 - T_i(x)) = \sum_i c_i T_i(x) + h.$$

Thus the finite lookup $g'(t) = g(t - h)$, for $0 \leq t \leq \mathrm{wt}(C)$, computes the same function. The new circuit C' has exactly $\mathrm{wt}(C)$ bottom gates and

$$W(C') = \sum_i a_i (f_i + 1) \leq (n+1)\,\mathrm{wt}(C) \leq \frac{n+1}{2n} \cdot \frac{n^4}{K_d \lambda_n^{17/4} \mu_n^{15/2}} \leq \frac{n^4}{K_d \lambda_n^{17/4} \mu_n^{15/2}}.$$

Since $W(C')$ is an integer, Theorem 1.1 applies. □

Here γ is a fixed verifier rejection gap; the factor five is the error weight of the local proof gadget. These constants and the logarithmic losses are not optimized. Consequently, by Theorem 1.1 and Corollary 1.2, the same F_d works eventually against every fixed total top weight $An^{3-\delta}$, and against every fixed wire count $An^{4-\delta}$, for $A, \delta > 0$.

Why wires. A gate of fan-in f costs f input wires and one connection into the output. Kane and Williams proved superquadratic wire lower bounds for depth-two threshold circuits [KW16]. The wire theorem also gives stronger bounds for sparse bottom gates. For example, if every bottom gate reads at most $\sqrt{n}$ inputs, a circuit with G bottom gates has $W \leq (1 + \sqrt{n})G$ wires. The wire theorem therefore excludes up to $n^{7/2}/\operatorname{polylog} n$ such gates, beyond the general near-cubic gate cap.

1.2 Prior work and significance

A symmetric output includes majority, parity, exact count, and any set of counts. Thus SYM $\circ$ THR contains MAJ $\circ$ THR, but differs from THR $\circ$ THR, whose output may have arbitrary unequal real weights. These depth-two models are central test cases for lower bounds against constant-depth threshold circuits. Proving superpolynomial lower bounds for either remains open.

Kane and Williams [KW16, Theorem 1.1] obtained average-case bounds of $\widetilde{\Omega}(n^{3/2})$ gates and $\widetilde{\Omega}(n^{5/2})$ wires against THR $\circ$ THR, for an explicit function in P. Tamaki [Tam16] and Alman, Chan, and Williams [ACW16] obtained subquadratic gate bounds for E^{NP}. Chen, Tal, and Wang [CTW26, Theorem 1.1] crossed the quadratic gate barrier with $n^{5/2-\varepsilon}$ gates, for every fixed $\varepsilon > 0$, for both SYM $\circ$ THR and THR $\circ$ THR. We raise the symmetric-top gate exponent to $3 - o(1)$ and prove the stronger $4 - o(1)$ wire exponent, with constant error at every large length.

Class and resource	Published benchmark	This paper	Guarantee here
SYM $\circ$ THR, gates	$n^{5/2-\varepsilon}$ [CTW26]	$n^3/\operatorname{polylog} n$	constant error, every large length
SYM $\circ$ THR, wires	$n^{5/2-\varepsilon}$ from the gate bound [CTW26]	$n^4/\operatorname{polylog} n$	constant error, every large length
THR $\circ$ THR, wires	$\widetilde{\Omega}(n^{5/2})$ [KW16]	no claim	average-case benchmark in P

The CTW benchmark is worst-case hardness in E^{NP}; our guarantee is constant error at every sufficiently large length. The wire benchmark in the second row follows because gate count never exceeds wire count; it is not a claim about a sharper standalone wire theorem. The Kane–Williams row concerns a different top class and a different hard-function class. We do not claim a near-cubic gate theorem for unrestricted THR $\circ$ THR. For wires, the best previous published bound for SYM $\circ$ THR is the $n^{5/2-\varepsilon}$ that follows from the gate bound of [CTW26]; Theorem 1.1 raises it to $n^{4-o(1)}$. The author's unrefereed preprints give wire bounds of $n^{3-o(1)}$ [Nag26a] and $n^{7/2-o(1)}$ [Nag26b] for both SYM $\circ$ THR and THR $\circ$ THR, with agreement at most $1/2 + o(1)$; both are formalized in Lean against their input theorems from the literature, in public repositories. For SYM $\circ$ THR, Theorem 1.1 has a larger cap and the weaker constant-error guarantee; it does not extend to THR $\circ$ THR (Section 13).

1.3 The algorithmic method

Williams' algorithmic method [Wil13] converts a fast algorithm that analyzes circuits into a lower bound for them. A nondeterministic machine for a hard computation guesses small circuits that

describe the bits of a probabilistically checkable proof (PCP), and uses a counting algorithm to estimate how often the verifier's local tests fail. Soundness must hold for every guess. If the proofs of every true statement had small circuits, some guess would verify each statement faster than the nondeterministic time hierarchy permits. The resulting contradiction produces hard proof data; the constructive refuter of Chen, Lyu, and Williams [CLW20] makes the hardness hold at every large length.

For SYM $\circ$ THR at constant error, the transfer below uses the weighted analysis [Nag26c, Section 11] of the MAX-2SAT gadget of Garey, Johnson, and Stockmeyer [GJS76]. It expresses the failure rate of the local tests through the acceptance probabilities of one circuit, of the exclusive-or of two circuits, or of a circuit exclusive-or an input parity. What the transfer needs is therefore a *certified counter*: an algorithm that reads such a query, called a *call*, and an auxiliary string, called a *witness*, returns an interval that contains the acceptance probability for *every* witness, returns an interval of radius at most a requested β for *some* witness, and runs in time $2^n/n^{\omega(1)}$. Our counter (Theorem 2.1) does this below a wire cap proportional to $\beta n^4/\operatorname{polylog} n$. Section 12 summarizes how the transfer, which was written for top weight, applies to the wire measure, and Appendix A carries it out step by step.

1.4 The idea: where $3/2 + 5/2 = 4$ comes from

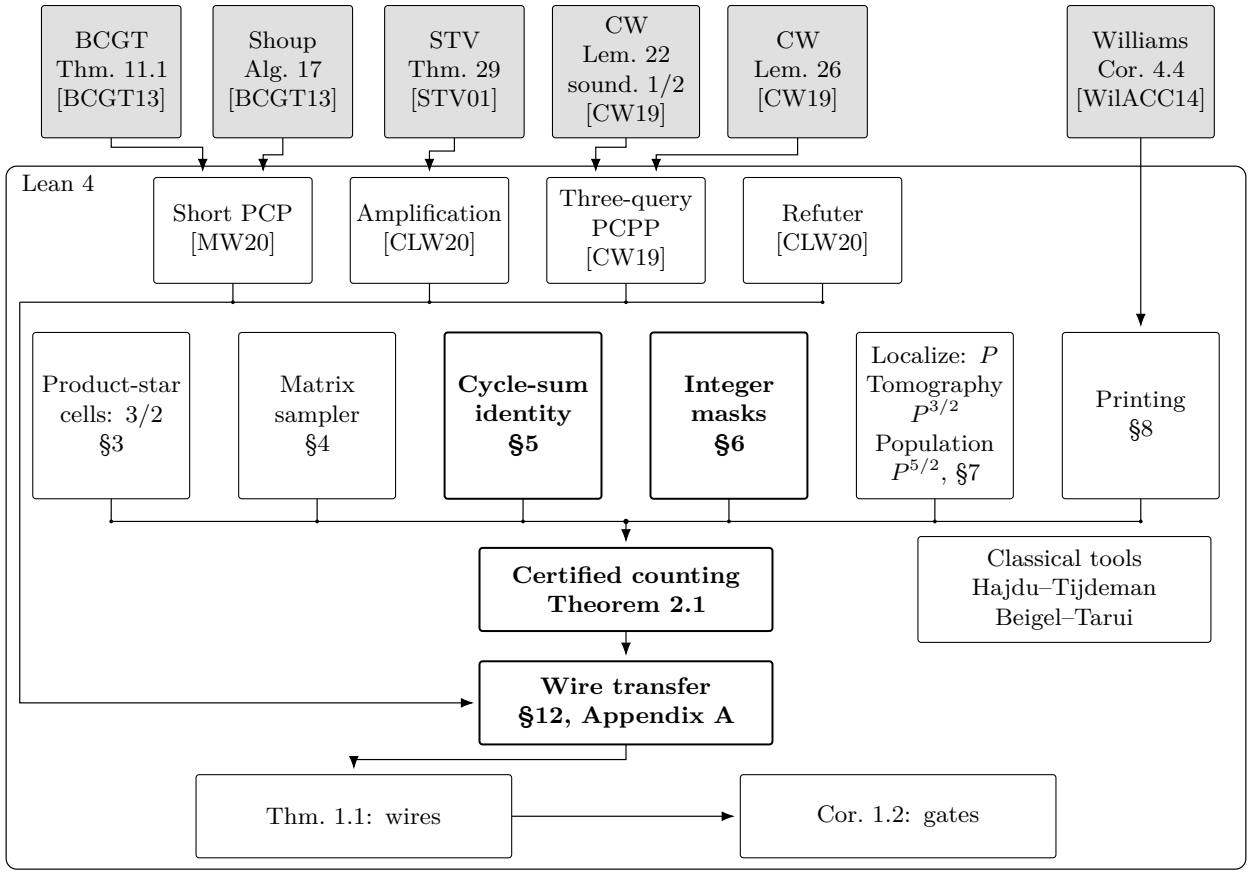
We describe the counter for a single circuit with lookup g; logarithmic factors are suppressed throughout this subsection.

Two earlier analyses by the author stop short of n^4. In [Nag26c], nested subsampling and tomography reconstruct the score distribution of a restricted circuit with up to about $P^{5/2}$ live gates, where $P \approx n$ is a prime scale; a restriction leaves about an $n^{-1/2}$ fraction of the gates live, which gives $n^{1/2}P^{5/2} \approx n^3$ gates. In [Nag26b], product-star cells reduce the expected mixed population to about $W/n^{3/2}$, but the circuit's value on a cell also depends on one-sided parts that a bilinear sampler must absorb; the analysis there applies the sampler to pointwise count lists and pays roughly the square root of the population, which gives $n^{7/2}$ wires. The new step is the cycle-sum identity (Lemma 5.1): summed over Fourier frequencies, the sampler's potential is a weighted count of closed walks whose signed residual sums vanish, and a walk sum, like a score, is an integer sum of per-gate contributions. The reconstruction of [Nag26c] then applies to walk sums, with masks that are exact integer polynomials valid modulo every prime (Section 6), and gives $n^{3/2} \cdot P \cdot P^{3/2} \approx n^4$ wires. Appendix C gives full attribution; supporting statements and proofs appear in Appendices D and E.

Cells that charge only joint dependence (3/2). The algorithm partitions $\{0,1\}^n$ into 2^{n-k} cells of 2^k inputs, where $k = \Theta(\log n \log\log n)$. Each cell is a product: its points are $c \oplus u \oplus v$, where c is a center, u ranges over $N = 2^{k/2}$ "left" error patterns and v over N "right" error patterns, supported on disjoint sets of coordinates. These cells come from the author's $n^{7/2}$-wire preprint [Nag26b, Appendix A]. A *star* in a block of $b = 2^s - 1$ coordinates is a point together with its b neighbors at Hamming distance one; the perfect Hamming code [Ham50] partitions each block into stars, and a cell is a product of stars in r blocks, $r/2$ of them on each side, so that $N = (b+1)^{r/2}$. On a cell the score $s(u,v) = \sum_i T_i(c \oplus u \oplus v)$ splits exactly as

$$s(u,v) = s^{\mathrm{L}}(u) + s^{\mathrm{R}}(v) + R(u,v), \qquad R = \sum_i \Delta_i, \quad \Delta_i = T_i(c\oplus u\oplus v) - T_i(c\oplus u) - T_i(c\oplus v) + T_i(c),$$

where s^{L} depends only on u and s^{R} only on v. The *mixed residual* Δ_i is 0 unless the gate T_i reads both a coordinate flipped by u and one flipped by v, and it always lies in $\{-1, 0, 1\}$. For a gate of



$$W/n^{3/2} \lesssim P \cdot P^{3/2} = P^{5/2}, \quad P \approx n \implies W \lesssim n^4.$$

Logarithmic factors suppressed: $3/2 + (1 + 3/2) = 4$.

Lean: `NearQuarticWires.Release.main_from_published`: `PublishedInputs` → `JointStatement4`

Figure 1: Proof architecture. Shaded: the six published results the formal proof assumes; bold: parts new relative to the author's preprints [Nag26b, Nag26c]; arrows: "is used to prove". Everything inside the frame is checked by the Lean kernel.
The P factor is localization; the $P^{3/2}$ window is tomography; their product bounds the allowed population. Classical tools: [HT01, BT94] and Appendix C.
Lean companion: github.com/devnag/wire4_gate3.

fan-in f, on a random cell and random point, $\Delta_i \neq 0$ with probability $O(f/n^{3/2})$, up to logarithmic factors (the 3/2 comes from a total-influence bound on one side and a support bound on the other). The *population* is

$$\mathrm{pop}(u, v) = \#\{i : \Delta_i(u, v) \neq 0\}, \qquad \mathbb{E}\,\mathrm{pop} \leq \widetilde{O}(W/n^{3/2}).$$

The expectation averages over partitions, cells and points; some partition therefore satisfies the same mean bound. Masks handle points with unusually large population.

A sampler that never sees one-sided behavior. The value of the circuit on the cell is $g(s^{\mathrm{L}}(u) + s^{\mathrm{R}}(v) + R(u, v))$, and we must estimate its average over the N^2 points. Expand the lookup in characters of a cyclic group $\mathbb{Z}/\Omega$, taking $\Omega = 4mG + 1$ for G bottom gates, where $m = 4$ is the moment order of the trace bound below. This exceeds the score range and the absolute value of

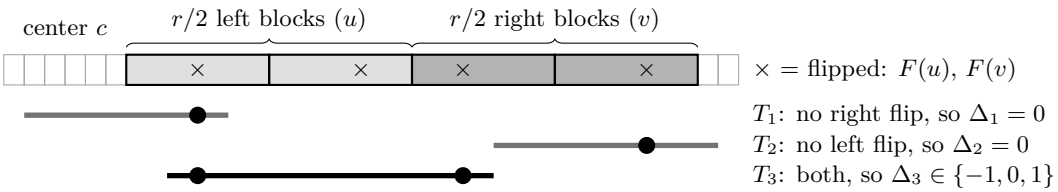


Figure 2: A product-star cell. The cell fixes a center c and, in each of r blocks of $b = 2^s - 1$ coordinates, lets the point move within a Hamming-code star (no flip or one flip); the left pattern u uses $r/2$ blocks and the right pattern v the other $r/2$. A gate contributes to the mixed residual only if it reads a coordinate flipped by u and one flipped by v.

every signed sum of $2m$ residuals: $g(s) = \sum_\theta \widehat{g}(\theta)\omega^{\theta s}$ with $\omega = e^{2\pi \mathrm{i}/\Omega}$. Then

$$\omega^{\theta s(u,v)} = \underbrace{\omega^{\theta s^{\mathrm{L}}(u)}}_{\text{depends on } u}\ \underbrace{\omega^{\theta s^{\mathrm{R}}(v)}}_{\text{depends on } v}\ \underbrace{\omega^{\theta R(u,v)}}_{=:M_\theta(u,v)}.$$

The first two factors are arbitrary unimodular *selectors*, one on each side; this separation is the standard finite Fourier expansion, as in [Nag26b, Equation (29)]. That preprint introduced a universal sampler: one list of Q sample pairs (U_j, V_j) that estimates $a^{\mathsf{T}} M b/N^2$ for all selectors a, b at once, provided the sampling error matrix of M has small operator norm. We apply the sampler separately to the real matrices $D_{\theta,0} = \mathrm{Re}\, M_\theta$ and $D_{\theta,1} = \mathrm{Im}\, M_\theta$. For each real error matrix E, its norm is controlled by $\mathrm{tr}((EE^{\mathsf{T}})^m)$. The sample list can be chosen deterministically, bit by bit, if the algorithm can compute conditional expectations of this potential exactly [Nag26b, Appendix F]. So the one-sided parts of the circuit, however complicated, never enter the object that must be computed; only the matrices M_θ do.

The cycle-sum identity. Expanding $\mathrm{tr}((EE^{\mathsf{T}})^m)$ gives a sum over closed walks w of length $2m$ that alternate between left and right patterns. Expanding the cosine and sine entries of the real matrices gives characters of *signed sums* of residuals along the $2m$ edges, and summing over all frequencies θ turns the character into an indicator: $\sum_\theta \omega^{\theta Y} = \Omega \cdot \mathbf{1}[Y = 0]$ when $|Y| < \Omega$. The result (Lemma 5.1) is that the potential, summed over frequencies, is a weighted count of closed walks whose signed residual sums

$$Y_\sigma(w) = \sum_{l=1}^{2m} \sigma_l R(e_l)$$

vanish, over sign patterns $\sigma \in \{\pm 1\}^{2m}$. The lookup has disappeared. What remains is one integer per circuit and walk, a sum of local contributions of the bottom gates. If each edge has population at most U, the total absolute contribution is at most $2mU$.

For example, a four-edge walk visits a rectangle of left and right choices. One sign pattern tests the alternating sum of its four residual values. Summing the characters turns this into the exact condition that this signed sum is zero. Lemma 5.1 gives the full identity, including all sign patterns, masks, and pair calls. For a pair call there are two walk sums, and the potential is a weighted sum of joint zero tests $\mathbf{1}[Y_1 = 0]\mathbf{1}[Y_2 = 0]$; after localization, Theorem 1.3 reconstructs their joint distribution.

Localize and reconstruct (5/2). Counting walks with $Y_\sigma(w) = 0$ is the problem that the author's near-cubic gate preprint [Nag26c, Sections 4–6] solves for the scores of restricted circuits.

Subsampling the bottom gates at nested rates writes the integer Y through a few short integer *digits*; once the digits are fixed, Y is confined to an interval about P times shorter than its population. Here P is a prime scale of order $n/\operatorname{polylog} n$: the algorithm evaluates polynomials of degree about P over all 2^{n-k} cells at once by Williams' rectangular matrix multiplication, so it can count residues modulo every prime in $(P/2, P]$; the Beigel–Tarui lifting polynomial [BT94] lifts the corresponding residue indicators to powers of p, letting the printer recover the counts exactly. Pair calls have two circuits and hence two scores; their localized joint distribution must be reconstructed. The printing procedure of Section 8 computes the modular count tables in batches, reducing each table to rectangular matrix multiplication.

Under the quantitative size condition of Lemma E.5, prime residue counts determine all short line projections, and hence the joint distribution on a box of side about $P^{3/2}$. To our knowledge, this is the first use of discrete tomography in circuit complexity; the method was introduced for this purpose in [Nag26c]. Within the algorithmic method, modular counting (e.g. [Wil13, ACW16]) computes residues of a single count; here many residue counts jointly determine a whole two-score distribution. Reconstructing a 0/1 lattice set from its line sums is NP-hard already for three directions [GGP99]. Here the unknown is an integer-valued function on a box of side about $P^{3/2}$, and Hajdu and Tijdeman's algebraic theory [HT01] shows that enough directions determine it uniquely. Recovery is therefore an exact linear system, with at most P^4 unknowns and 0/1 coefficients, solved in polynomial time by fraction-free elimination [Bar68]. Thus walk sums with population up to $U \approx P^{5/2}$ can be handled exactly.

Theorem 1.3 (Modular discrete tomography)**.** *Let $\mathcal{P}$ be a finite nonempty set of primes, $D' = 1+\sum_{p\in\mathcal{P}}(p-1)$, and $S(q)=\sum_{m=1}^{q} m\varphi(m)$, where φ is Euler's totient. For $a,b\in\mathbb{Z}$ and $L\in\mathbb{N}$ with $2\le L\le S(\lfloor (D'-1)/(L-1)\rfloor)$, every $f : B\to\mathbb{Q}$ on the known box $B=([a,a+L)\times[b,b+L))\cap\mathbb{Z}^2$ is determined by its exact rational sums $F_p(r,s)=\sum_{(x,y)\in B,\ (x,y)\equiv(r,s) \bmod p} f(x,y)$, for $p\in\mathcal{P}$, $r,s\in\mathbb{Z}/p\mathbb{Z}$. Exact elimination recovers f in time polynomial in L, $\sum_{p\in\mathcal{P}} p^2$, and the total binary input length, including the box endpoints and rational sums.*

The uniqueness and exact-inverse assertions are formalized as `NearCubicGates.Reconstruction.pairMatrix_injective` and `pair_reconstruction_complete` in the Lean companion.

Proof. Lemma E.5 proves uniqueness; fraction-free elimination [Bar68] on the 0/1 residue matrix gives recovery with polynomial bit cost, as in Proposition 10.3. □

Corollary 1.4 (Prime-window tomography)**.** *There are absolute constants $c>0$ and P_0 such that, for every integer $P\ge P_0$, taking $\mathcal{P}$ to be all primes in $(P/2,P]$, Theorem 1.3 applies to every known integer box of side $2\le L\le cP^{3/2}/(\log P)^{3/4}$.*

Proof. Lemmas E.6 and E.7 give an admissible side of order $D'^{3/4}=\Theta(P^{3/2}/(\log P)^{3/4})$. The size condition is preserved when $L\ge 2$ decreases. □

For one score, exact residue sums for the same prime set determine a rational function on a known interval of at most D' consecutive integers (Lemma E.4; Lean: `NearCubicGates.ScalarWindow.frameResidueData_injective`).

The exponent. Combining the two estimates,

$$\frac{W}{n^{3/2}} \lesssim U \lesssim P^{5/2} \quad\Longrightarrow\quad W \lesssim \underbrace{n^{3/2}}_{\text{star cells}}\ \underbrace{P}_{\text{localization}}\ \underbrace{P^{3/2}}_{\text{reconstruction}} \approx n^4.$$

The gain is charged per wire, so sparse gates are handled according to the inputs they read.

One mask for every prime. Points of abnormally large population must be discarded by a *mask*, and the mask enters the polynomials whose residues are counted modulo many odd primes. The masks of [Nag26b] are polynomials over $\mathbb{F}_2$ whose integer lifts are exact only modulo powers of two, so they cannot be used there. Our mask (Section 6) is an exact integer polynomial: if X counts the subsampled bottom gates with nonzero residual at a point, then

$$\mathbf{1}[X \le \tau] = \sum_{a=0}^{X_{\max}} c_a \binom{X}{a} \qquad (0 \le X \le X_{\max})$$

with integers c_a given by Newton forward differences, and $\binom{X}{a}$ is the elementary symmetric polynomial of degree a in the Boolean residual indicators. A single moment test certifies $X \le X_{\max}$ on a whole cell. The identity holds over $\mathbb{Z}$, so the mask is correct modulo every integer at once.

Main components. Lean names below have prefix `NearQuarticWires.`; the full hypotheses are in the indicated statements.

Tool	Statement	Where	Lean name
Modular tomography	Joint residue tables determine a rational box function under an explicit window criterion.	Thm. 1.3	`Tomography.residueCounts_determine`
Integer threshold masks	Exact bounded-population cutoff, with integer coefficients, valid modulo every integer.	Lem. 6.1	`Mask.cutoff_eq_newton_sum_of_le`
Cycle-sum identity	A Fourier trace sum counts zero signed walk sums; Lean checks $m = 4$.	Lem. 5.1	`Cycle.cycle_sum_identity`
Product-star population	Some partition has mean mixed population at most $6r^{3/2}W/n^{3/2}$.	Lem. 3.3	`Cells.mean_population`
Certified counting	Every witness is sound; a short witness is precise below $a\beta\mathcal{A}_4(n)$.	Thm. 2.1	`Program.counterExists4`
Wire transfer	The full quartic counter contract yields a.e. constant-error E^{NP} hardness.	Thm. 12.1	`Transfer.transfer4_official`

Where the method stops.

- Two-score reconstruction handles walk sums of population up to about $P^{5/2}$ (the window condition (6)), and the degree of the printed polynomials keeps P below about n (by (5)). The $P^{3/2}$ window is sharp up to logarithmic factors for recovering arbitrary box distributions from residue data (Remark E.8).
- Arbitrary real top weights lack this bounded integer-score guarantee; the fixed-error transfer gives no vanishing correlation. Extensions remain open; these are limits of this method (Section 13).

Published foundations. The formal proof assumes exactly six published results from four papers, listed with exact locators in Section 1.5. The BCGT13 verifier input is Theorem 11.1, at the parameter tuple $(2, 4, 0, 1)$ used in this paper [BCGT13]. Everything else the formal proof uses is proved in Lean, including the short PCP of Murray–Williams [MW20], the refuter and amplification interfaces of Chen–Lyu–Williams [CLW20], integer normalization of threshold gates, concentration, and the prime counts from Chebyshev-type bounds. Appendix E restates and proves the PCP, refuter and amplification components, with their origins credited; the classical normalization, concentration and prime-count facts are cited to their published sources in Appendix D and Lemma E.7.

1.5 The formal statement

The formalization is available at github.com/devnag/wire4_gate3 (release tag `v1.0`). The following verbatim definitions in `Proof/Quartic/Statement/Joint.lean` state Theorem 1.1 and Corollary 1.2 with shared constants and the rejection gap γ quantified existentially (some rational $0 < \gamma < 1$); `wireCap` and `gateCap` are the two boxed caps, and `InENP` defines E^{NP} by an explicit oracle machine:

```
def WiresAt (K : ℕ) (F : Language) (d : ℝ) (n : ℕ) : Prop :=
  ∀ C : SymmetricThresholdCircuit n, wires C ≤ wireCap K n → agr C F < 1 - d

def SymGatesAt (K : ℕ) (F : Language) (d : ℝ) (n : ℕ) : Prop :=
  ∀ C : SymmetricThresholdCircuit n, C.bottomCount ≤ gateCap K n → agr C F < 1 - d

def GatesAt (K : ℕ) (F : Language) (d : ℝ) (n : ℕ) : Prop :=
  ∀ C : WeightedSymmetricCircuit n, C.topWeight ≤ gateCap K n → agreement C F < 1 - d

def JointAt4 (gamma : ℚ) : Prop :=
  ∀ d : ℝ, 0 < d → d < (gamma : ℝ) / 5 →
    ∃ K : ℕ, 0 < K ∧ ∃ F : Language, InENP F ∧
      ∃ n₀ : ℕ, 4 ≤ n₀ ∧ ∀ n : ℕ, n₀ ≤ n → WiresAt K F d n ∧ SymGatesAt K F d n ∧
            GatesAt K F d n

def JointStatement4 : Prop :=
  ∃ gamma : ℚ, 0 < gamma ∧ gamma < 1 ∧ JointAt4 gamma
```

The release type is `ReleaseStatement := PublishedInputs → JointStatement4`. Its six inputs are transcribed in their source models:

Source	Exact published locator
Williams [WilACC14]	Cor. 4.4, p. 2:17 (also Cor. C.2, p. 2:29).
BCGT [BCGT13]	Theorem 11.1, at the parameter tuple $(2, 4, 0, 1)$ used in this paper; p. 51 (ECCC rev. 3).
BCGT [BCGT13]	Alg. 17, p. 124 (ECCC rev. 3).
Sudan–Trevisan–Vadhan [STV01]	Thm. 29, p. 17 (ECCC rev. 2).
Chen–Williams [CW19]	Lem. 22, p. 19:15, soundness 1/2.
Chen–Williams [CW19]	Lem. 26, p. 19:16.

Running `lake env lean Check.lean` prints the axiom list:

```
[propext, Classical.choice, Quot.sound]
```

The companion's source mapping quotes each published statement beside its transcription and records the modeling conventions; this faithfulness comparison is separate from kernel checking. The trusted files are `Statement.lean` and `Final.lean`, `Joint.lean`, and `Release.lean` in `Proof/Quartic/Statement/`. They define the machine and circuit models, measures, published premises and conclusion; everything else is checked by the kernel. A full build is a large computation; the companion's README lists its requirements.

Generative AI tools were used substantively throughout this work, including proof search, parameter selection, drafting, and the construction of the Lean formalization; see the AI methodology statement at the end of the paper. The correctness of both theorems, as formally stated, rests on the kernel-checked proof described above.

1.6 Organization

Figure 1 gives the architecture.

§2 The counting theorem (matrix concentration is in Appendix D).
§3 Product-star cells and their mixed residuals.
§4 Sampling bounded real matrices.
§5 The cycle-sum identity and its computation.
§6 Exact integer polynomial masks.
§7 Localization and certification of walk sums.
§8 Exact printing of the modular tables.
§9 The counting algorithm.
§10 Soundness, completeness and running time.
§11 The parameter choices and quartic wire cap.
§12 The almost-everywhere lower-bound transfer.
§13 Scope.
The appendices give the wire-measure transfer, notation, attribution, and the statements and proofs from earlier work.

2 The counting theorem

This section states the counting guarantee. Supporting statements and their complete proofs are in Appendices D and E. The published foundations are identified in Section 1.4.

2.1 Normalized calls and the counting theorem

A *call* asks for the acceptance probability of one of three predicates on $\{0,1\}^n$: one circuit C^1; the exclusive-or $C^1 \oplus C^2$ of two circuits; or $C^1 \oplus \mathrm{PAR}_\Sigma$, where $\mathrm{PAR}_\Sigma(x) = \bigoplus_{i\in\Sigma} x_i$ is an explicit input parity. We call these *single*, *pair* and *parity* calls and write pred for the predicate. Let $n \geq 4$ and

$$B_0 = 4n\lceil \log_2(n+1)\rceil.$$

A call is *normalized* if its circuits are SYM $\circ$ THR circuits on n inputs whose bottom weights and thresholds are integers of magnitude below 2^{B_0}, with explicit lookup tables, and whose total wire count satisfies $W = W(C^1) + W(C^2) \leq 2n^4$ (with $W(C^2) = 0$ for single and parity calls). The call also contains a rational radius β with $n^{-B_1} \leq \beta \leq 1/16$, encoded in at most n^{B_2} bits, for fixed positive integers B_1, B_2.

Every single, pair or parity call made from SYM $\circ$ THR circuits with total wire count at most $2n^4$ can be put into the integer-parameter form above, with the same bottom gates and no more wires. Indeed, apply supporting statement D.10 to each bottom gate as a function of the $f \geq 1$ inputs it reads (a gate of fan-in zero is a constant and is written $\mathbf{1}[0 \geq \theta]$ with $\theta \in \{0,1\}$): its new weights and threshold are integers in $\pm f^f$, and $f^f \leq n^n < 2^{B_0}$. No new input is read, so the wire count does not increase. Running times are counted in steps of a deterministic multitape Turing machine.

Theorem 2.1 (Certified counting below the quartic wire cap)**.** *Fix positive integers B_1, B_2. There are a constant $a = a_{B_1,B_2} > 0$ and one deterministic algorithm whose input is a call on n inputs and a binary string, the* witness*, with the following properties.*

(i) (Soundness) For every witness, it returns dyadic rationals $(\widehat{p}, \widehat{\kappa})$ with $|\widehat{p} - \Pr_x[\mathrm{pred}(x) = 1]| \leq \widehat{\kappa}$.

(ii) (Completeness) For all sufficiently large n, on every normalized call, if $W \leq a\,\beta\,\mathcal{A}_4(n)$, some witness of polynomial length gives $\widehat{\kappa} \leq \beta$.

(iii) (Time) For all sufficiently large n, on every normalized call and every witness it halts within $2^n/n^{\mu_n}$ steps, reads a prefix of polynomial length of the witness, and uses no oracle. Hence any prescribed fixed polynomial number of calls has total workload $2^n n^{O(1)-\mu_n}$.

(iv) (Interface) Its outputs are dyadic with denominators dividing 2^{n+1} and encoded length polynomial in n. On an input that is not a normalized call it returns $(1/2, 1/2)$ in time polynomial in the encoded length of the call, independently of the witness.

Lean: `NearQuarticWires.Program.counterExists4`, from Williams' rectangular-product input.

The same cap handles all three call types; this is sufficient for the transfer. Sections 3 to 11 prove Theorem 2.1.

The standard matrix concentration inequality used below is supporting statement D.1 in Appendix D; throughout, $\|\cdot\|$ is the operator norm.

3 The call on product-star cells

This section rewrites a call on one cell. Each score splits into a part that depends only on the left pattern, a part that depends only on the right pattern, and a mixed residual (Lemma 3.1). The mixed residual is rare (Lemmas 3.2 and 3.3), and a Fourier expansion moves everything except the residual into one-sided selectors (Lemma 3.4).

Fix a normalized call and one of the 2^k partitions of supporting statement D.2; its index is part of the witness. Recall from there that a cell has an address $z \in \{0,1\}^{n-k}$, its *column*, a center $c(z)$, and points $c(z) \oplus u \oplus v$; the left pattern $u \in [N]$ flips at most one coordinate in each of $r/2$ blocks of $b = 2^s - 1$ coordinates, the right pattern v likewise in the other $r/2$ blocks, and $F(u)$ is the set of coordinates flipped by u. Let $\mathcal{J} \subseteq \{1,2\}$ be the set of circuits of the call ($\mathcal{J} = \{1,2\}$ for pair calls and $\mathcal{J} = \{1\}$ otherwise). Circuit $j \in \mathcal{J}$ has bottom gates $T_{j,1}, \dots, T_{j,G_j}$ and lookup g_j; put $G_2 = 0$ if $2 \notin \mathcal{J}$. For a column z, write $c = c(z)$ and define, for $u, v \in [N]$,

$$\Delta_{j,i}(z,u,v) = T_{j,i}(c \oplus u \oplus v) - T_{j,i}(c \oplus u) - T_{j,i}(c \oplus v) + T_{j,i}(c), \qquad R_j = \sum_i \Delta_{j,i},$$

$$s_j^{\mathrm{L}}(z,u) = \sum_i \big(T_{j,i}(c \oplus u) - T_{j,i}(c)\big), \qquad s_j^{\mathrm{R}}(z,v) = \sum_i T_{j,i}(c \oplus v),$$

$$s_j(z,u,v) = \sum_i T_{j,i}(c \oplus u \oplus v), \qquad \mathrm{pop}(z,u,v) = \sum_{j \in \mathcal{J}} \sum_i |\Delta_{j,i}(z,u,v)|.$$

We call s_j the *score* of circuit j, $\Delta_{j,i}$ the *mixed residual* of the bottom gate (j,i), and pop the *population* of the point. The values $T_{j,i}(c \oplus x)$ for $x \in \{0, u, v, u \oplus v\}$ are the *atoms* of the bottom gate at the point.

Throughout, $\Delta_{j,i}$ is used only as the integer linear form displayed above in four atoms, and $|\Delta_{j,i}| = \Delta_{j,i}^2$ (by supporting statement D.3) only as an integer quadratic form. No representation of a residual over $\mathbb{F}_2$ is used.

Lemma 3.1 (Exact score identity)**.** *$s_j = s_j^{\mathrm{L}} + s_j^{\mathrm{R}} + R_j$ at every point, $s_j \in [0, G_j]$, and $|R_j| \le G_j$.*

Proof. Summing the definitions gives $s_j^{\mathrm{L}} + s_j^{\mathrm{R}} + R_j = \sum_i T_{j,i}(c \oplus u \oplus v)$. The range claims follow from $T \in \{0,1\}$ and $\Delta \in \{-1, 0, 1\}$. □

Lemma 3.2 (Mixed dependence)**.** *If $\Delta_{j,i}(z,u,v) \neq 0$, then $T_{j,i}$ has a nonzero weight on some coordinate of $F(u)$ and on some coordinate of $F(v)$. Consequently, if d_l is the number of bottom gates, of all circuits of the call, with nonzero weight on coordinate l, then for every column z*

$$\operatorname{pop}(z,u,v) \le \min\Big\{ \sum_{l\in F(u)} d_l,\ \sum_{l\in F(v)} d_l \Big\}.$$

Proof. If $T = T_{j,i}$ has zero weight on all of $F(u)$, then $T(c\oplus u\oplus v) = T(c\oplus v)$ and $T(c\oplus u) = T(c)$, so $\Delta = 0$; symmetrically for $F(v)$. Each bottom gate with nonzero residual is therefore incident to $F(u)$, and also to $F(v)$. □

Lemma 3.3 (Product-star mean population)**.** *Let r be even with $2 \le r \le n/4$, set $s = \lfloor \log_2(n/r+1) \rfloor$, $b = 2^s - 1$, and $k = rs$. Use r disjoint blocks of b coordinates, split equally between left and right, and a binary Hamming code whose parity-check columns are the nonzero vectors of $\mathbb{F}_2^s$. Choosing a code coset in each block gives 2^k partitions into products of stars, each with 2^{n-k} cells and 2^k points per cell; other coordinates belong to the center. Let $T_1,\dots,T_G$ be integer-parameter Boolean threshold gates with fan-ins f_i, counted with repetition, and $\sum_i f_i \le W$. Define*

$$\Delta_i(c,u,v) = T_i(c\oplus u\oplus v) - T_i(c\oplus u) - T_i(c\oplus v) + T_i(c), \qquad \operatorname{pop}(z,u,v) = \sum_i |\Delta_i(c(z),u,v)|.$$

For a uniform partition index σ, uniform cell z, and independent uniform left and right star patterns u, v,

$$\mathbb{E}_{\sigma,z,u,v} \operatorname{pop}(z,u,v) \le 6r^{3/2}W/n^{3/2}.$$

Consequently some fixed partition has the same bound on $\mathbb{E}_{z,u,v} \operatorname{pop}(z,u,v)$.

Lean: `NearQuarticWires.Cells.average_population_le` and `Cells.mean_population`. This is the product-star estimate from [Nag26b, Appendix A], used here by the collective cycle analysis.

Proof. Averaged over the partition index and the column, which together give a uniform center independent of the patterns (supporting statement D.2), supporting statement D.4 gives $\Pr[\Delta_i \neq 0] \le 2f_i/(b+1)^{3/2} \le 2f_i(2r/n)^{3/2} \le 6r^{3/2}f_i/n^{3/2}$, using $b+1 > n/(2r)$. Sum over bottom gates and choose a partition index whose mean is at most the average. □

3.1 Fourier decomposition with the mask inside the matrix

Fix the trace order $m = 4$. Put $\Omega_j = 4mG_j + 1$ for $j \in \mathcal{J}$ and $\Omega_2 = 1$ if $2 \notin \mathcal{J}$, let $\Theta = \mathbb{Z}/\Omega_1 \times \mathbb{Z}/\Omega_2$, $\omega_j = e^{2\pi i/\Omega_j}$, and let $M_F = \lceil \sqrt{|\Theta|}\, \rceil$, the least integer with $M_F^2 \ge |\Theta|$. The modulus $\Omega_j > 2mG_j$ exceeds every score, which lies in $[0, G_j]$, and the absolute value of every walk sum used later, which is at most $2mG_j$ (Lemma 5.1).

Lemma 3.4 (Fourier decomposition)**.** *For each call there are complex numbers $\widehat{F}(\theta,\nu)$, $(\theta,\nu) \in \Theta \times \{0,1\}$, with $\sum_{\theta,\nu} |\widehat{F}(\theta,\nu)| \le M_F$, and for every column z functions $a_{z,\theta,\nu}, b_{z,\theta,\nu} : [N] \to \mathbb{C}$ of modulus at most one, such that for every (u,v) and every $\mu_0 \in \{0,1\}$*

$$\operatorname{pred}\big(c(z)\oplus u\oplus v\big)\,\mu_0 = \sum_{\theta,\nu} \widehat{F}(\theta,\nu)\, a_{z,\theta,\nu}(u)\, b_{z,\theta,\nu}(v)\, \omega_1^{\theta_1 R_1(z,u,v)} \omega_2^{\theta_2 R_2(z,u,v)}\, \mu_0.$$

Proof. Single and pair calls. Define F on Θ by $F(t_1,t_2)=g_1(t_1)$ (single) or $g_1(t_1)\oplus g_2(t_2)$ (pair) for $(t_1,t_2)\in[0,G_1]\times[0,G_2]$, and $F=0$ elsewhere; this is well defined because $\Omega_j>G_j$. Put $\widehat{F}(\theta,0)=|\Theta|^{-1}\sum_t F(t)\omega_1^{-\theta_1 t_1}\omega_2^{-\theta_2 t_2}$ and $\widehat{F}(\theta,1)=0$. Character orthogonality on the finite group Θ gives $F(t)=\sum_\theta \widehat{F}(\theta,0)\omega_1^{\theta_1 t_1}\omega_2^{\theta_2 t_2}$, Parseval gives $\sum_\theta|\widehat{F}(\theta,0)|^2=|\Theta|^{-1}\sum_t F(t)^2\le 1$, and the Cauchy–Schwarz inequality gives $\sum_\theta|\widehat{F}|\le\sqrt{|\Theta|}\le M_F$. By Lemma 3.1, $\mathrm{pred}=F(s_1,s_2)$ and $\omega_j^{\theta_j s_j}=\omega_j^{\theta_j s_j^{\mathrm{L}}}\omega_j^{\theta_j s_j^{\mathrm{R}}}\omega_j^{\theta_j R_j}$. Take $a_{z,\theta,0}(u)=\prod_j\omega_j^{\theta_j s_j^{\mathrm{L}}(z,u)}$ and $b_{z,\theta,0}(v)=\prod_j\omega_j^{\theta_j s_j^{\mathrm{R}}(z,v)}$.

Parity calls. By linearity, $\mathrm{PAR}_\Sigma(c\oplus u\oplus v)=\mathrm{PAR}_\Sigma(c)\oplus\mathrm{PAR}_\Sigma(u)\oplus\mathrm{PAR}_\Sigma(v)$. Let $\chi_0=(-1)^{\mathrm{PAR}_\Sigma(c(z))}$, $\chi_{\mathrm{L}}(u)=(-1)^{\mathrm{PAR}_\Sigma(u)}$, $\chi_{\mathrm{R}}(v)=(-1)^{\mathrm{PAR}_\Sigma(v)}$, and let $\Gamma_g(t)=(-1)^{g_1(t)}$ on $[0,G_1]$ and $\Gamma_g=0$ elsewhere on $\mathbb{Z}/\Omega_1$. Then

$$\mathrm{pred}=\tfrac12-\tfrac12\,\chi_0\chi_{\mathrm{L}}(u)\chi_{\mathrm{R}}(v)\,\Gamma_g(s_1).$$

Let $\widehat{F}(0,0)=\frac12$ with $a_{z,0,0}=b_{z,0,0}=1$, $\widehat{F}(\theta,0)=0$ for $\theta\neq0$, and $\widehat{F}(\theta,1)=-\frac12\widehat{\Gamma}_g(\theta)$ with $a_{z,\theta,1}=\chi_0\chi_{\mathrm{L}}\,\omega_1^{\theta_1 s_1^{\mathrm{L}}}$ and $b_{z,\theta,1}=\chi_{\mathrm{R}}\,\omega_1^{\theta_1 s_1^{\mathrm{R}}}$. As above $\sum_\theta|\widehat{\Gamma}_g(\theta)|\le\sqrt{\Omega_1}$, so the total mass is at most $\frac12+\frac12\sqrt{\Omega_1}\le\sqrt{\Omega_1}\le M_F$. □

For a 0/1 matrix $\mathrm{mask}(z)$ on $[N]^2$, defined in Section 6, put

$$M_\theta(z)[u,v]=\omega_1^{\theta_1 R_1(z,u,v)}\omega_2^{\theta_2 R_2(z,u,v)}\mathrm{mask}(z,u,v),\qquad D_{\theta,0}(z)=\mathrm{Re}\,M_\theta(z),\quad D_{\theta,1}(z)=\mathrm{Im}\,M_\theta(z).$$

These $I=2|\Theta|$ real matrices have entries in $[-1,1]$. They depend on the circuits, the partition and the mask, but not on the lookups or the parity set: the lookups and the parity enter only through the coefficients $\widehat{F}$ and the selectors. The second index of D specifies the real or imaginary part, independently of the selector index in Lemma 3.4.

4 Sampling bounded real matrices

A list of Q sample pairs estimates the average of a matrix D against selectors a,b with an error governed by the operator norm of one error matrix, uniformly in the selectors. This section shows that the fixed-moment analysis of [Nag26b, Lemma F.1], written there for 0/1 matrices, holds for real matrices with entries in $[-1,1]$, and records the two consequences we use. In this section m is any positive integer; the algorithm uses $m=4$.

For a list of Q pairs $(U_j,V_j)\in[N]^2$ let

$$\Gamma(x)=Q-N^2\,\#\{j:(U_j,V_j)=x\}\qquad(x\in[N]^2).$$

For a real $N\times N$ matrix D, the error matrix $E_D=QD-N^2\sum_j D(U_j,V_j)e_{U_j}e_{V_j}^{\mathsf T}$ satisfies

$$E_D[x]=D[x]\,\Gamma(x)\qquad(x\in[N]^2),\tag{1}$$

since the x entry of the sum counts the samples at x. Let $\mathcal{H}(E)=\left(\begin{smallmatrix}0&E\\E^{\mathsf T}&0\end{smallmatrix}\right)$; for real E with singular values s_i, $\mathrm{tr}\,\mathcal{H}(E)^{2m}=2\sum_i s_i^{2m}$.

Lemma 4.1 (Trace bound for entries in $[-1,1]$)**.** *If D is real with entries in $[-1,1]$ and the pairs are independent and uniform, then $\mathbb{E}\,\mathrm{tr}\,\mathcal{H}(E_D)^{2m}\le 4N^2((4QN^3)^m m!+(8N^2/3)^{2m}(2m)!)$.*

Proof. $E_D=\sum_{j=1}^Q X_j$ with independent $X_j=D-N^2D(U_j,V_j)e_{U_j}e_{V_j}^{\mathsf T}$. We check the hypotheses of supporting statement D.1. (1) $\mathbb{E}[N^2D(U,V)e_Ue_V^{\mathsf T}]=\sum_{u,v}D(u,v)e_ue_v^{\mathsf T}=D$, so $\mathbb{E}X_j=0$. (2) $\|D\|\le\|D\|_F\le N$, so $\|X_j\|\le N+N^2\le 2N^2$. (3) $\mathbb{E}X_jX_j^{\mathsf T}=N^2\,\mathrm{diag}_u(\sum_v D(u,v)^2)-DD^{\mathsf T}\preceq N^3\mathrm{Id}$,

because $\sum_v D(u,v)^2 \le N$ and $DD^{\mathsf T} \succeq 0$; likewise $\mathbb{E}X_j^{\mathsf T}X_j \preceq N^3\mathrm{Id}$. With $L = 2N^2$ and $\sigma^2 \le QN^3$, supporting statement D.1 gives

$$\Pr[\|E_D\| \ge x] \le 2N\exp\Big(-\frac{x^2}{2(QN^3 + 2N^2x/3)}\Big) \le 2N\big(e^{-x^2/(4QN^3)} + e^{-3x/(8N^2)}\big),$$

since $2(a+b) \le 4\max\{a,b\}$. As $\operatorname{tr}\mathcal{H}(E)^{2m} \le 2N\|E\|^{2m}$,

$$\begin{aligned}
\mathbb{E}\operatorname{tr}\mathcal{H}(E_D)^{2m} &\le 2N\int_0^\infty 2mx^{2m-1}\Pr[\|E_D\| > x]\,dx \\
&\le 4N^2\cdot 2m\Big(\int_0^\infty x^{2m-1}e^{-x^2/(4QN^3)}dx \\
&\qquad + \int_0^\infty x^{2m-1}e^{-3x/(8N^2)}dx\Big) \\
&= 4N^2\cdot 2m\Big(\tfrac12(4QN^3)^m(m-1)! + (8N^2/3)^{2m}(2m-1)!\Big) \\
&= 4N^2\big((4QN^3)^m m! + (8N^2/3)^{2m}(2m)!\big).
\end{aligned}$$

Only (1)–(3) were used about the summands. □

The Lean companion proves the $m = 4$ bound it uses by a direct moment computation, without Tropp's inequality (`NearQuarticWires.Sampler.mean_traceMoment_le`).

Lemma 4.2 (Uniform error over selectors)**.** *For every column z, sample list and mask, put $\varepsilon_z = \max_{\theta,\nu}\|E_{D_{\theta,\nu}(z)}\|/(QN)$. Then*

$$\Big|\frac{1}{N^2}\sum_{u,v}(\mathrm{pred}\cdot\mathrm{mask})(u,v) - \frac{1}{Q}\sum_j(\mathrm{pred}\cdot\mathrm{mask})(U_j,V_j)\Big| \le 2M_F\,\varepsilon_z,$$

$$\Big|\frac{1}{N^2}\sum_{u,v}\mathrm{mask}(u,v) - \frac{1}{Q}\sum_j\mathrm{mask}(U_j,V_j)\Big| \le \varepsilon_z,$$

where $(\mathrm{pred}\cdot\mathrm{mask})(u,v) = \mathrm{pred}(c(z)\oplus u\oplus v)\mathrm{mask}(z,u,v)$.

Proof. For $a,b \in \mathbb{C}^N$ with coordinates of modulus at most one and a real matrix E, $|a^{\mathsf T}Eb| \le \|E\|\,\|a\|_2\|b\|_2 \le N\|E\|$; the operator norm of a real matrix is the same over $\mathbb{R}$ and over $\mathbb{C}$. For each (θ,ν), take $a = a_{z,\theta,\nu}$ and $b = b_{z,\theta,\nu}$,

$$\frac{a^{\mathsf T}M_\theta b}{N^2} - \frac{1}{Q}\sum_j a_{U_j}b_{V_j}M_\theta(U_j,V_j) = \frac{a^{\mathsf T}(E_{D_{\theta,0}} + \mathrm{i}E_{D_{\theta,1}})b}{QN^2},$$

of modulus at most $2N\cdot QN\varepsilon_z/(QN^2) = 2\varepsilon_z$. Multiply by $\widehat{F}(\theta,\nu)$, sum, and apply Lemma 3.4 with $\mu_0 = \mathrm{mask}(z,u,v)$ at each point. For the second line use $\theta = 0$, $M_0 = \mathrm{mask} = D_{0,0}$ and $a = b = \mathbf{1}$. □

Lemma 4.3 (From a potential bound to an average error)**.** *Let Z be a nonempty finite set of columns, let $m \ge 1$, let $0 < \eta \le 1$, and let $Q \ge 1024Nm\eta^{-2}(4N^2I)^{1/m}$. If real matrices $E_t(z)$, $z \in Z$, $t \le I$, satisfy*

$$\sum_{z\in Z}\sum_{t\le I}\operatorname{tr}\mathcal{H}(E_t(z))^{2m} \le |Z|\,I\,4N^2\big((4QN^3)^m m! + (8N^2/3)^{2m}(2m)!\big),$$

then $|Z|^{-1}\sum_{z\in Z}\max_t\|E_t(z)\|/(QN) \le \eta$.

Proof. Since $\operatorname{tr}\mathcal{H}(E)^{2m} = 2\sum_i s_i^{2m} \geq 2\|E\|^{2m}$, $\max_t \|E_t(z)\|^{2m} \leq \frac{1}{2}\sum_t \operatorname{tr}\mathcal{H}(E_t(z))^{2m}$. By Jensen's inequality for $x \mapsto x^{1/(2m)}$,

$$\frac{1}{|Z|}\sum_z \max_t \|E_t(z)\| \leq \Big(2N^2 I\big((4QN^3)^m m! + (8N^2/3)^{2m}(2m)!\big)\Big)^{1/(2m)}$$
$$\leq (4N^2 I)^{1/(2m)}\Big(\sqrt{4QN^3 m} + \tfrac{16N^2 m}{3}\Big),$$

using $(x+y)^{1/(2m)} \leq x^{1/(2m)} + y^{1/(2m)}$, $m! \leq m^m$ and $(2m)! \leq (2m)^{2m}$. Dividing by QN, the bound is $(4N^2 I)^{1/(2m)}(2\sqrt{Nm/Q} + 16Nm/(3Q))$. Under the hypothesis on Q the first term is at most $\eta/16$ and the second at most $\eta/192$. □

This is the argument of [Nag26b, Lemma F.1], with an arbitrary set Z of columns in place of all columns.

5 The cycle-sum identity

The sampler of Section 4 needs a sample list with a small potential $\sum_{\theta,\nu} \operatorname{tr}\mathcal{H}(E_{D_{\theta,\nu}(z)})^{2m}$. We will choose it bit by bit, and for that we must compute the potential's conditional expectations. This section shows that the potential is a weighted count of closed walks with vanishing signed residual sums.

A *closed walk* is $w = (u_1, \ldots, u_m, v_1, \ldots, v_m) \in [N]^{2m}$; its *edges* are $e_{2i-1} = (u_i, v_i)$ and $e_{2i} = (u_{i+1}, v_i)$ for $1 \leq i \leq m$, with $u_{m+1} = u_1$. There are $N^{2m} = 2^{mk}$ closed walks. For a sign pattern $\sigma \in \{\pm 1\}^{2m}$ put

$$Y_{j,\sigma}(z,w) = \sum_{l=1}^{2m} \sigma_l R_j(z, e_l), \qquad \kappa_\sigma = 1 + (-1)^m \prod_l \sigma_l \in \{0, 2\},$$

and let $\Gamma_w = \prod_l \Gamma(e_l)$ and $\operatorname{mask}_w(z) = \prod_l \operatorname{mask}(z, e_l)$. If $2 \notin \mathcal{J}$, then $Y_{2,\sigma} \equiv 0$. We call $Y_{j,\sigma}$ a *walk sum.*

Lemma 5.1 (Cycle-sum identity)**.** *Let $m, N \geq 1$ and $Q, G_1, G_2 \geq 0$ be integers, let $R_j : [N]^2 \to \mathbb{Z}$ satisfy $|R_j(e)| \leq G_j$, and let $M : [N]^2 \to \mathbb{R}$ be any common real weight. For any sample list $S = (S_1, \ldots, S_Q)$, allowing repetitions, put*

$$\Omega_j = 4mG_j + 1, \quad \Theta = \mathbb{Z}/\Omega_1 \times \mathbb{Z}/\Omega_2, \quad \Gamma(e) = Q - N^2|\{t : S_t = e\}|,$$
$$\psi_\theta(e) = 2\pi(\theta_1 R_1(e)/\Omega_1 + \theta_2 R_2(e)/\Omega_2),$$
$$E_{\theta,0}(e) = \Gamma(e)M(e)\cos\psi_\theta(e), \quad E_{\theta,1}(e) = \Gamma(e)M(e)\sin\psi_\theta(e).$$

Let $\mathcal{H}(E) = \left(\begin{smallmatrix} 0 & E \\ E^{\mathsf{T}} & 0 \end{smallmatrix}\right)$. A closed alternating walk $w \in [N]^{2m}$ has edges $e_{2i-1} = (u_i, v_i)$, $e_{2i} = (u_{i+1}, v_i)$, where $u_{m+1} = u_1$. For $\sigma \in \{\pm 1\}^{2m}$, set

$$Y_{j,\sigma}(w) = \sum_{l=1}^{2m} \sigma_l R_j(e_l), \quad \kappa_\sigma = 1 + (-1)^m \prod_l \sigma_l, \quad \Gamma_w = \prod_l \Gamma(e_l), \quad M_w = \prod_l M(e_l).$$

Then the frequency-summed real/imaginary trace potential is the signed weighted walk count

$$\sum_{\theta\in\Theta}\sum_{\nu=0}^{1} \operatorname{tr}\mathcal{H}(E_{\theta,\nu})^{2m} = \frac{2|\Theta|}{4^m}\sum_w \Gamma_w M_w \sum_{\sigma\in\{\pm1\}^{2m}} \kappa_\sigma \mathbf{1}[Y_{1,\sigma}(w) = 0]\mathbf{1}[Y_{2,\sigma}(w) = 0].$$

Lean: `NearQuarticWires.Cycle.cycle_sum_identity` checks the $m = 4$ specialization (eight edges); the displayed algebraic proof holds for every $m \geq 1$. For circuit applications take $M = \mathrm{mask}$. The sample and common weight must be the same at every frequency.

Proof. For real E, $\mathcal{H}(E)^2 = \mathrm{diag}(EE^{\mathsf{T}}, E^{\mathsf{T}}E)$, so $\operatorname{tr}\mathcal{H}(E)^{2m} = 2\operatorname{tr}((EE^{\mathsf{T}})^m)$, and expanding the matrix product, $\operatorname{tr}((EE^{\mathsf{T}})^m) = \sum_w \prod_{i=1}^m E(u_i, v_i)E(u_{i+1}, v_i) = \sum_w \prod_{l=1}^{2m} E(e_l)$. By definition, $E_{\theta,\nu}(e) = \Gamma(e)M(e)\,\mathrm{cs}_\nu(\psi_\theta(e))$ with $\mathrm{cs}_0 = \cos$, $\mathrm{cs}_1 = \sin$ and $\psi_\theta(e) = 2\pi(\theta_1 R_1(e)/\Omega_1 + \theta_2 R_2(e)/\Omega_2)$. For reals $x_1, \ldots, x_{2m}$, expanding $\cos x = (e^{\mathrm{i}x} + e^{-\mathrm{i}x})/2$ and $\sin x = (e^{\mathrm{i}x} - e^{-\mathrm{i}x})/(2\mathrm{i})$ gives

$$\prod_l \cos x_l = 4^{-m} \sum_\sigma e^{\mathrm{i}\sum_l \sigma_l x_l}, \qquad \prod_l \sin x_l = (2\mathrm{i})^{-2m} \sum_\sigma \Big(\prod_l \sigma_l\Big) e^{\mathrm{i}\sum_l \sigma_l x_l},$$

and $(2\mathrm{i})^{-2m} = (-1)^m 4^{-m}$. Adding, $\prod_l \cos x_l + \prod_l \sin x_l = 4^{-m}\sum_\sigma \kappa_\sigma e^{\mathrm{i}\sum_l \sigma_l x_l}$. With $x_l = \psi_\theta(e_l)$ we have $\sum_l \sigma_l x_l = 2\pi(\theta_1 Y_{1,\sigma}/\Omega_1 + \theta_2 Y_{2,\sigma}/\Omega_2)$, and

$$\sum_{\theta\in\Theta} e^{2\pi\mathrm{i}(\theta_1 Y_1/\Omega_1 + \theta_2 Y_2/\Omega_2)} = |\Theta|\,\mathbf{1}[Y_1 \equiv 0 \pmod{\Omega_1}]\,\mathbf{1}[Y_2 \equiv 0 \pmod{\Omega_2}].$$

Finally $|Y_{j,\sigma}| \leq \sum_l |R_j(e_l)| \leq 2mG_j < \Omega_j$ by the assumed bound, so $Y_{j,\sigma} \equiv 0 \pmod{\Omega_j}$ if and only if $Y_{j,\sigma} = 0$. Collect the factor 2 from the trace and the factor $\Gamma_w M_w$. □

For a circuit column z, take $M(e) = \mathrm{mask}(z, e)$ and $R_j(e) = R_j(z, e)$. By (1) this gives the original column potential

$$\Phi_z := \sum_{\theta\in\Theta}\sum_{\nu\in\{0,1\}} \operatorname{tr}\mathcal{H}(E_{D_{\theta,\nu}(z)})^{2m} = \frac{2|\Theta|}{4^m}\sum_w \Gamma_w\,\mathrm{mask}_w(z)\sum_\sigma \kappa_\sigma\,\mathbf{1}[Y_{1,\sigma}(z,w) = 0]\,\mathbf{1}[Y_{2,\sigma}(z,w) = 0].$$

The identity holds for every column, sample list and mask, with no assumption on populations. The one-sided scores never enter the matrices; they sit in the selectors of Lemma 3.4. Both sides are finite sums of real numbers, and the left side is nonnegative, although individual walk weights Γ_w may be negative.

The same identity gives exact conditional expectations under the short seed: Corollary E.16 records the formula and its computation, with proof in Section E.24.

6 Masks that are exact integer polynomials

The mask must discard the points where the population is too large to localize, and it must enter the residue counts of Section 8, which are computed modulo p^e for every odd prime $p \in (P/2, P]$; reconstruction needs the *same* 0/1 mask for every prime. The masks of [Nag26b, Appendix D] are polynomials over $\mathbb{F}_2$, converted to integers by a binomial lift that is exact only modulo powers of two [Nag26b, Lemma C.7], so they cannot be multiplied into these counts. The mask below is an exact integer polynomial wherever it is used, so no modulus and no lift is involved.

Parameters. Integers $\tau \geq 1$, $U \geq \tau$ and $X_{\max} \geq 3r\tau$ are fixed in Section 9.1; put $\pi_0 = \tau/(4U) \leq 1/4$ and $d^* = \lfloor 2X_{\max}U/(3r\tau)\rfloor \geq 1$. The witness contains a set $\mathcal{S}$ of bottom gate labels (j, i).

Definition 6.1 (Mask)**.** Let $H_c = \{l : d_l \geq d^*\}$, with d_l as in Lemma 3.2. Let $\mathrm{smask}(u, v) = 0$ if both $F(u) \cap H_c \neq \emptyset$ and $F(v) \cap H_c \neq \emptyset$, and $\mathrm{smask}(u, v) = 1$ otherwise. Let $X(z, u, v) = \sum_{(j,i)\in\mathcal{S}} |\Delta_{j,i}(z, u, v)|$ and

$$\mathrm{mask}(z, u, v) = \mathrm{smask}(u, v) \cdot \mathbf{1}[X(z, u, v) \leq \tau].$$

The *structural mask* smask is a fixed 0/1 matrix on $[N]^2$, the same in every column; it depends only on the circuits' incidences and the partition.

Lemma 6.1 (Integer threshold masks)**.** *For nonnegative integers τ, D, put*

$$c_a = \sum_{i=0}^{\min(a,\tau)} (-1)^{a-i} \binom{a}{i} \in \mathbb{Z} \quad (0 \le a \le D).$$

Then $|c_a| \le 2^a$, and for every integer $0 \le X \le D$,

$$\mathbf{1}[X \le \tau] = \sum_{a=0}^{D} c_a \binom{X}{a}.$$

Equivalently, for Boolean variables $y_1, \dots, y_G$, $P_{\tau,D}(y) = \sum_{a=0}^{D} c_a e_a(y)$ is a multilinear integer polynomial of degree at most D, each coefficient has magnitude at most 2^D, and

$$P_{\tau,D}(y) = \mathbf{1}\Big[\sum_i y_i \le \tau\Big] \quad \text{whenever } \sum_i y_i \le D.$$

The same identity holds modulo every integer. Writing $X = \sum_i y_i$, if $s \in \{0,1\}$, then $sP_{\tau,D}(y) = s\mathbf{1}[X \le \tau]$ whenever $X \le D$ or $s = 0$.

For the circuit mask, take $D = X_{\max}$, set $b_{j,i} = \Delta_{j,i}^2$, an integer polynomial of degree at most two in the four atoms of $T_{j,i}$. Define the integer polynomial

$$\mathrm{Mpoly}(z,u,v) = \mathrm{smask}(u,v) \sum_{a=0}^{X_{\max}} c_a \, e_a\big((b_{j,i}(z,u,v))_{(j,i)\in\mathcal{S}}\big),$$

where e_a is the elementary symmetric polynomial of degree a. Then $\mathrm{Mpoly}(z,u,v) = \mathrm{mask}(z,u,v)$ whenever $X(z,u,v) \le X_{\max}$ or $\mathrm{smask}(u,v) = 0$. Its degree in the atoms is at most $2X_{\max}$, and with separate atom variables for each bottom gate, after multilinearization its coefficients have $O(X_{\max} + \lambda_n)$ bits.

The scalar and Boolean identities and the bound $|c_a| \le 2^a$ are formalized by `NearQuarticWires.Mask.cutoff_eq_newton_sum_of_le`, `Mask.newtonCoeff_eq_sum_min`, `Mask.newtonCoeff_abs_le`, and `Mask.indicatorPolynomial_eq_mask`.

Proof. The $b_{j,i}$ take values in $\{0,1\}$ and sum to X, so $e_a(b) = \binom{X}{a}$. For $f : \{0, \dots, X_{\max}\} \to \mathbb{Z}$ and an integer $0 \le x \le X_{\max}$, Newton's forward-difference formula $f(x) = \sum_{a=0}^{x} \binom{x}{a} (\Delta^a f)(0)$ holds (write the shift operator as $1 + \Delta$ and expand $(1+\Delta)^x$ by the binomial theorem); since $\binom{x}{a} = 0$ for $a > x$, the sum may run to $X_{\max}$. For $f = \mathbf{1}[\cdot \le \tau]$, $(\Delta^a f)(0) = \sum_{i \le a} (-1)^{a-i} \binom{a}{i} f(i) = c_a$. Hence $\sum_{a \le X_{\max}} c_a \binom{X}{a} = \mathbf{1}[X \le \tau]$ for $0 \le X \le X_{\max}$. The degree bound holds because e_a has degree a in the b's, and $|c_a| \le 2^a$. For Boolean y with $X = \sum_i y_i$, $e_a(y) = \binom{X}{a}$, giving the generic identity; reducing this integer equality gives every modulus. In the independent indicator variables, a squarefree monomial indexed by I has coefficient $c_{|I|}$, proving the degree and coefficient claims. With separate atom variables for each bottom gate, a monomial identifies its selected gates; each substituted quadratic has bounded coefficient mass, giving the stated bit bound. □

Unlike a probabilistic or sign representation [ACW16], the bounded-range identity here is deterministic over $\mathbb{Z}$, so the same mask can be used in every residue computation. Hence Mpoly represents the mask modulo every integer simultaneously, in particular modulo p^e for every prime p, as soon as $X \le X_{\max}$ is certified (Lemma 7.3).

Lemma 6.2 (Population under the structural mask)**.** *If* $\mathrm{smask}(u,v)=1$*, then* $\mathrm{pop}(z,u,v)<(r/2)d^*$ *for every column* z*.*

Proof. One of $F(u), F(v)$ avoids H_c; it has at most $r/2$ coordinates, each with $d_l < d^*$. Apply Lemma 3.2. □

Lemma 6.3 (Filter guarantees)**.** *Let* $\mathcal{S}$ *contain each bottom gate label independently with probability* π_0*. For every fixed* (z,u,v)*:*

(a) if $\mathrm{pop}(z,u,v)\le U$*, then* $\Pr[X>\tau]\le(e/4)^{\tau+1}$*;*

(b) if $\mathrm{pop}(z,u,v)\ge 16U$*, then* $\Pr[X\le\tau]\le e^{-9\tau/8}$*;*

(c) if $\mathrm{smask}(u,v)=1$*, then* $\Pr[X\ge X_{\max}/2]\le 2^{-X_{\max}/2}$*.*

Proof. X counts the $\mathrm{pop}(z,u,v)$ labels with $|\Delta|=1$ that lie in $\mathcal{S}$, so X is binomial with parameters (pop,π_0) and mean $\bar{X}=\pi_0\,\mathrm{pop}$. We use the Chernoff bounds $\Pr[X\ge t]\le(e\bar{X}/t)^t$ for $t>\bar{X}$ and $\Pr[X\le(1-\delta)\bar{X}]\le e^{-\delta^2\bar{X}/2}$ for $0<\delta<1$. (a) $\bar{X}\le\tau/4$; take $t=\tau+1$. (b) $\bar{X}\ge 4\tau$; with $(1-\delta)\bar{X}=\tau$, $\delta^2\bar{X}=(\bar{X}-\tau)^2/\bar{X}$, which is increasing in $\bar{X}\ge\tau$, hence at least $(3\tau)^2/(4\tau)=9\tau/4$. (c) By Lemma 6.2, $\bar{X}<\pi_0(r/2)d^*\le\frac{\tau}{4U}\cdot\frac{r}{2}\cdot\frac{2X_{\max}U}{3r\tau}=X_{\max}/12$; take $t=\lceil X_{\max}/2\rceil$, giving $(e/6)^{X_{\max}/2}\le 2^{-X_{\max}/2}$. □

Lemma 6.4 (Structural masked mass)**.** *If* $W\ge\sum_l d_l$*, then* $\Pr_{u,v}[\mathrm{smask}(u,v)=0]\le\big(W/(d^*(b+1))\big)^2$*.*

Proof. $|H_c|\le\sum_l d_l/d^*\le W/d^*$. In each block, u chooses one of the $b+1$ options $\{0,e_1,\dots,e_b\}$ uniformly, so a fixed coordinate lies in $F(u)$ with probability at most $1/(b+1)$; a union bound gives $\Pr_u[F(u)\cap H_c\neq\emptyset]\le|H_c|/(b+1)$, and likewise for v. The patterns u and v are independent. □

7 Walk sums: localization and certification

The walk sum $Y_{j,\sigma}$ is too wide to be identified by its residues modulo primes of size P. Following [Nag26c, Section 5], nested subsampling writes it through short integer digits, which residues do identify, and a final error, which confines Y to a short box once the digits are known. Moment tests certify all of this on a whole column for an arbitrary guessed subsample, so soundness never depends on the subsample being honest.

7.1 Sampled sums and digits

For circuit j, bottom gate $t\in[G_j]$, walk w, pattern σ and column z let

$$c_{j,t}(z,w,\sigma)=\sum_{l=1}^{2m}\sigma_l\Delta_{j,t}(z,e_l)\in[-2m,2m],\qquad\text{so that}\qquad Y_{j,\sigma}(z,w)=\sum_t c_{j,t}(z,w,\sigma).$$

The witness contains vectors $\xi_r^{(j)}\in\{0,1\}^{G_j}$ for $1\le r\le J$ (soundness would hold for any integer vectors). Define

$$K_{r,\sigma}^{(j)}(z,w)=\sum_t\xi_{r,t}^{(j)}\,c_{j,t}(z,w,\sigma),\quad E_{0,\sigma}^{(j)}=K_{1,\sigma}^{(j)},\quad E_{r,\sigma}^{(j)}=K_{r+1,\sigma}^{(j)}-a_rK_{r,\sigma}^{(j)}\ (1\le r<J),$$

$$\varphi_\sigma^{(j)}=M_1K_{J,\sigma}^{(j)}-A\,Y_{j,\sigma},$$

with a_r, A and M_1 from the chain of supporting statement D.12, run with the parameters (T, h, ℓ_0) of Section 9.1. We call $E^{(j)}_{r,\sigma}$ the *digits* and $\varphi^{(j)}_{\sigma}$ the *final error*. All of these are integer linear forms in the atoms at the corners of the walk's edges. The same witness vectors serve every column, walk and pattern; the two circuits have separate vectors.

Lemma 7.1 (Localization of one walk sum)**.** *Fix z, w, σ and j, and suppose $\mathrm{pop}(z, e_l) \le 16U$ for all l. Put $\ell_0 = 32mU$, and let $\bar{\ell}$ and $s(\cdot)$ be as in supporting statement D.12. Let $\xi^{(j)}_{r,t} = \mathbf{1}[\upsilon_t \le \pi_r]$ with independent uniform $\upsilon_t \in [0,1]$. With probability at least $1 - 2Je^{-T}$,*

$$|E^{(j)}_{r,\sigma}(z,w)| \le mh \quad (0 \le r < J), \qquad |\varphi^{(j)}_{\sigma}(z,w)| \le 172\, m\, A\, T\bar{\ell}/h.$$

Proof. Write $c_t = c_{j,t}(z,w,\sigma)$ and $Y = Y_{j,\sigma}(z,w)$. Then $|c_t| \le 2m$ and $\sum_t |c_t| \le \sum_l \sum_t |\Delta_{j,t}(e_l)| \le \sum_l \mathrm{pop}(e_l) \le \ell_0$; in particular $|Y| \le \ell_0 \le \bar{\ell}$ and $\sum_t c_t^2 \le 2m\ell_0$. For a stage r, $K_r - \pi_r Y = \sum_t (\mathbf{1}[\upsilon_t \le \pi_r] - \pi_r) c_t$. The variables $Y_t = (\mathbf{1}[\upsilon_t \le \pi_r] - \pi_r)c_t/(2m)$ are independent, centered and bounded by one, with $\sum_t \mathbb{E}Y_t^2 \le \pi_r \sum_t c_t^2/(4m^2) \le \pi_r \ell_0/(2m) \le \pi_r \bar{\ell}$. Supporting statement D.11 gives $|K_r - \pi_r Y| \le 2m\, s(\pi_r)$ with probability at least $1 - 2e^{-T}$; take a union bound over the J stages. On this event, by supporting statement D.12(ii), $|E_0| = |K_1| \le \pi_1 |Y| + 2ms(\pi_1) \le 2m(\pi_1 \bar{\ell} + s(\pi_1)) \le mh$. For $r \ge 1$, since $\pi_{r+1} = a_r \pi_r$, $E_r = (K_{r+1} - \pi_{r+1} Y) - a_r (K_r - \pi_r Y)$, so $|E_r| \le 2m(s(\pi_{r+1}) + a_r s(\pi_r)) \le mh$. Since $M_1 \pi_J = A$, $\varphi = M_1(K_J - \pi_J Y)$, and by supporting statement D.12(iv) $|\varphi| \le 2mM_1 s(\pi_J) = 2mA\, s(\pi_J)/\pi_J \le 172 mAT\bar{\ell}/h$. □

This is the argument of [Nag26c, Lemma 5.2], with summands bounded by $2m$ instead of one; every stage error scales by $2m$.

7.2 Certification

Lemma 7.2 (Moment certificates)**.** *Let $K \ge 1$ be an integer, let $\mathcal{V}$ be a finite set with $|\mathcal{V}| \le 2^K$, let $v : \mathcal{V} \to \mathbb{Z}$, and let $R_0 \ge 1$ be an integer. If $\sum_{x \in \mathcal{V}} v(x)^{2K} < R_0^{2K}$, then $|v(x)| < R_0$ for all $x \in \mathcal{V}$. Conversely, if $|v| \le R'$ on $\mathcal{V}$ and $\sqrt{2}R' < R_0$, then $\sum_{x \in \mathcal{V}} v(x)^{2K} < R_0^{2K}$.*

Proof. One value of modulus at least R_0 contributes R_0^{2K} alone. Conversely, the sum is at most $2^K R'^{2K} = (\sqrt{2}R')^{2K} < R_0^{2K}$. □

This is [Nag26c, Lemma 5.3], stated for an arbitrary index set. Let $\mathsf{W}_{\mathrm{s}} = \{w : \prod_l \mathrm{smask}(e_l) = 1\}$, let h_E and Λ be as in Section 9.1, and write $\mathrm{Mpoly}_w(z) = \prod_l \mathrm{Mpoly}(z, e_l)$. Call a column z *certified* if the following strict tests pass:

$$\sum_{(u,v):\,\mathrm{smask}(u,v)=1} X(z,u,v)^{2k} < (X_{\max} + 1)^{2k}; \tag{2}$$

$$\sum_{w \in \mathsf{W}_{\mathrm{s}}} \mathrm{Mpoly}_w(z)\, E^{(j)}_{r,\sigma}(z,w)^{2mk} < h_E^{2mk} \quad \text{for all } j \in \mathcal{J},\ r < J,\ \sigma \text{ with } \kappa_\sigma \ne 0; \tag{3}$$

$$\sum_{w \in \mathsf{W}_{\mathrm{s}}} \mathrm{Mpoly}_w(z)\, \varphi^{(j)}_{\sigma}(z,w)^{2mk} < \Lambda^{2mk} \quad \text{for all } j \in \mathcal{J},\ \sigma \text{ with } \kappa_\sigma \ne 0. \tag{4}$$

Test (2) is evaluated first; if it fails, the column is not certified and (3), (4) are not consulted. Certification depends on the call, the partition, $\mathcal{S}$ and the vectors ξ, but not on the sampler seed.

Lemma 7.3 (Consequences of certification)**.** *On a certified column z, for every sign pattern σ with $\kappa_\sigma \ne 0$:*

(a) $X \le X_{\max}$ at every point with $\mathrm{smask} = 1$, so $\mathrm{Mpoly}_w(z) = \mathrm{mask}_w(z)$ for every walk w;

(b) for every walk with $\mathrm{mask}_w(z) = 1$, all digits satisfy $|E^{(j)}_{r,\sigma}| < h_E$ and $|\varphi^{(j)}_{\sigma}| < \Lambda$;

(c) *for every prime* $p \in (P/2, P]$ *and every integer vector* δ *with entries in* $(-h_E, h_E)$*, on such walks* $E^{(j)}_{\cdot,\sigma} \equiv \delta \pmod p$ *holds if and only if* $E^{(j)}_{\cdot,\sigma} = \delta$*;*

(d) *on the walks with* $\mathrm{mask}_w(z) = 1$ *and digit vector* $E^{(j)}_{\cdot,\sigma} = \delta^{(j)}$*, the walk sum* $Y_{j,\sigma}$ *lies in*

$$\mathrm{Box}_j(\delta^{(j)}) = \left[\left\lceil \tfrac{M_1 K(\delta^{(j)}) - \Lambda + 1}{A} \right\rceil, \left\lfloor \tfrac{M_1 K(\delta^{(j)}) + \Lambda - 1}{A} \right\rfloor\right] \cap [-2mG_j, 2mG_j],$$

where $K(\delta) = \sum_{r<J} A_r \delta_r$*, and* $\#\mathrm{Box}_j(\delta) \le L_w$*.*

Proof. (a) Apply Lemma 7.2 with $\mathcal{V}$ the points where $\mathrm{smask} = 1$ ($|\mathcal{V}| \le N^2 = 2^k$, $K = k$) to get $X \le X_{\max}$; then apply Lemma 6.1. (b) By (a), the sums in (3) and (4) run over the walks with $\mathrm{mask}_w = 1$, since the other walks have weight 0; there are at most 2^{mk} of them. Apply Lemma 7.2 with $K = mk$. (c) Two distinct integers in $(-h_E, h_E)$ differ by at most $2h_E - 2 < P/2 < p$. (d) The telescoping identity of supporting statement D.12 gives $K^{(j)}_{J,\sigma} = K(\delta^{(j)})$ on these walks, so $AY = M_1 K(\delta^{(j)}) - \varphi \in (M_1 K(\delta^{(j)}) - \Lambda,\ M_1 K(\delta^{(j)}) + \Lambda)$; AY is an integer, which gives the interval, and $|Y| \le 2mG_j$ by Lemma 3.1. Finally $\#\mathrm{Box}_j \le (2\Lambda - 2)/A + 1 \le L_w$, because $\Lambda - 1 \le (L_w - 1)A/2$. □

8 Printing the tables

The algorithm needs, for every column, the three certification sums of Section 7 and, for each seed decision, weighted counts of walks by the residues of their digits and walk sums. This section writes each of these as an integer polynomial in threshold atoms and prints its value at all 2^{n-k} columns at once.

Fix σ with $\kappa_\sigma \ne 0$, a prime $p \in \mathcal{P}$, residue vectors $\rho_E \in (\mathbb{Z}/p)^{J|\mathcal{J}|}$ and $\rho_Y \in (\mathbb{Z}/p)^{|\mathcal{J}|}$, and rational walk weights $c(w)$ with $2^S c(w) \in \mathbb{Z}$. Define, for each column z,

$$\begin{aligned}
\mathrm{Fer}(z,w) &= \prod_{j\in\mathcal{J},\, r<J} \Big(1 - \big(E^{(j)}_{r,\sigma}(z,w) - \rho_{E,j,r}\big)^{p-1}\Big) \prod_{j\in\mathcal{J}} \Big(1 - \big(Y_{j,\sigma}(z,w) - \rho_{Y,j}\big)^{p-1}\Big),\\
\Pi(z) &= \sum_{w\in\mathsf{W_s}} 2^S c(w)\, \mathrm{Mpoly}_w(z)\, \mathrm{Amp}_e(\mathrm{Fer}(z,w)),\\
\mathcal{N}(z) &= \sum_{w\in\mathsf{W_s}} 2^S c(w)\, \mathrm{mask}_w(z)\, \mathbf{1}\big[E_{\cdot,\sigma}(z,w) \equiv \rho_E,\ Y_{\cdot,\sigma}(z,w) \equiv \rho_Y \pmod p\big].
\end{aligned}$$

Lemma 8.1 (Residue counts)**.** *If* $X \le X_{\max}$ *at every point of column* z *with* $\mathrm{smask} = 1$*, then* $\Pi(z) \equiv \mathcal{N}(z) \pmod{p^e}$*. If moreover* $p^e > 2 \cdot 2^S \sum_w |c(w)|$*, then* $\mathcal{N}(z)$ *is the unique integer in* $(-p^e/2, p^e/2]$ *congruent to* $\Pi(z)$ *modulo* p^e*.*

Proof. By Lemma 6.1, $\mathrm{Mpoly}_w(z) = \mathrm{mask}_w(z)$ exactly. By Fermat's theorem each factor of Fer is congruent modulo p to the indicator of the corresponding congruence, so Fer is congruent to the joint indicator, which is 0 or 1; by supporting statement D.14, $\mathrm{Amp}_e(\mathrm{Fer})$ is congruent modulo p^e to that same 0/1 value. Multiply by the integers $2^S c(w)\mathrm{mask}_w(z)$ and sum. Finally $|\mathcal{N}(z)| \le 2^S \sum_w |c(w)| < p^e/2$. □

This is [Nag26c, Lemma 6.7] with signed walk weights and the exact mask. No object over $\mathbb{F}_2$ occurs in Π: the residual forms are integer linear forms, the mask is an exact integer polynomial, and the only modular step is the reduction of the exact integer $\Pi(z)$ modulo p^e.

Lemma 8.2 (Printing). *Let $N_w = (4m+1)(G_1+G_2)$, $m_c = n(B_0+\lambda_n+3)$, and let $U_{\mathrm{occ}} \geq 1+N_w m_c$ be an integer. In part (a), assume $2^S c(w)$ are integers with bit length polynomial in n, as for the conditional weights of Corollary E.16. Each of the following tables over all columns $z \in \{0,1\}^{n-k}$ can be printed in $2^{n-k}\,\mathrm{poly}(n)$ bit operations, plus $2^{mk}(1+N_w)^{2D}\,\mathrm{poly}(n)$ for constructing its list of equations, provided $mk + 2 + D\log_2 U_{\mathrm{occ}} \leq (n-k)/100$, where D is the stated degree:*

(a) $\Pi(z)$ of Lemma 8.1, with $D = 4mX_{\max} + (2e-1)(|\mathcal{J}|J + |\mathcal{J}|)(p-1)$;

(b) the moment sums in (3) and (4), with $D = 4mX_{\max} + 2mk$;

(c) the moment sum in (2), with $D = 4k$.

Proof. Fix a walk w. Its edges have corners $c(z)\oplus x$ with x in $\{0\}\cup\{u_i, v_i, u_i\oplus v_i, u_{i+1}\oplus v_i : i \leq m\}$, at most $4m+1$ offsets, so every quantity attached to w is an integer polynomial in the N_w atoms $T_{j,i}(c(z)\oplus x)$. Expand and multilinearize, which is valid because the atoms are Boolean. The degree is at most D: Mpoly has degree $2X_{\max}$ per edge and there are $2m$ edges (Lemma 6.1); E, Y and φ are linear in the atoms; $\mathrm{Amp}_e(\mathrm{Fer})$ has degree $(2e-1)\deg \mathrm{Fer}$; and X is quadratic. There are at most $\sum_{d\leq D}\binom{N_w}{d}$ monomials, with integer coefficients of polynomially many bits.

Because the call is normalized, every atom is a threshold gate on at most n inputs with integer parameters of magnitude below 2^{B_0}. Writing $\sum_l w_l x_l \geq \theta$ as $\sum_l w_l x_l > \theta - 1$, supporting statement D.5 applies with $\ell \leq B_0 + \lambda_n + 2$, so the atom is a disjoint sum of at most m_c exact children. By supporting statement D.6, each child at the offset x is a separated equation in $(z_{\mathrm{L}}, z_{\mathrm{R}})$. A monomial of degree d is therefore a sum of at most m_c^d products of child indicators, each of which is a single stacked separated equation by supporting statement D.7. Hence the list of equations for w has at most $\sum_{d\leq D}\binom{N_w}{d} m_c^d \leq (1+N_w m_c)^D \leq U_{\mathrm{occ}}^D$ entries. Summing over at most 2^{mk} walks (for (c), over at most 2^k points) gives at most $2^{mk}U_{\mathrm{occ}}^D \leq 2^{(n-k)/100}$ entries. Supporting statement D.8 with $n_{\mathrm{out}} = n-k$ prints the table. Dense expansion of each walk's polynomial costs at most $(1+N_w)^{2D}\,\mathrm{poly}(n)$. □

9 The counting algorithm

9.1 Parameters, with exact finite rules

All choices depend only on the normalized call, through (n, β, W, G_1, G_2), and never on the witness. Floors and ceilings of expressions with square roots or mth roots are computed exactly by comparing integer powers; all other comparisons are exact rational comparisons. If the input is not a normalized call (a polynomial-time check) or $n < 4$, the algorithm returns the sound default $(1/2, 1/2)$. If $W = 0$, the circuits are constant and the call is evaluated exactly (a nonempty parity has probability $1/2$). Otherwise:

1. $\lambda = \lceil \log_2 n \rceil \geq 2$, $\mu = \lceil \log_2 \lambda \rceil \geq 1$, $m = 4$, $r = 16\mu$. Require $r \leq n/4$; then $s = \lfloor \log_2(n/r+1) \rfloor$, $b = 2^s - 1$, $k = rs$ and $N = 2^{k/2}$.

2. $J_{\max} = \mu + 1$; T is the least positive integer with $2^T \beta \geq 2^{mk+2m+5} J_{\max}$.

3. $\tau = k + \lceil \log_2(16/\beta) \rceil$; $U = \max\{\tau, \lceil 96 r^{3/2} W/(\beta n^{3/2}) \rceil\}$; $\pi_0 = \tau/(4U)$; $X_{\max}$ is the least integer with $X_{\max}^2 \geq 9\tau^2 nr$; $d^* = \lfloor 2X_{\max} U/(3r\tau) \rfloor$.

4. $\Omega_j = 4mG_j + 1$ for $j \in \mathcal{J}$ ($\Omega_2 = 1$ if $2 \notin \mathcal{J}$), Θ, $I = 2|\Theta|$; M_F is the least integer with $M_F^2 \geq |\Theta|$, and $\eta = \beta/(16 M_F)$. Q is the least power of two with $Q \geq 1024 N m \eta^{-2} (4N^2 I)^{1/m}$. $\ell_f = \max\{k, \log_2 Q + 1\}$ and $S = 2m\ell_f$.

5. $N_{\mathrm{at}} = (4m+1)\cdot 2n^4$, $m_c = n(B_0+\lambda+3)$, $U_{\mathrm{occ}} = 1 + N_{\mathrm{at}} m_c$; $D_{\max}$ is the largest nonnegative integer with $100(mk + 2 + D_{\max}\lceil \log_2 U_{\mathrm{occ}} \rceil) \le n - k$. If none exists, return the sound default.

6. $B_N = S + mk + 2m\lceil \log_2(N^2 Q)\rceil + 1$. For an integer $P \in [100, n]$ let e_P be the least $e \ge 1$ with $(P/2)^e \ge 2^{B_N}$. Let P be the largest integer in $[100, n]$ with

$$(2e_P - 1)(2J_{\max} + 2)(P-1) + 4mX_{\max} \le D_{\max}, \tag{5}$$

returning the sound default if no such P exists; then set $e = e_P$. Require $4mX_{\max} + 2mk \le D_{\max}$. Let $\mathcal{P}$ be the set of primes in $(P/2, P]$ (require $\mathcal{P} \neq \emptyset$), compute D' and L_2 as in supporting statement D.13, and let $L_w = L_2$ for a pair call and $L_w = D'$ otherwise.

7. $h_E = \lfloor P/4 \rfloor$ and $h = \lfloor h_E/(2m) \rfloor$; require $h \ge 64T$. Put $\ell_0 = 32mU$. Build the chain of supporting statement D.12 with (T, h, ℓ_0) in exact rational arithmetic (a_r is the largest integer with $16a_r^2 x_r \le 1$); require $J \le J_{\max}$. Put $\Lambda = \lfloor (L_w - 1)A/2 \rfloor + 1$.

If a requirement fails, the algorithm returns the sound default $(1/2, 1/2)$.

Two consequences of these rules are used below. First, $N_{\mathrm{at}} \ge N_w$ because $G_1 + G_2 \le W \le 2n^4$, so $U_{\mathrm{occ}} \ge 1 + N_w m_c$, and every table printed below has degree at most $D_{\max}$; hence Lemma 8.2 applies to all of them. Second, since N^2 and Q are powers of two, $2^{B_N} = 2 \cdot 2^S \cdot 2^{mk}(N^2Q)^{2m}$ exactly, and since $p > P/2$ for every $p \in \mathcal{P}$,

$$p^e > (P/2)^e \ge 2^{B_N} = 2 \cdot 2^S \cdot 2^{mk}(N^2Q)^{2m} \ge 2 \cdot 2^S \sum_w |c_\pi(w)|$$

for every seed prefix π, by Corollary E.16 and because there are at most 2^{mk} walks. This is the hypothesis of Lemma 8.1.

9.2 Witness

A witness consists of (W1) a partition index in $[2^k]$; (W2) the set $\mathcal{S}$, one bit per bottom gate label; and (W3) vectors $\xi_r^{(j)} \in \{0,1\}^{G_j}$ for $j \in \mathcal{J}$ and $1 \le r \le J$. Its length is $k + (J+1)(G_1 + G_2)$, which is polynomial. Missing or malformed bits are read as zeros.

9.3 Steps

Step 1. Compute the incidences d_l, the set H_c, the matrix smask and the walk list $\mathsf{W_s}$.

Step 2. Print the tables of Lemma 8.2(c), then those of Lemma 8.2(b). Let Z_c be the set of certified columns.

Step 3. (Seed fixing.) Let π be the empty prefix. For $i = 1, \dots, S$, and for each bit $\iota \in \{0,1\}$: compute $c_{\pi\iota}(w)$ for $w \in \mathsf{W_s}$ (Corollary E.16); for every σ with $\kappa_\sigma \neq 0$, every $p \in \mathcal{P}$ and all residue vectors, print Π with $c = c_{\pi\iota}$ (Lemma 8.2(a)) and, at each $z \in Z_c$, recover $\mathcal{N}(z)$ (Lemma 8.1). For each $z \in Z_c$, each such σ, and each digit vector $\delta = (\delta^{(j)})_{j\in\mathcal{J}}$ with entries in $(-h_E, h_E)$, let $\mathrm{Box}(\delta) = \prod_{j\in\mathcal{J}} \mathrm{Box}_j(\delta^{(j)})$ and solve exactly over $\mathbb{Q}$ for a function $\widetilde{H}$ supported on $\mathrm{Box}(\delta)$ with $\widetilde{H}_p(\rho_Y) = \mathcal{N}(z)$ evaluated at $(\rho_E, \rho_Y) = (\delta \bmod p, \rho_Y)$, for all $p \in \mathcal{P}$ and all ρ_Y. Put $v_{z,\sigma,\delta} = \widetilde{H}(0)$ if $0 \in \mathrm{Box}(\delta)$, and $v_{z,\sigma,\delta} = 0$ otherwise. Let

$$\widetilde{\Psi}(\pi\iota) = \sum_{z\in Z_c} \sum_{\sigma} \kappa_\sigma \sum_{\delta} v_{z,\sigma,\delta}.$$

Replace π by $\pi\iota$ for the bit ι with the smaller $\widetilde{\Psi}$ (on ties, $\iota = 0$). If some system is inconsistent, return the default; by Proposition 10.1 this never happens.

Step 4. (Main pass.) With the final seed, list the samples (U_j, V_j). For each $z \in Z_c$ compute by direct evaluation $\widehat{V}(z) = Q^{-1}\sum_j(\text{pred}\cdot\text{mask})(z, U_j, V_j)$ and $\widehat{\mu}(z) = Q^{-1}\sum_j \text{mask}(z, U_j, V_j)$; each term evaluates at most $4(G_1 + G_2)$ threshold gates.

Step 5. (Output.) With $n_{\text{col}} = 2^{n-k}$ and $n_c = |Z_c|$, let

$$\text{Lo} = n_{\text{col}}^{-1}\Big[\textstyle\sum_{z\in Z_c}\widehat{V}(z) - 2M_F n_c\eta\Big],$$
$$\text{Hi} = n_{\text{col}}^{-1}\Big[\textstyle\sum_{z\in Z_c}(\widehat{V}(z) + 1 - \widehat{\mu}(z)) + (2M_F+1)n_c\eta + (n_{\text{col}} - n_c)\Big].$$

These are rational, since $M_F\eta = \beta/16$. Round Lo down and Hi up to multiples of 2^{-n}, and return $\widehat{p} = (\text{Lo} + \text{Hi})/2$ and $\widehat{\kappa} = (\text{Hi} - \text{Lo})/2$, computed from the rounded values.

10 Soundness, completeness and running time

Proposition 10.1 (Soundness for every witness)**.** *For every normalized call and every witness, the returned interval contains* $p_{\text{true}} = \Pr_x[\text{pred}(x) = 1]$.

Proof. Fix the witness, and with it the partition, $\mathcal{S}$, the vectors ξ and the set Z_c of certified columns; none of these depends on the sampler seed.

(1) Exact conditional potentials. Let $z \in Z_c$, let π' be a prefix and let σ be a pattern. By Lemma 7.3(a) and Lemma 8.1, the recovered $\mathcal{N}(z)$ equals $\sum_w 2^S c_{\pi'}(w)\text{mask}_w(z)\mathbf{1}[E \equiv \rho_E,\, Y \equiv \rho_Y]$. By Lemma 7.3(b,c), at $\rho_E = \delta \bmod p$ with the entries of δ in $(-h_E, h_E)$, this is the push-forward modulo p of

$$H_{z,\sigma,\delta}(y) = \sum_{w:\ \text{mask}_w(z)=1,\ E(w)=\delta} 2^S c_{\pi'}(w)\,\mathbf{1}[Y_{\cdot,\sigma}(z,w) = y],$$

which by Lemma 7.3(d) is supported in $\text{Box}(\delta)$, a box with sides at most L_w. By supporting statement D.13 (part (a) for one score, where $L_w = D'$; part (b) for two scores, where $L_w = L_2$), the system has the unique solution $\widetilde{H} = H_{z,\sigma,\delta}$; in particular it is consistent. Every walk with $\text{mask}_w(z) = 1$ has its digit vector in the enumerated range, so summing $H_{z,\sigma,\delta}(0)$ over δ gives $\sum_w 2^S c_{\pi'}(w)\text{mask}_w(z)\mathbf{1}[Y_{1,\sigma} = 0]\mathbf{1}[Y_{2,\sigma} = 0]$. By Corollary E.16, $\widetilde{\Psi}(\pi') = 2^S\frac{4^m}{2|\Theta|}\Psi(\pi')$ with $\Psi(\pi') = \sum_{z\in Z_c}\mathbb{E}[\Phi_z \mid \pi']$.

(2) Greedy choice. By supporting statement D.9 the two children of a prefix average to the parent, so Ψ of the chosen child is at most Ψ of the parent. After S steps the seed is complete and $\Psi(\text{final}) = \sum_{z\in Z_c}\Phi_z$ for the final sample list, while $\Psi(\emptyset) = \sum_{z\in Z_c}\mathbb{E}\Phi_z \le n_c\cdot I\cdot 4N^2((4QN^3)^m m! + (8N^2/3)^{2m}(2m)!)$ by Corollary E.16 (independent expectation) and Lemma 4.1 applied to each of the I matrices $D_{\theta,\nu}(z)$.

(3) Sampler error. If $Z_c = \emptyset$ there is nothing to prove. Otherwise, by (2) and Lemma 4.3 with $Z = Z_c$, the choice of Q gives $n_c^{-1}\sum_{z\in Z_c}\varepsilon_z \le \eta$, with ε_z as in Lemma 4.2.

(4) Per-column intervals. Let $V(z) = N^{-2}\sum_{u,v}\text{pred}$, $V_{\text{m}}(z) = N^{-2}\sum_{u,v}\text{pred}\cdot\text{mask}$ and $\mu(z) = N^{-2}\sum_{u,v}\text{mask}$. Then $V_{\text{m}} \le V \le V_{\text{m}} + 1 - \mu$. By Lemma 4.2, $|\widehat{V} - V_{\text{m}}| \le 2M_F\varepsilon_z$ and $|\widehat{\mu} - \mu| \le \varepsilon_z$ for $z \in Z_c$. Hence $\widehat{V} - 2M_F\varepsilon_z \le V(z) \le \widehat{V} + 1 - \widehat{\mu} + (2M_F+1)\varepsilon_z$. For $z \notin Z_c$, $0 \le V(z) \le 1$. Since a uniform column and a uniform point give a uniform input (supporting statement D.2), $p_{\text{true}} = n_{\text{col}}^{-1}\sum_z V(z)$. Summing and using (3) gives $\text{Lo} \le p_{\text{true}} \le \text{Hi}$, and outward rounding preserves this. □

Proposition 10.2 (Existence of a useful witness)**.** *Suppose that all parameter requirements hold, that* $k \ge \log_2(16/\beta)$ *and* $X_{\max} \ge 2k + 2\log_2(64/\beta)$*, and that the window condition*

$$487\, m\, T\, \bar{\ell}/h < L_w - 1 \tag{6}$$

holds, with $\bar{\ell} = \max\{\ell_0, h^2/(256T)\}$*. Then some witness gives* $\widehat{\kappa} \le 7\beta/16 + 2^{-n}$*.*

Proof. Choose the partition index of Lemma 3.3. Draw $\mathcal{S}$ with rate π_0 and the vectors $\xi_{r,t}^{(j)} = \mathbf{1}[\upsilon_t \le \pi_r]$ with independent uniform υ's, independently of $\mathcal{S}$ and between circuits.

Masked mass. $\Pr_{z,u,v}[\text{mask} = 0]$ is at most $\Pr[\text{smask} = 0] + \Pr[\text{pop} > U] + \Pr[\text{pop} \le U,\ X > \tau]$. By Lemma 6.4, using $d^* \ge X_{\max}U/(3r\tau)$ (as $2X_{\max}U/(3r\tau) \ge 2$), $W/U \le \beta n^{3/2}/(96r^{3/2})$ and $b + 1 > n/(2r)$,

$$\frac{W}{d^*(b+1)} \le \frac{3r\tau W}{X_{\max}U(b+1)} \le \frac{3r\tau\beta n^{3/2}\cdot 2r}{96r^{3/2}X_{\max}n} = \frac{\tau\beta\sqrt{nr}}{16X_{\max}} \le \frac{\beta}{48}.$$

By Lemma 3.3 and Markov's inequality, $\Pr[\text{pop} > U] \le 6r^{3/2}W/(n^{3/2}U) \le \beta/16$. By Lemma 6.3(a), the last term has expectation at most $(e/4)^{\tau+1}$, and $(e/4)^{\tau} \le 2^{-0.55\tau} \le \beta/16$ because $\tau \ge 2\log_2(16/\beta)$. So the expected masked mass is at most $\beta/7$.

Uncertified columns. Fix a column and consider three events. (i) Some point with smask $= 1$ has $X \ge X_{\max}/2$: by Lemma 6.3(c) its probability is at most $2^k 2^{-X_{\max}/2} \le \beta/64$. (ii) Some point with mask $= 1$ has pop $> 16U$: by Lemma 6.3(b), at most $2^k e^{-9\tau/8} \le 2^{k-\tau} \le \beta/16$. (iii) Conditioned on $\mathcal{S}$ and on (ii) failing, some walk w with $\text{mask}_w = 1$, some σ and some j violate the conclusion of Lemma 7.1: at most $2^{mk}\cdot 4^m \cdot 2 \cdot 2Je^{-T} \le 2^{mk+2m+2}J_{\max}2^{-T} \le \beta/8$. If none of the three occurs, the column is certified. Test (2) passes by Lemma 7.2, since $\sqrt{2}\,X_{\max}/2 < X_{\max} + 1$. The digit tests pass since $|E| \le mh \le h_E/2$ and $\sqrt{2}h_E/2 < h_E$. The final tests pass since $\sqrt{2}\cdot 172\, mAT\bar{\ell}/h < \frac{487}{2}mAT\bar{\ell}/h < (L_w - 1)A/2 < \Lambda$ by (6). So the expected uncertified fraction is at most $\beta/4$.

Hence some witness has masked mass plus uncertified fraction at most $\beta/2$. For it, using $\sum_{z\in Z_c}(1 - \widehat{\mu}(z)) \le \sum_z (1 - \mu(z)) + n_c\eta$ (Lemma 4.2 and step (3) of the proof of Proposition 10.1), the unrounded radius satisfies

$$\frac{\text{Hi} - \text{Lo}}{2} \le \tfrac{1}{2}\Pr[\text{mask} = 0] + \tfrac{1}{2}\cdot\tfrac{n_{\text{col}} - n_c}{n_{\text{col}}} + (2M_F + 1)\eta \le \tfrac{\beta}{4} + \tfrac{(2M_F+1)\beta}{16M_F} \le \tfrac{\beta}{4} + \tfrac{3\beta}{16} = \tfrac{7\beta}{16}.$$

Rounding to multiples of 2^{-n} adds at most 2^{-n} to the radius. □

Proposition 10.3 (Running time)**.** *For all sufficiently large* n*, on every normalized call and every witness, the algorithm halts within* $2^n/n^{\mu}$ *steps, reads a witness prefix of polynomial length, and returns dyadic outputs of* $O(n)$ *bits.*

Proof. Parameter preparation is polynomial: the search over $P \le n$, the primes, L_2 by a totient sieve, and the chain. Step 1 costs $(N^2 + 2^{mk})\,\text{poly}(n)$. By Lemma 8.2, each printed table costs $2^{n-k}\,\text{poly}(n)$ plus $2^{mk}U_{\text{occ}}^{2D_{\max}}\,\text{poly}(n) \le 2^{(n-k)/50}\,\text{poly}(n)$ for its construction. Step 2 prints $1 + 2\cdot 4^m(J+1)$ tables. Step 3 has $2S$ child evaluations. Each computes $|\mathsf{W}_s| \le 2^{mk}$ coefficients in $(Q+1)^{2m}\,\text{poly}(n)$ time each, prints at most $4^m|\mathcal{P}|P^{2J+2} \le 4^m P^{2J+3}$ tables, and solves at most $n_c 4^m(2h_E)^{2J} \le 2^{n-k}4^m P^{2J}$ exact systems. Each system has at most $L_w^2 \le D'^2 \le P^4$ unknowns, at most P^3 equations, 0/1 matrix entries and right sides of $O(B_N)$ bits, so fraction-free elimination [Bar68] solves it in polynomial time, as in the proof of [Nag26c, Proposition 8.3]. Step 4 costs $2^{n-k}Q\,\text{poly}(n)$. Since $P \le n$, $J \le J_{\max} = \mu + 1$ and $S = O(k + \lambda)$, the total is at most

$$2^{n-k}n^{2J_{\max}+c_3} + 2^{n-k}Q\,n^{c_3} + 2^{10k+c_3\lambda}$$

for a constant c_3 depending only on B_1, B_2 and the fixed polynomial envelopes. Here every integer that is stored or operated on has bit length bounded by a fixed polynomial in n, and each equation record, reconstruction system and witness has polynomial description size. The bottom gate-list construction costs are included above; the superpolynomial counts (the $Q \le 2^{3k/4} n^{c_Q}$ samples, with $c_Q = O(B_1)$ by Lemma 11.1, the at most 2^{mk} walks, and the 2^{n-k} columns) appear explicitly in the display. Now $k = rs \ge 16\mu \cdot \lambda/2 = 8\mu\lambda$ for large n, since $s \ge \lambda - \log_2 r - 2 \ge \lambda/2$; so $2^{-k} \le n^{-8\mu}$ and $2^{-k/4} \le n^{-2\mu}$. Hence the first term is at most $2^n n^{2\mu+2+c_3-8\mu}$, the second at most $2^n n^{c_Q+c_3-2\mu}$, and the third is $2^{O(\mu\lambda)}$; for large n the sum is below $2^n/n^{\mu}$. Sorting and gathering records on a multitape machine add polynomial factors only, as in [Nag26c, Proposition 8.3]. □

11 Parameters and the quartic cap

Lemma 11.1 (Asymptotics)**.** *Fix B_1, B_2. Uniformly over normalized calls with $W > 0$, as $n \to \infty$: $r = \Theta(\mu)$; $k, T, \tau = \Theta(\lambda\mu)$; $X_{\max} = \Theta(\lambda\mu^{3/2}\sqrt{n})$; $\log_2 Q \le 3k/4 + c_Q\lambda$; $B_N = \Theta(k)$; $D_{\max} = \Theta(n/\lambda)$; $e = \Theta(\mu)$; $P = \Theta(n/(\lambda\mu^2))$; $h = \Theta(P)$; $D' = \Theta(P^2/\lambda)$; and $L_2 = \Theta(P^{3/2}/\lambda^{3/4})$. All parameter requirements and the side conditions $k \ge \log_2(16/\beta)$ and $X_{\max} \ge 2k + 2\log_2(64/\beta)$ of Proposition 10.2 hold for large n.*

Proof. $r = 16\mu$ and $s = \log_2 n - O(\log_2 \mu)$ give $k = rs = \Theta(\lambda\mu)$. Since $\log_2(1/\beta) \le B_1\lambda$, $T = mk + O(B_1\lambda)$ and $\tau = k + O(B_1\lambda)$. With $W \le 2n^4$, $|\Theta| \le (16 \cdot 2n^4 + 1)^2$, so η^{-2} and I are $n^{O(B_1)}$ and $Q \le 2 \cdot 1024 N m \eta^{-2} (4N^2 I)^{1/4} \le 2^{3k/4} n^{c_Q}$. Then $\ell_f = k$ eventually, $S = 8k$, and $20k \le B_N \le 8k + 4k + 8(k + 3k/4 + c_Q\lambda) + 1$, so $B_N = \Theta(k)$. Next, $\log_2 U_{\mathrm{occ}} = 6\lambda + O(\log_2 \lambda)$ gives $D_{\max} = \Theta(n/\lambda)$. For $P \in [n^{1/2}, n]$, $e_P = \lceil B_N / \log_2(P/2) \rceil = \Theta(k/\lambda) = \Theta(\mu)$. The candidate $P_0 = \lfloor cn/(\lambda\mu^2) \rfloor$ satisfies (5) for a small constant c, because $(2e-1)(2J_{\max}+2)(P_0 - 1) = O(c\,n/\lambda)$ and $4mX_{\max} = O(\lambda\mu^{3/2}\sqrt{n}) = o(n/\lambda)$; conversely (5) forces $P = O(D_{\max}/(e\mu)) = O(n/(\lambda\mu^2))$. Hence $\ln P = \Theta(\lambda)$, and supporting statement D.13(c) gives D' and L_2. Further, $h = \Theta(P/m)$ and $T = \Theta(\lambda\mu)$ give $h \ge 64T$; $4mX_{\max} + 2mk = o(D_{\max})$; $h/T \le n$ gives $J \le \lceil \log_2 \log_2 n \rceil + 1 \le J_{\max}$ by supporting statement D.12(iii); $\mathcal{P} \ne \emptyset$ by Bertrand's postulate [Erd32]; and $r \le n/4$. Finally $k \ge 8\mu\lambda \ge \log_2(16/\beta)$ and $X_{\max} = \Omega(\sqrt{n}) \ge 2k + 2\log_2(64/\beta)$ eventually. □

Proof of Theorem 2.1. Soundness is Proposition 10.1 and the running time is Proposition 10.3. For completeness it remains to verify (6) under $W \le a\beta\mathcal{A}_4(n)$; then Proposition 10.2 gives $\widehat{\kappa} \le 7\beta/16 + 2^{-n} \le \beta$ once $2^{-n} \le \beta/2$, which holds for large n since $\beta \ge n^{-B_1}$.

If $\bar{\ell} = h^2/(256T)$, the left side of (6) is $487mh/256 = O(P)$, while $L_w - 1 \ge L_2 - 1 = \Theta(P^{3/2}/\lambda^{3/4})$, so (6) holds for large n. Otherwise $\bar{\ell} = \ell_0 = 32mU$, and (6) reads $U < U^* := h(L_w - 1)/(487 \cdot 32m^2T)$. Every call type has $L_w \ge L_2$: pair calls have $L_w = L_2$, and single and parity calls have $L_w = D' \ge L_2$ (Section D.3). Hence $U^* \ge U_2^* := h(L_2 - 1)/(487 \cdot 32m^2T)$, and by Lemma 11.1

$$U_2^* = \Theta\Big(\frac{P \cdot P^{3/2}\lambda^{-3/4}}{\lambda\mu}\Big) = \Theta\Big(\frac{n^{5/2}}{\lambda^{17/4}\mu^6}\Big)$$

uniformly. (For single and parity calls U^* itself is larger, $\Theta(n^3/(\lambda^5\mu^7))$; only the lower bound U_2^* is used.) Since $\tau = O(\lambda\mu) < U_2^*$ eventually, it suffices that $96r^{3/2}W/(\beta n^{3/2}) \le U_2^* - 1$, that is,

$$W \le \frac{\beta n^{3/2}(U_2^* - 1)}{96 r^{3/2}} = \Theta\Big(\frac{\beta n^4}{\lambda^{17/4}\mu^6 \cdot \mu^{3/2}}\Big) = \Theta(\beta\mathcal{A}_4(n)).$$

Take $a = a_{B_1,B_2}$ strictly below the uniform constant implicit here. □

Remark 11.2 (Where the logarithms go). Up to constants, $\mathcal{A}_4 = n^{3/2} \cdot U_2^*/r^{3/2}$, with $U_2^* \asymp P\, L_2/T$; $P \asymp n/(\lambda\mu^2)$, from the printer budget n/λ divided by $e \asymp \mu$ and $J \asymp \mu$; $L_2 \asymp P^{3/2}/\lambda^{3/4}$; $T \asymp \lambda\mu$, from the union bound over 2^{mk} walks; and $r \asymp \mu$. This gives the powers $\lambda^{1+3/2+3/4+1} = \lambda^{17/4}$ and $\mu^{2+3+1+3/2} = \mu^{15/2}$. For comparison, the near-cubic gate cap is $\mathcal{A}_3 = n^3/(\lambda^{21/4}\mu^7)$ [Nag26c, Theorem 3.1]: the live-gate factor $k/\sqrt{n}$ of [Nag26c, Lemma 3.3] is replaced here by the mixed-population factor $r^{3/2}/n^{3/2}$ of [Nag26b, Theorem A.4]. None of the constants is optimized.

12 From counting to almost-everywhere hardness

Theorem 12.1 (Black-box near-quartic wire transfer)**.** *For $n \geq 4$, put $\lambda_n = \lceil \log_2 n \rceil$, $\mu_n = \lceil \log_2 \lambda_n \rceil$, and $\mathcal{A}_4(n) = n^4/(\lambda_n^{17/4}\mu_n^{15/2})$. A circuit is $C(x) = g(\sum_i T_i(x))$, where the T_i are threshold gates and g is an arbitrary Boolean lookup, with $W(C) = \sum_i (1 + |\operatorname{supp} T_i|)$, counting repetitions. A call asks for the uniform acceptance probability of C^1, $C^1 \oplus C^2$, or $C^1 \oplus \mathrm{PAR}_\Sigma$, and pays the total wires of its circuits. Suppose that for every fixed pair of positive integers B_1, B_2, an oracle-free deterministic algorithm in the multitape model of Theorem 2.1 takes a call and a finite binary witness and satisfies the following contract. A normalized call has $n \geq 4$, explicit lookup tables, integer bottom weights and thresholds of magnitude below $2^{4n\lceil \log_2(n+1) \rceil}$, at most $2n^4$ total wires, and a rational requested radius $n^{-B_1} \leq \beta \leq 1/16$ encoded in at most n^{B_2} bits; the parity uses at most n input indices.*

(i) On every encoded call and every witness it returns $(\widehat{p}, \widehat{\kappa})$ with $|\widehat{p} - \Pr[\mathrm{call} = 1]| \leq \widehat{\kappa}$.

(ii) For some $a = a_{B_1,B_2} > 0$, every sufficiently large normalized call with $W \leq a\beta\mathcal{A}_4(n)$ has a witness of polynomial length in n giving $\widehat{\kappa} \leq \beta$.

(iii) At every sufficiently large length, every normalized call and every witness take at most $2^n/n^{\mu_n}$ steps, whether or not the completeness cap holds.

(iv) Outputs are dyadic with denominators dividing 2^{n+1} and encoded length polynomial in n. On calls outside the normalized domain, time is polynomial in the encoded call length, independently of witness length.

Under the published foundations of this paper, there is a rational $0 < \gamma < 1$ such that, for each fixed real $0 < d < \gamma/5$, there are an integer $K_d > 0$, a language $F_d \in \mathrm{E}^{\mathrm{NP}}$, and an integer $n_0 \geq 4$ for which every real-weight SYM ∘ THR *circuit C on $n \geq n_0$ inputs satisfies*

$$W(C) \leq \lfloor \mathcal{A}_4(n)/K_d \rfloor \quad \Longrightarrow \quad \Pr_x[C(x) = F_d(x)] < 1 - d.$$

Lean: `NearQuarticWires.Transfer.transfer4_official`.
The closed joint theorem is `NearQuarticWires.Release.main_from_published`.

Proof. This section and Appendix A prove the implication using only the stated counter contract. The size test, simulation and length-dilation bounds are Lemmas A.6 and A.8 and Proposition A.11. □

The transfer isolates the counting algorithm as a replaceable component. Its checked interface fixes $\mathcal{A}_4$; transferring a different cap additionally requires the corresponding size test, simulation and length-dilation bounds.

We derive the wire lower bound from the certified counter. The transfer uses only complementation, restrictions, short circuit descriptions, and polynomial general-circuit simulation, together with the derived proof-system, refuter, and amplification interfaces. Appendix A gives every target-class

step, and Appendix E reproduces the supporting constructions. This transfer originates in [Nag26c, Section 11]; all needed arguments are included in these appendices.

Fix $0 < d < \gamma/5$ and use the fixed-error schedule of Section A.2: a fixed $\epsilon = 2^{-e_0}$, for a positive integer e_0, with $\epsilon < \gamma/10$ and $d < \gamma/5 - \epsilon$. With the PCP width q defined there, put

$$P_q = 128/\epsilon, \qquad \omega_q = \epsilon/128, \qquad b(q) = \lfloor \eta_0 \epsilon \, \mathcal{A}_4(q) \rfloor,$$

where η_0 is a fixed rational with $0 < \eta_0 < \min\{1, a_{1,1}/512\}$, and $a_{1,1}$ is the constant of Theorem 2.1 for $B_1 = B_2 = 1$. The number $b(q)$ is the *verifier-length wire cap*: the weak machine of Section A.4 guesses target circuits with at most $b(q)$ wires.

(S1) *Calls.* The expected gadget penalty decomposes into single, pair and parity calls on guessed circuits or their complements, by (9). All calls are made at the PCP width q: each guessed circuit computes one proof coordinate as a function of the random string $x \in \{0,1\}^q$. The complement of a SYM $\circ$ THR circuit is obtained by replacing its lookup g by $1 - g$, with the same wire count. No circuit computing an exclusive-or is built; the exclusive-or is part of the call.

(S2) *Cap.* Two guessed circuits of at most $b(q)$ wires each form a call with $W \le 2b(q) \le 2\eta_0\epsilon\mathcal{A}_4(q) < (a_{1,1}/256)\epsilon\mathcal{A}_4(q) < a_{1,1}\omega_q\mathcal{A}_4(q)$, and $W \le 2q^4$. The radius $\omega_q = \epsilon/128 \le 1/16$ is at least q^{-1} and has an encoding of at most q bits for large q. So Theorem 2.1 with $B_1 = B_2 = 1$, at input length q, supplies a useful witness of radius at most ω_q for every call the machine makes.

(S3) *Guessing and the size test.* The weak machine guesses, for each auxiliary proof coordinate, one normalized SYM $\circ$ THR circuit on q inputs, and accepts its size if and only if $W^4\lambda_q^{17}\mu_q^{30} \le (\eta_0\epsilon)^4 q^{16}$. For an integer W this comparison is equivalent to $W \le b(q)$. Every circuit with at most $b(q)$ wires has a normalized description with no more wires (Section 2.1): supporting statement D.10, applied to each gate on its own inputs, gives integers of magnitude at most $q^q < 2^{B_0(q)}$.

(S4) *Simulation.* A SYM $\circ$ THR circuit with at most q^4 wires on at most Aq inputs has a general Boolean circuit of size $O_A(q^5 \log q)$, hence at most q^6 for large q: normalize each gate on its own inputs (integers of $O(Aq \log(Aq))$ bits), evaluate each linear form by an adder tree and one comparison, count the at most q^4 gate outputs, and look up g. The exponent 6 does not depend on A. Choose the general-circuit exponent $G^\star$ with $G^\star \ge b_{\mathrm{amp}}$ and $G^\star/b_{\mathrm{amp}} > 7$, where b_{amp} is the amplification constant of supporting statement A.3.

(S5) *Restrictions.* Cofactors (fixing the address inputs of a circuit) and averaging over an ignored suffix of inputs are literal restrictions. They do not increase the wire count, and the restricted circuits are normalized again after restriction, as in (S3).

(S6) *Lengths.* For $q \le n \le B_{\mathrm{len}}q$, with the constant B_{len} of Section A.6, $\mathcal{A}_4(n) \le (n/q)^4\mathcal{A}_4(q) \le B_{\mathrm{len}}^4\mathcal{A}_4(q)$, because $\lambda_n \ge \lambda_q$ and $\mu_n \ge \mu_q$. Choose $K_d > 2B_{\mathrm{len}}^4/(\eta_0\epsilon)$. Then $\lfloor \mathcal{A}_4(n)/K_d \rfloor < \eta_0\epsilon\mathcal{A}_4(q)/2 \le b(q)$ once $\eta_0\epsilon\mathcal{A}_4(q) \ge 2$.

(S7) *Envelopes and time.* A normalized circuit with at most $b(q) \le q^4$ wires has a description of $q^{O(1)}$ bits, so all guesses together are $o(N)$ for the source length N, since $q = \Theta(\log N)$. The polynomially many calls take $2^q q^{O(1)-\mu_q}$ steps by Theorem 2.1(iii) at length q, which is $o(\mathcal{T}(N))$ for the clock $\mathcal{T}$ of that section.

(S8) *Order of constants.* In the order of constants of [Nag26c, the paragraph after Lemma C.3], the exponent $G^\star$ is chosen from the simulation exponent and b_{amp}, and η_0 depends only on

the counting theorem. Here the simulation exponent is the absolute constant 6 of (S4), and η_0 depends only on Theorem 2.1; the order is unchanged.

Proof of Theorem 1.1. This is Proposition A.11, which runs the proof of [Nag26c, Theorem 1.2] in [Nag26c, Sections 11.1–11.6] with the fixed-error schedule, Theorem 2.1 in place of [Nag26c, Theorem 3.1], and the substitutions (S1)–(S8). The counter's guarantees used there are soundness for every witness, a useful witness at radius ω_q under the pair cap, running time $2^q/q^{\mu_q}$ at length q, a witness of polynomial length, and no oracle; these are exactly the guarantees of Theorem 2.1. Lemmas A.5 and A.6 prove the properties of the wire count and of the cap used in (S1)–(S8); Lemmas A.7 to A.10 carry out the certified estimate, the weak machine and the extraction. The conclusion is that F_d has agreement less than $1-\gamma/5+\epsilon<1-d$ with every SYM $\circ$ THR circuit with at most $\lfloor \mathcal{A}_4(n)/K_d \rfloor$ wires, at every sufficiently large length n. □

The author's near-cubic gate preprint [Nag26c] also proves a refinement in which one language approaches the boundary error $\gamma/5$ [Nag26c, Theorem 11.3 and Corollary 11.4]. We do not state or claim its wire analogue.

13 Scope

The earlier pointwise count-list analysis pays roughly the square root of the mixed population. The gain here comes from summing a closed walk before reconstructing its score distribution (Lemma 5.1), rather than from a stronger per-gate population estimate. This comparison concerns those two analyses; it is not an obstruction to every gate-by-gate method.

For the present parameters, the binding completeness requirement is the pair-call window condition (6). When $\bar{\ell}=32mU$, it is exactly

$$U<U_2^*:=\frac{h(L_2-1)}{487\cdot 32m^2T}.$$

The alternative branch $\bar{\ell}=h^2/(256T)$ is asymptotically slack. The degree constraint (5) fixes the selected scale $P=\Theta(n/(\lambda\mu^2))$; hence $U_2^*=\Theta(n^{5/2}/(\lambda^{17/4}\mu^6))$, and the sufficient wire cap is

$$W\le\frac{\beta n^{3/2}(U_2^*-1)}{96r^{3/2}}=\Theta(\beta\mathcal{A}_4(n)).$$

The rank bound in Remark E.8 shows that the $P^{3/2}$ power of the two-score window is sharp for recovering arbitrary full-box distributions from these residue data, up to logarithms. This does not rule out exploiting additional structure in circuit score distributions or a different counting method.

Theorem 1.1 is a constant-error lower bound. We do not know whether vanishing-correlation bounds hold at the near-quartic wire scale. The wire theorem concerns ordinary symmetric tops; its Fourier moduli $\Omega_j=4mG_j+1$ use the unweighted count of bottom-gate outputs. The gate corollary covers integer top weights by charging every signed top coefficient through its absolute value. It gives no bound on the unweighted number of gates in an unrestricted real-weight THR $\circ$ THR circuit: converting large unequal top weights to repeated connections may greatly increase the charged resource. The constants, including the logarithmic powers of Remark 11.2, are not optimized, and the length from which the theorem holds is not computed. The proof uses only the six published results of Section 1.5; the transfer interfaces of Appendix A are derived from them (Appendix E), and Appendix A re-derives every transfer step that involves the target class.

The formal verification is described in Section 1.5.

A The almost-everywhere transfer for the wire measure

This appendix proves Theorem 1.1 from Theorem 2.1. Every step involving the target circuit class is restated and proved for the wire measure. The class-independent source, proof-table, and machine constructions are reproduced in Appendix E; citations below identify their origin. The transfer originates in [Nag26c, Sections 11.1–11.6]. The following derived interfaces are proved from the published foundations of Section 1.4.

Supporting statement A.1 (Short PCP: [MW20, Section 5, proof of Theorem 1.4 and footnote 5])**.** Fix a verifier $V(x, y)$ with running time $T(a) \geq a$ on explicit inputs $x \in \{0,1\}^a$, for $a \geq 1$, where $T(a)$ is computable in $\mathrm{poly}(a, \log T(a))$ time. In the same polynomial time one can construct an oracle circuit C_x^O of $\mathrm{poly}(a, \log T(a))$ size, with $\ell = \log_2 T(a) + O(\log\log T(a))$ input bits and oracle gates of arity ℓ, such that:

(a) if $V(x, y)$ accepts some $y \in \{0,1\}^{T(a)}$, some oracle $O : \{0,1\}^\ell \to \{0,1\}$ makes C_x^O unsatisfiable;

(b) if it rejects every such y, then for every oracle O, at least a $1 - 1/a$ fraction of inputs satisfy C_x^O.

The full ordinary-machine construction, including the explicit clock convention, is proved in Section E.25. Satisfying assignments of C_x^O are rejecting random strings, so an accepted oracle makes the circuit *unsatisfiable*. The explicit-input dependence in the circuit size is retained.

Supporting statement A.2 (Fixed-machine refuter: [CLW20, Theorem 1.13, proved as Theorem 4.6])**.** For a time-constructible clock $N \leq \mathcal{T}(N) \leq 2^{\mathrm{poly}(N)}$, there are a fixed language $L_{\mathcal{T}} \in \mathrm{NTIME}[\mathcal{T}(N)]$ and a deterministic SAT-oracle procedure $R_{\mathcal{T}}$ with this property. For each fixed nondeterministic machine M that runs in $o(\mathcal{T}(N))$ time and guesses at most $N/10$ bits, $R_{\mathcal{T}}(M, 1^N)$ returns, at every sufficiently large N, a word of length N on which M and $L_{\mathcal{T}}$ disagree. Its time is $\mathrm{poly}(\mathcal{T}(N))$.

The multitape and framed formulations are proved in Lemma E.13 and Corollary E.14. The weak machine below guesses at most $\lfloor N/16 \rfloor$ bits and is fixed before its source input is chosen.

Supporting statement A.3 (General-circuit amplification: [CLW20, Lemma 3.9])**.** There is an absolute constant $b_{\mathrm{amp}} \geq 1$ such that, for every fixed integer $G^\star \geq b_{\mathrm{amp}}$, there is a constant $a_{G^\star} \geq 1$ with the following property. A Boolean function on q inputs with no general circuit of size $q^{G^\star}$ determines, from its truth table in $2^{O(q)}$ time, a function on at most $a_{G^\star} q$ inputs that has agreement less than $1/2 + q^{-G^\star/b_{\mathrm{amp}}}$ with every general circuit of size $q^{G^\star/b_{\mathrm{amp}}}$.

The full-range construction is proved in Section E.34, here specialized to the size function $S(q) = q^{G^\star}$. The constant b_{amp} is chosen before $G^\star$; the arity and running-time constants may depend on $G^\star$. Amplification is used only when a source function or an outer proof oracle is hard for general circuits.

Supporting statement A.4 (Pointwise three-query PCPP: [CW19, Lemma 24])**.** There are a rational constant $0 < \gamma < 1$, a constant-rate binary linear code Enc with a polynomial-time encoder, and a polynomial-time transformation taking a general circuit D on q inputs to a polynomial-size list of three-literal clauses on systematic and auxiliary variables, such that for every $x \in \{0,1\}^q$, with the systematic variables fixed to $(x, \mathrm{Enc}(x))$:

(a) if $D(x) = 1$, a polynomial-time computable auxiliary assignment satisfies every clause;

(b) if $D(x) = 0$ and $q \geq 2$, every auxiliary assignment leaves at least a γ fraction of clauses false.

Each systematic coordinate is a parity of input coordinates, and the number of clauses may be taken to be a power of two.

The uniform interface and its clause conversion are proved in Section E.34; Section E.18 reproduces the decoder-based pointwise argument of [Nag26c]. The formal proof uses only the encoder: it applies the proximity construction to the circuit that accepts exactly the systematic words of inputs accepted by D. Positive relative distance separates a rejected systematic word from every accepted one. Perfect completeness and an efficient honest proof give part (a); the proximity gap gives part (b). The underlying published proximity and encoder theorems are the two Chen–Williams inputs of Section 1.4.

Conventions. A *target circuit* is a SYM $\circ$ THR circuit (Definition 1.1). A *general circuit* is a Boolean circuit of fan-in-two AND, OR and NOT gates, and its size is its number of gates. To avoid clashes with the counter's notation, this appendix writes:

- $\zeta \in \{0,1\}^N$ for the source input, written z in [Nag26c];
- $\mathcal{U}$ for the certified estimate of the gadget penalty, written U there;
- $G^\star$ for the general-circuit exponent, written G there;
- M_{cl} for the number of clauses, written M there.

Where the target class enters. In the proof of [Nag26c, Theorem 1.2], target circuits occur in exactly four places:

- the counter evaluates calls on them (Section A.3);
- the weak machine guesses them, one per auxiliary proof coordinate (Section A.4);
- the extraction step compares the hard functions with them (Section A.5);
- the padding step restricts them (Section A.6).

The source circuit, the proof-oracle circuit, the short-PCP circuit, the outer verifier and the amplifier's circuits are general circuits. Their sizes are fixed polynomials in the native width q and do not depend on the target class. A target circuit undergoes only four operations: complementation of its output, literal restriction, integer normalization, and simulation by a general circuit (Lemma A.5(e)). It is never composed with another circuit, duplicated inside one, padded, or substituted into one. Lemma A.5 proves the properties of the wire count that these operations need.

A.1 The wire count

A *normalized description* of a target circuit on a inputs lists a and the number G of bottom gates. For each bottom gate it lists the fan-in f_i, the indices and integer weights of the f_i inputs read, and an integer threshold. It ends with the lookup table $(g(0), \dots, g(G))$. All integers are written in a fixed self-delimiting binary code that uses $O(\log(|v|+2))$ bits for an integer v. A bottom gate of fan-in zero is written $\mathbf{1}[0 \geq \theta]$ with $\theta \in \{0,1\}$.

Lemma A.5 (Properties of the wire count)**.** *Let C be a target circuit on $a \geq 1$ inputs.*

(a) Complement. *Replacing the lookup g by $1-g$ gives a target circuit that computes $1-C$ and has the same bottom gates, so its wire count is $W(C)$.*

(b) Restriction. *Fixing any set of inputs to constants gives a target circuit C' on the remaining inputs. It has the same number of bottom gates and the same lookup, and $W(C') \le W(C)$.*

(c) Averaging. *Write the inputs as (x, y). If $F(x, y)$ does not depend on y, some assignment ρ of y satisfies* $\operatorname{agr}(C(\cdot, \rho), F(\cdot, \rho)) \ge \operatorname{agr}(C, F)$.

(d) Normalization. *C has a normalized description with the same lookup and at most the same wire count. In it, every weight and threshold of a bottom gate of fan-in $f \ge 1$ is an integer in $\pm f^f$. Its length is at most $c_0(W(C)+1)\, a \lceil \log_2(a+1) \rceil$ bits, for an absolute constant c_0.*

(e) Simulation. *For every fixed $A \ge 1$ there is q_A such that for all $q \ge q_A$ the following holds. If $a \le Aq$ and $W(C) \le q^4$, then a general circuit of size at most q^6 computes C.*

Proof. (a) The bottom gates are unchanged, and $(1-g)(s) = 1 - g(s)$ for every count s.

(b) Fix $x_l = c_l$ for $l \in R$. The bottom gate T_i becomes

$$\mathbf{1}\Big[\sum_{l \notin R} w_{i,l} x_l \ge \theta_i - \sum_{l \in R} w_{i,l} c_l\Big].$$

This is a threshold gate on the remaining inputs. Its nonzero weights are among those of T_i, so its fan-in is at most f_i. A bottom gate all of whose inputs are fixed becomes a gate of fan-in zero and is still counted once. The bottom gates and the lookup are kept, so $W(C') = G + \sum_i f_i' \le W(C)$.

(c) $\operatorname{agr}(C, F) = \mathbb{E}_\rho \Pr_x[C(x, \rho) = F(x, \rho)]$, and some ρ attains at least the mean.

(d) Apply supporting statement D.10 to each bottom gate of fan-in $f \ge 1$, as a function of the f inputs it reads. This gives integer weights and a threshold in $\pm f^f$ that compute the same function on those inputs. A weight that becomes zero only lowers the fan-in. Keep the lookup. If the support shrinks, repeat normalization on the smaller support; this terminates with the stated bound for the resulting fan-in. Encode any resulting constant by the fan-in-zero convention.

Length: a bottom gate of fan-in $f \ge 1$ writes f input indices and $f+1$ integers of magnitude at most f^f. That takes $O((f+1)(f \log_2(f+1) + \log_2(a+1))) \le O((f+1)\, a \log_2(a+1))$ bits, since $f \le a$. A bottom gate of fan-in zero takes $O(1)$ bits. Since $\sum_i (f_i + 1) = W(C)$, the bottom gates take $O(W(C)\, a \log_2(a+1))$ bits. The lookup takes $G + 1 \le W(C) + 1$ bits and the header takes $O(\log_2(a+1) + \log_2(G+2))$ bits.

(e) Normalize C as in (d). If $G = 0$, C is a constant and has a circuit of three gates, for example $x_1 \wedge \neg x_1$ or its negation. Otherwise build the circuit in three stages.

- *Bottom gates.* Consider a bottom gate T_i of fan-in $f_i \ge 1$ with weights $w_{i,l}$ and threshold θ_i. Put $L_i = \lceil f_i \log_2(f_i+1) \rceil + \lceil \log_2(f_i+2) \rceil + 2$. Every partial sum of the $f_i + 1$ integers $w_{i,l} x_l$ (l read by T_i) and $-\theta_i$ has magnitude at most $(f_i+1) f_i^{f_i} < 2^{L_i - 1}$. So the sum is computed without overflow in L_i-bit two's complement:
  - each term $w_{i,l} x_l$ is the bit pattern of $w_{i,l}$ ANDed with x_l;
  - f_i ripple-carry adders of width L_i add the terms;
  - T_i is the negation of the sign bit.

  This uses at most $c_1(f_i+1)L_i$ gates for an absolute c_1. Bottom gates of fan-in zero are constants.
- *Count.* A tree of ripple-carry adders of width $\lceil \log_2(G+1) \rceil$ computes the count $s = \sum_i T_i$ in at most $c_1 G \log_2(G+2)$ gates.

- *Lookup.* A binary tree of two-to-one multiplexers on the bits of s selects $g(s)$ from the table, padded with zeros to $2^{\lceil \log_2(G+1) \rceil} \le 2G+2$ entries. It uses at most $c_1(G+1)$ gates.

The constants 0 and 1 are produced once, with three gates. Since $f_i \le a \le Aq$, we have $L_i \le c_2 Aq \log_2(Aq+2)$ for an absolute c_2. Using $\sum_i (f_i+1) = W(C) \le q^4$ and $G \le W(C)$, the size is at most

$$c_1 c_2 \, W(C) \, Aq \log_2(Aq+2) + c_1 G \log_2(G+2) + c_1(G+1) + 3 = O_A(q^5 \log q),$$

which is at most q^6 for all $q \ge q_A$. □

A.2 Parameters and the verifier-length wire cap

Fix $0 < d < \gamma/5$. Use the fixed-error schedule of [Nag26c, Section 11.4]: a fixed $\epsilon = 2^{-e_0}$, for a positive integer e_0, with $\epsilon < \gamma/10$ and $d < \gamma/5 - \epsilon$. Put

$$P_q = 128/\epsilon, \qquad \omega_q = \epsilon/128, \qquad b(q) = \lfloor \eta_0 \epsilon \, \mathcal{A}_4(q) \rfloor.$$

Here η_0 is a fixed rational with $0 < \eta_0 < \min\{1, a_{1,1}/512\}$, and $a_{1,1}$ is the constant of Theorem 2.1 for $B_1 = B_2 = 1$. This is [Nag26c, Equation (14)] with $\mathcal{A}_4$ in place of $\mathcal{A}_3$.

Lemma A.6 (Verifier-length wire cap)**.** *For all $q \ge 4$:*

(i) $b(q) \le q^4$;

(ii) $2b(q) < a_{1,1} \, \omega_q \, \mathcal{A}_4(q)$;

(iii) for every integer $W \ge 0$, $W \le b(q)$ if and only if $W^4 \lambda_q^{17} \mu_q^{30} \le (\eta_0 \epsilon)^4 q^{16}$, a comparison of exact rationals;

(iv) if $q \ge 2^{e_0+8}$, then $q^{-1} \le \omega_q \le 1/16$, and ω_q has a binary representation of at most q bits;

(v) $\mathcal{A}_4(n) \le (n/q)^4 \mathcal{A}_4(q)$ *for every $n \ge q$.*

Proof. (i) $\eta_0 \epsilon < 1$ and $\mathcal{A}_4(q) \le q^4$.

(ii) $2b(q) \le 2\eta_0 \epsilon \mathcal{A}_4(q) < (a_{1,1}/256) \, \epsilon \, \mathcal{A}_4(q) = (a_{1,1}/2) \, \omega_q \mathcal{A}_4(q)$.

(iii) $b(q) = \lfloor x \rfloor$ with $x = \eta_0 \epsilon q^4 / (\lambda_q^{17/4} \mu_q^{15/2}) > 0$. For an integer $W \ge 0$, $W \le \lfloor x \rfloor$ if and only if $W \le x$, if and only if $W^4 \le x^4 = (\eta_0 \epsilon)^4 q^{16} / (\lambda_q^{17} \mu_q^{30})$.

(iv) $\omega_q = 2^{-e_0-7} \in [q^{-1}, 1/16]$ for these q, and it is written with $e_0 + 8 \le q$ bits.

(v) λ and μ are nondecreasing. □

A.3 The gadget and the certified estimate

[Nag26c, Sections 11.1–11.2] supply the following, all of which are statements about Boolean values and do not involve the target class:

- the pointwise three-query PCPP of supporting statement A.4, with rejection gap γ, systematic coordinates that are parities of the input, and auxiliary coordinates;
- the Garey–Johnson–Stockmeyer penalty $Q(x_1, x_2, x_3, y)$ (8) ([Nag26c, Equation (10)]). It is nonnegative on Boolean arguments, at least one when the original clause is false, and zero when the clause is true and $y = x_1 x_2 x_3$;
- its weighted stability, Lemma E.12;

- the identity (9)

$$Q = 1 + \tfrac{1}{2}\Big(y - \sum_i x_i - \sum_{i<j}(x_i \oplus x_j) + \sum_i (x_i \oplus y)\Big),$$

with ten probability terms, each of coefficient $\pm\frac{1}{2}$.

Fix q and a list of M_{cl} three-literal clauses, with fresh helper names, as produced by the weak machine (Section A.4). For each auxiliary name X, including each helper, fix a target circuit C_X on q inputs. A literal is $C_X(x)$, a systematic parity $\mathrm{PAR}_\Sigma(x)$, or the complement of either. Write $\mathbb{E}Q$ for the exact average of Q over the gadgets and over $x \in \{0,1\}^q$, with each auxiliary coordinate X evaluated as $C_X(x)$.

Define $\mathcal{U}$ as in [Nag26c, Section 11.2]:

- each exact term of the identity contributes its value;
- for each nonexact term, let $(\widehat{p}, \widehat{\kappa})$ be the counter's estimate and radius, and let $\sigma \in \{\pm 1\}$ be the term's sign; substitute $\sigma\widehat{p} + \widehat{\kappa}$ for the signed probability;
- average over the gadgets.

Lemma A.7 (Certified estimate for guesses below the wire cap)**.** *Suppose that q is large, that every C_X is given by a normalized description, and that $W(C_X) \le b(q)$. Then:*

(i) every nonexact term is a normalized single, pair or parity call of length q, with total wire count at most $2b(q)$ and requested radius ω_q, which Theorem 2.1 with $B_1 = B_2 = 1$ admits;

(ii) for every choice of witnesses, $\mathcal{U} \ge \mathbb{E}Q$;

(iii) some choice of witnesses gives $\mathcal{U} \le \mathbb{E}Q + 10\omega_q$.

Proof. (i) Classify the ten terms. A complemented auxiliary literal $1 - C_X$ is the target circuit of Lemma A.5(a), with the same wire count. The complement of a parity is the parity XOR the constant one.

- *One literal.* If it is a parity or its complement, its probability is exact: $1/2$ for a nonempty support, and 0 or 1 for an empty one. Otherwise it is a single call on C_X or on $1 - C_X$.
- *Two auxiliary literals.* If they have the same name, the XOR is the constant 0 or 1. Otherwise, absorbing complements into the circuits, the term is a pair call $C^1 \oplus C^2$.
- *One auxiliary and one systematic literal.* Absorbing a complement into the circuit, the term is a parity call $C^1 \oplus \mathrm{PAR}_\Sigma$.
- *Two systematic literals.* The XOR is $\mathrm{PAR}_{\Sigma \triangle \Sigma'}$, possibly XOR 1, and is exact.

In every call each circuit has a normalized description on q inputs whose integers have magnitude at most $q^q < 2^{B_0(q)}$ (Lemma A.5(d)). The total wire count is at most $2b(q) \le 2q^4$ (Lemma A.6(i)), and the radius meets the encoding conditions with $B_1 = B_2 = 1$ (Lemma A.6(iv)). So the call is normalized in the sense of Section 2.1.

(ii) By Theorem 2.1(i), every witness gives $|\widehat{p} - p| \le \widehat{\kappa}$ for the true probability p, so $\sigma\widehat{p} + \widehat{\kappa} \ge \sigma p$. Summing the terms of one gadget with their coefficients bounds the gadget's expected penalty from above. Averaging over the gadgets gives $\mathcal{U} \ge \mathbb{E}Q$.

(iii) By Lemma A.6(ii), every call has $W \le 2b(q) < a_{1,1}\omega_q \mathcal{A}_4(q)$. So Theorem 2.1(ii), at length q, gives each call a witness of polynomial length with $\widehat{\kappa} \le \omega_q$. Then $\sigma\widehat{p} + \widehat{\kappa} \le \sigma p + 2\omega_q$. Each gadget has at most ten nonexact terms, each of coefficient $\frac{1}{2}$, so $\mathcal{U} \le \mathbb{E}Q + 10\omega_q$. □

This is [Nag26c, Equations (12) and (13)], with Theorem 2.1 in place of [Nag26c, Theorem 3.1].

A.4 The weak machine

The weak machine $\mathsf{M}_{\mathrm{wire}}$ is the machine of [Nag26c, Section 11.4] with target circuits measured by wires. Its constants are fixed in advance in the order of Section A.7. Steps marked (V) are class-independent; their constructions are given in Appendix E.

(M1) (V) *Fixed objects*, all as in [Nag26c, Section 11.4]:

- the clock $\mathcal{T}(N) = N^{C_{\mathrm{clk}}}$;
- the hierarchy language $L_{\mathcal{T}}$ and its multitape verifier (Lemma E.13);
- the succinct verifier $\mathcal{V}_{G^\star}$ and its time bound $T_{G^\star}$ Lemma E.15;
- $L_\nu = \nu^{G'}$ and the native width $q = q(N)$.

Only source lengths $N = 2^\nu$ are used.

(M2) (V) *Guards.*

- Reject unless $N = 2^\nu$.
- Compute a fixed polynomial envelope for all guesses (Lemma A.8(i)) and reject if it exceeds $\lfloor N/16 \rfloor$. Bound every subsequent guess by its allotted envelope, rejecting if its description would exceed that bound.
- Check the precision guards of Lemma A.6(iv) and $\gamma/L_\nu < \epsilon$.
- Reject if the guessed source circuit's description has more than L_ν bits.

(M3) (V) *Source circuit.* Guess a general circuit B_{src} with q inputs and at most $q^{G^\star}$ gates. Check (12), that is, $B_{\mathrm{src}}(\mathrm{bin}_\nu(i)0^{q-\nu}) = \zeta_i$ for every $i < N$.

(M4) (V) *Clauses.*

- Construct $C^{O}_{\langle B_{\mathrm{src}} \rangle_\nu}$ by the short PCP.
- Guess a general circuit B_{pf} with q inputs and at most $q^{G^\star}$ gates, and form the outer verifier D_B.
- Apply the PCPP, and repeat the clause list to exactly $M_{\mathrm{cl}} = 2^{\lceil d_{G^\star} \log_2(q+2) \rceil}$ clauses, rejecting if it is longer.
- Introduce one fresh helper name per clause.

(M5) *Target guesses.* For each distinct auxiliary name X, guess a normalized description of a target circuit C_X on q inputs. Reject if it is malformed, if some integer has magnitude at least $2^{B_0(q)}$, or if $W(C_X)^4 \lambda_q^{17} \mu_q^{30} > (\eta_0 \epsilon)^4 q^{16}$. By Lemma A.6(iii), the last test rejects exactly when $W(C_X) > b(q)$. Systematic coordinates are the prescribed parities.

(M6) *Estimate.* Guess witness prefixes for the nonexact calls of Section A.3, each of the polynomial length that Theorem 2.1(iii) reads. Run the counter of Theorem 2.1 with $B_1 = B_2 = 1$ and radius ω_q on each call, and form $\mathcal{U}$. Accept if and only if $\mathcal{U} < \gamma - \epsilon$.

Lemma A.8 (The weak machine)**.** *For all sufficiently large ν, with $N = 2^\nu$:*

(i) $\mathsf{M}_{\mathrm{wire}}$ *guesses at most q^c bits, for a constant c fixed before N, hence at most $\lfloor N/16 \rfloor$;*

(ii) it halts within $o(\mathcal{T}(N))$ steps on every branch;

(iii) on every $\zeta \notin L_{\mathcal{T}}$, every branch rejects.

Consequently, for all sufficiently large ν, the refuter returns a word $\zeta_N \in L_{\mathcal{T}} \cap \{0,1\}^N$ on which every branch of $\mathsf{M}_{\mathrm{wire}}$ rejects.

Proof. (i) The source and proof-oracle circuits have descriptions of $q^{O(G^\star)}$ bits [Nag26c, Section 11.4]. There are at most $4M_{\mathrm{cl}} \le 8(q+2)^{d_{G^\star}}$ auxiliary names: at most three literal names and one helper per clause. The explicit description format, $W \le b(q) \le q^4$, and the checked integer bound $B_0(q) = O(q\log(q+1))$ give a fixed bound of $O((q^4+1)q\log(q+1))$ bits for every admitted target description. Use this as its guess envelope. There are at most $10M_{\mathrm{cl}}$ calls, each with a witness prefix of polynomial length. Since $q = \Theta(\log N)$, the total is $o(N)$.

(ii) Steps (M1)–(M4) take at most $N^{d_0} + q^{O(1)}$ steps with $d_0 = 2$ [Nag26c, Section 11.4]. Step (M5) is arithmetic on integers of polynomial bit length. Step (M6) makes at most $10M_{\mathrm{cl}} = q^{O(1)}$ calls at length q. By Theorem 2.1(iii), together they take $2^q q^{O(1)-\mu_q}$ steps, and polynomial-bit arithmetic adds a factor $q^{O(1)}$. Since $2^q/\mathcal{T}(N) = q^{O(1)}$ [Nag26c, Section 11.4 and Appendix C], this is $\mathcal{T}(N)\, q^{O(1)-\mu_q} = o(\mathcal{T}(N))$. Finally $N^{d_0} = o(\mathcal{T}(N))$ because $C_{\mathrm{clk}} > d_0 + 2$.

(iii) This is the argument of [Nag26c, Section 11.4]. Consider a branch that passes the source check; every other branch rejects.

- $\mathrm{tt}_N(B_{\mathrm{src}}) = \zeta \notin L_{\mathcal{T}}$, so $\mathcal{V}_{G^\star}$ rejects $\langle B_{\mathrm{src}}\rangle_\nu$ with every witness Lemma E.15.
- By the short PCP, $C^{O_{B_{\mathrm{pf}}}}_{\langle B_{\mathrm{src}}\rangle_\nu}$ is satisfied by at least a $1-1/L_\nu$ fraction of its inputs. So $D_B(x) = 1$ for at most a $1/L_\nu$ fraction of x.
- For every other x, the PCPP leaves at least a γ fraction of the clauses false under every auxiliary assignment, in particular under $(C_X(x))_X$. Each false clause has penalty at least one for either helper value, and all penalties are nonnegative.

Hence $\mathbb{E}Q \ge \gamma(1 - 1/L_\nu) > \gamma - \epsilon$ by the guard $\gamma/L_\nu < \epsilon$. By Lemma A.7(ii), $\mathcal{U} \ge \mathbb{E}Q$ for every witness, so the branch rejects. This step uses only the soundness of the counter. It holds for every Boolean guess, whatever its size.

The machine is one fixed nondeterministic multitape machine. By (i) and (ii), the refuter of supporting statement A.2 applies to it: for all sufficiently large N it returns a word on which $\mathsf{M}_{\mathrm{wire}}$ and $L_{\mathcal{T}}$ disagree. By (iii) the machine rejects every word outside $L_{\mathcal{T}}$, so that word lies in $L_{\mathcal{T}}$ and the machine rejects it. □

A.5 Extracting a hard function

Fix a large ν, $N = 2^\nu$, $q = q(N)$, and the rejected word $\zeta = \zeta_N \in L_{\mathcal{T}}$ of Lemma A.8. The canonical choices of [Nag26c, Section 11.5, items (i)–(iii)] use only ζ, the source check, the short PCP and general circuits with at most $q^{G^\star}$ gates. Their construction is reproduced in Section E.22. They give one of two outcomes.

- NoSource or NoProof: a function H on q inputs with general circuit complexity greater than $q^{G^\star}$.
- Tautology: a pair $B = (B^*_{\mathrm{src}}, B^*_{\mathrm{pf}})$ whose source circuit passes the check and for which D_B is a tautology. Every auxiliary name X has an honest function Π_X on q inputs: the efficient honest PCPP proof for an original name, and the AND of its clause's three honest literal values for a helper. For a systematic name X, let Π_X be its prescribed parity. The honest table T of [Nag26c, Section 11.5] has $q_0(q) = q + \log_2 M_{\mathrm{cl}} + \log_2 P_q$ inputs (13). Its first q inputs are the random string x. Its remaining inputs are an address α: a gadget and one of P_q

positions, which cycle through that gadget's five-entry list $(X_{a,1}, X_{a,2}, X_{a,3}, Y_a, Y_a)$ of names. At (x, α) the table returns $\Pi_X(x)$ for the name X at that position.

Lemma A.9 (NoSource and NoProof)**.** *Let H', on $a' \le a_{G^\star} q$ inputs, be the function that general-circuit amplification (supporting statement A.3) produces from H. For all large q, every target circuit C on a' inputs with $W(C) \le q^4$ has $\mathrm{agr}(C, H') < \frac{1}{2} + q^{-G^\star/b_{\mathrm{amp}}}$.*

Proof. The amplification interface applies because $G^\star \ge b_{\mathrm{amp}}$. By Lemma A.5(e) with $A = a_{G^\star}$, C is computed by a general circuit of size at most $q^6 \le q^{G^\star/b_{\mathrm{amp}}}$, since $G^\star/b_{\mathrm{amp}} > 7$. The amplified function has agreement less than $\frac{1}{2} + q^{-G^\star/b_{\mathrm{amp}}}$ with every general circuit of that size. □

Lemma A.10 (Tautology)**.** *Every target circuit C on $q_0(q)$ inputs with $W(C) \le b(q)$ has $\mathrm{agr}(C, \mathsf{T}) < 1 - \gamma/5 + \epsilon$.*

Proof. Let $\delta = 1 - \mathrm{agr}(C, \mathsf{T})$, and suppose that $\delta \le \gamma/5 - \epsilon$.

Cofactors. For each auxiliary name X, choose, among the addresses α whose list entry is X, one address α_X that minimizes the error of the cofactor $C(\cdot, \alpha_X)$ against Π_X. Put $C_X = C(\cdot, \alpha_X)$. By Lemma A.5(b), C_X is a target circuit on q inputs with $W(C_X) \le W(C) \le b(q)$. By Lemma A.5(d), it has a normalized description with no more wires, which passes the tests of step (M5). So $\mathsf{M}_{\mathrm{wire}}$ has a branch that guesses B, passes the source check, and guesses these descriptions.

Penalty. The weighted-cofactor incidence argument of Section E.22 gives, for this branch, $\mathbb{E}Q \le (P_q/\lfloor P_q/5 \rfloor)\,\delta \le (5+\epsilon)\,\delta$. Its proof uses only Lemma E.12 and the choice of one minimizing cofactor per name, and it holds for any Boolean functions C_X. The weighted incidence calculation is reproduced in Section E.22 (paragraph "Weighted-cofactor incidence").

Estimate. By Lemma A.7(iii), some witnesses give

$$\mathcal{U} \le (5+\epsilon)\Big(\frac{\gamma}{5} - \epsilon\Big) + 10\omega_q \le \gamma - 4\epsilon + \frac{10\epsilon}{128} < \gamma - \epsilon,$$

using $\gamma < 1$. So this branch accepts ζ, contradicting Lemma A.8. Hence $\delta > \gamma/5 - \epsilon$. □

A.6 Every sufficiently large length

[Nag26c, Section 11.6] sets up the final lengths as follows.

- Pad the hard function (H' in NoSource and NoProof, T in Tautology) to exactly $m(q) = K_{\mathrm{amp}} q_0(q)$ inputs, ignoring a suffix, for a fixed integer $K_{\mathrm{amp}} \ge a_{G^\star}$.
- For a final length n, choose the largest ν with $m(q(2^\nu)) \le n/2$, and let F_d at length n be the padded function of that ν, ignoring its remaining inputs. When no ν qualifies, which happens for only finitely many n, the slice is zero.
- There is a fixed B_{len} such that eventually $q \le n \le B_{\mathrm{len}} q$.

None of this refers to the target class.

Proposition A.11 (The main theorem)**.** *Let K_d be an integer with $K_d > 2B_{\mathrm{len}}^4/(\eta_0\epsilon)$. For every sufficiently large n, every target circuit C on n inputs with $W(C) \le \lfloor \mathcal{A}_4(n)/K_d \rfloor$ has $\mathrm{agr}(C, F_d) < 1 - d$. Moreover, $F_d \in \mathrm{E}^{\mathrm{NP}}$.*

Proof. Let ν and q be the parameters selected for n. As $n \to \infty$, both tend to infinity, so every statement above for large q applies for large n.

Cap. By Lemma A.6(v), $\mathcal{A}_4(n) \le (n/q)^4 \mathcal{A}_4(q) \le B_{\mathrm{len}}^4 \mathcal{A}_4(q)$. So

$$\lfloor \mathcal{A}_4(n)/K_d \rfloor < \tfrac{1}{2}\eta_0 \epsilon\, \mathcal{A}_4(q) \le b(q)$$

once $\eta_0 \epsilon \mathcal{A}_4(q) \ge 2$, because $\lfloor x \rfloor \ge x/2$ for $x \ge 1$.

Restriction. At length n, F_d depends only on the $a' \le m(q)$ inputs of H', or on the $q_0(q)$ inputs of T, where it equals that function. By Lemma A.5(c), applied to all other inputs at once, some restriction C' of C to those inputs satisfies $\mathrm{agr}(C', H') \ge \mathrm{agr}(C, F_d)$, respectively $\mathrm{agr}(C', \mathsf{T}) \ge \mathrm{agr}(C, F_d)$. By Lemma A.5(b), $W(C') \le W(C) < b(q) \le q^4$.

Conclusion. By Lemmas A.9 and A.10, $\mathrm{agr}(C, F_d) < \max\{\frac{1}{2} + q^{-G^\star/b_{\mathrm{amp}}},\, 1 - \gamma/5 + \epsilon\}$. For large q this maximum is $1 - \gamma/5 + \epsilon$, which is less than $1 - d$ because $d < \gamma/5 - \epsilon$.

Uniformity. [Nag26c, Section 11.6, final paragraph] computes F_d at length n in time $2^{O(q)}$ with a SAT oracle, in five steps:

- run the refuter on the fixed machine, here $\mathsf{M}_{\mathrm{wire}}$, in time $\mathrm{poly}(\mathcal{T}(N)) = 2^{O(q)}$, to obtain ζ;
- recover the first source circuit B^*_{src}, or its absence, by prefix self-reduction on a SAT formula of size $Nq^{O(G^\star)}$;
- recover the first oracle O^* by prefix self-reduction on a formula of size $2^{O(q)}$; if no such oracle exists at an exceptional initial length, set the slice to zero, as in the cited construction;
- recover the first B^*_{pf}, or its absence, in the same way;
- compute H' in time $2^{O(q)}$, or compute T from the honest PCPP proof and the AND helpers in polynomial time.

These formulas involve the source circuit, the oracle and the proof-oracle circuit, and no target circuit. The target class enters only through the fixed machine handed to the refuter. Since $q \le n$, $F_d \in \mathrm{E}^{\mathrm{NP}}$. □

A.7 Order of constants

The constants are chosen in the order of [Nag26c, the paragraph after Lemma C.3], with two substitutions: the counter is Theorem 2.1, and the simulation exponent is the absolute constant 6 of Lemma A.5(e).

1. $\epsilon = 2^{-e_0}$, from d and γ.
2. η_0, from $a_{1,1}$ of Theorem 2.1.
3. $d_0 = 2$.
4. $G^\star$ with $G^\star \ge b_{\mathrm{amp}}$ and $G^\star/b_{\mathrm{amp}} > 7$, from b_{amp} and the simulation exponent 6. This fixes $a_{G^\star}$.
5. $G' \ge 2$, from $G^\star$ and the description format of general circuits.
6. $C_{\mathrm{clk}} > d_0 + 2$. This fixes $\mathcal{V}_{G^\star}$ and $T_{G^\star}$.
7. The constants of the short PCP for $\mathcal{V}_{G^\star}$, then A_{pad}, as in [Nag26c, Appendix C].
8. The clause-count exponent $d_{G^\star}$.
9. $K_{\mathrm{amp}} \ge a_{G^\star}$, then B_{len}, then $K_d > 2B_{\mathrm{len}}^4/(\eta_0\epsilon)$.

Only two entries refer to the target class: η_0, which depends only on Theorem 2.1, and the exponent 6, which is absolute and depends on no multiplier A. Lemma A.5(e) is used only with $A = a_{G^\star}$, which is fixed at step 4. No constant depends on one chosen after it.

B Reference tables

Symbol	Meaning	Set in
n; λ, μ	input length of the counter; $\lceil \log_2 n \rceil$, $\lceil \log_2 \lambda \rceil$	§1.1
$\mathrm{wt}(C)$	total top weight $\sum_i \lvert c_i \rvert$ of the integer-weighted extension	§1.1
W; G_j; $f_{j,i}$	total wires of a call; bottom gates of circuit j; fan-in of bottom gate (j, i)	§2.1
$\mathcal{A}_4$	the cap $n^4/(\lambda^{17/4}\mu^{15/2})$	§1.1
r, s, b, k; N	blocks; $b = 2^s - 1$; $k = rs$; $N = 2^{k/2}$ patterns per side	§D.2
z; $c(z)$; u, v; $F(u)$	column (cell address); its center; left and right patterns; coordinates flipped by u	§D.2
$\Delta_{j,i}$, R_j, pop	mixed residual of a bottom gate; of circuit j; population	§3
m; Ω_j, Θ; M_F	trace order 4; Fourier moduli and group; Fourier mass bound $\lceil \sqrt{\lvert\Theta\rvert} \rceil$	§3.1
Q; η; Γ; S	samples; sampler accuracy; sample-count function; seed length	§4, §5
w, e_l; $Y_{j,\sigma}$; κ_σ	closed walk and its edges; signed walk sum; sign-pattern weight	§5
$\mathcal{S}$; X; $\tau, U, X_{\max}$	subsampled bottom gates; their residual count; mask parameters	§6
$\xi_r^{(j)}$; $E_{r,\sigma}^{(j)}$, $\varphi_\sigma^{(j)}$	witness selection vectors; digits and final error of a walk sum	§7
T, h, ℓ_0; a_r, A, M_1, J	chain inputs; chain ratios, product, first denominator, length	§D.3
P, $\mathcal{P}$, p, e	prime scale; primes in $(P/2, P]$; a prime; lifting exponent	§9.1
D', L_2, L_w	one-score window; two-score window; window in use	§D.3
h_E, Λ	digit bound $\lfloor P/4 \rfloor$; final-error bound	§9.1
β; $(\widehat{p}, \widehat{\kappa})$	requested radius; returned estimate and radius	§2.1

Table 1: Recurring notation. Subscripts $j \in \{1, 2\}$ index the circuits of a call.

C Attribution in detail

Throughout the paper four levels of ownership are distinguished. *Classical* tools are used as published. *Imported* statements are restated, credited to their origins and proved in Appendix E. *Adapted* statements are restated because a hypothesis changes, and the change is proved here. *New* components are the constructions and applications introduced in this paper, rather than the standard identities used to prove them. Table 2 assigns each component of the proof to one of these levels.

Joint Lean companion: github.com/devnag/wire4_gate3.

The companion's source mapping documents how each printed algorithm of [BCGT13] is transcribed.

The algorithmic method and the transfer. The algorithm-to-lower-bound program is Williams' [Wil13]. The almost-everywhere transfer used here follows Chen–Lyu–Williams [CLW20], including the constructive refuter, which adapts the almost-everywhere hierarchy of Fortnow and Santhanam [FS16], and the general-circuit hardness amplification of Sudan, Trevisan, and Vadhan [STV01], in the form of [CLW20, Lemma 3.9]. The transfer also uses the PCP refinements of Chen and Williams [CW19] and of Bathie and Williams [BW24], the proximity construction of Ben-Sasson, Goldreich, Harsha, Sudan, and Vadhan [BGHSV06], and the short PCP in the form stated by Murray and Williams [MW20, Section 5], credited there to Ben-Sasson and Viola [BV14]. The author's preprint [Nag26c] assembles these into a transfer from certified pair counting to almost-everywhere constant-error hardness, through a weighted analysis of the MAX-2SAT gadget of Garey–Johnson–Stockmeyer [GJS76]; its Appendix D proves the short PCP in the form it uses, from the construction of Ben-Sasson, Chiesa, Genkin, and Tromer [BCGT13]. The full transfer is given here with the wire measure; its supporting constructions are reproduced in Appendix E. We claim no new PCP, amplification or hierarchy theorem.

Relation to [Nag26b]. The author's preprint [Nag26b] introduced the partition into products of Hamming-code stars [Ham50], the mixed-residual population bound, and the universal fixed-moment sampler with exact short-seed conditional expectations, whose concentration step is Tropp's matrix Bernstein inequality [Tro12, Theorem 1.6]. Its counter applies the sampler to Boolean state matrices built from pointwise count lists and $\mathbb{F}_2$ filters, and it pays roughly the square root of the population. We keep its cells and its sampler and replace everything in between: the sampled matrices are Fourier characters of the residual, the potential is computed by walk-sum tomography instead of count lists, and the masks are exact integer polynomials instead of $\mathbb{F}_2$ filters. No count list, radix state, polarization or Boolean lift of [Nag26b] is used. We also import its separated exact printer and the lemmas that feed it.

Relation to [Nag26c]. The author's preprint [Nag26c] reconstructs exact score distributions of restricted circuits by nested subsampling, moment certificates and modular tomography, building on Hajdu and Tijdeman's factorization method for discrete tomography [HT01]. We apply the same machinery to a different integer, the signed residual sum of a closed walk; the localization lemma is re-proved for walk sums, whose summands are bounded by $2m$ rather than one. We also use its transfer, as described above.

Classical tools. Integer normalization of real threshold functions is classical, due to Muroga, Toda, and Takasu [MTT61], in the form of [CTW26, Lemma 3.2]. The decomposition of a threshold

function into disjoint exact thresholds goes back to Hansen and Podolskii [HP10]; the first-crossing construction used is that of Chen and Williams [CW19, Proposition 18(2) and Appendix B]. Batch evaluation uses Williams' rectangular matrix multiplication [WilACC14, Corollary 4.4], based on Coppersmith's algorithm [Cop82], with Matoušek's bucketing [Mat91]. The modulus-amplifying polynomial is that of Beigel and Tarui [BT94]. Exact rational linear systems are solved by fraction-free elimination [Bar68]. Newton's forward-difference formula, the Chernoff and Bernstein bounds, and finite Fourier analysis are standard.

C.1 Formal counterparts of the proof diagram

Table 3 maps the nodes of Figure 1 to formal declarations. The adapter `Adapters.sources_to_nine` derives, from the six published results, the intermediate interfaces used by the proof (normalization, restriction, concentration, rectangular product, short PCP, refuter, amplification, three-query PCPP). Wire names are relative to `NearQuarticWires`; source-adapter names are relative to `NearCubicGates`. The theorem `Release.main_from_published` proves the joint statement from `Close.PublishedInputs`, with the common constants in `Statement.JointStatement4`.

Table 3: Formal counterparts of the proof diagram.

Diagram component	Exact formal declaration
Williams	`Close.PublishedInputs.williams_rectangularProductLeaf`
BCGT13	`Close.PublishedInputs.bcgt13_theorem11_1`
Shoup/BCGT	`Close.PublishedInputs.shoup_findIrrPolyAlgorithm17`
STV	`Close.PublishedInputs.stv_theorem29`
CW proximity	`Close.PublishedInputs.cw19Proximity_lemma22Half`
CW code	`Close.PublishedInputs.cw19Code_lemma26`
Published-source adapter	`Adapters.sources_to_nine`
Refuter	`Adapters.Assembly.field7`
Short PCP	`Adapters.Assembly.Fixed.field6OrdinaryFixed`
Amplification	`Adapters.Amplification.field8`
Three-query PCPP/code	`Adapters.Assembly.field9Encoder`
Product-star cells	`Cells.partition_average`; `Cells.mean_population`
Bounded-matrix sampler	`Sampler.mean_traceMoment_le`; `Sampler.paper_fourth_root_budget`
Cycle-sum identity	`Cycle.cycle_sum_identity`; `Cycle.conditional_cycle_sum_identity`
Integer masks	`Mask.maskPoly_eq_mask`; `Mask.indicatorPolynomial_eq_mask`
Localization/tomography	`Walks.chain_localization_failure_bound`; `Walks.signed_residue_recovery`
Printing	`Cells.compiledTable_value`; `Cells.compiledTable_weightedDominance`
Certified counter	`Program.counterExists4`
Wire transfer	`Transfer.transfer4_official`; `Program.finalStatement4`
Wires and gate outputs	`Statement.WiresAt`; `Statement.SymGatesAt`; `Statement.GatesAt`; `GateCorollary.jointStatement4_of_final`
Joint conclusion	`Release.main_from_published`; `Statement.JointStatement4`

D Statements from earlier work

This appendix records the supporting statements used in the counter. Their origins are credited at each statement; proofs are reproduced in Appendix E. Classical normalization, concentration, and lifting are cited directly to their published sources.

Component (where used)	Level	Source and change
Signed-top gate corollary (Corollary 1.2)	Classical	Replication and finite-cube threshold complementation; wire cost at most $(n+1)\,\mathrm{wt}(C)$
Product-star partition (supporting statement D.2)	Imported	[Nag26b, Lemma A.1]; Hamming codes [Ham50]
Range of the mixed residual (supporting statement D.3)	Imported	[Nag26b, Lemma A.2]; we use only the integer form of Δ, never its $\mathbb{F}_2$ form
Mixed-residual population (supporting statement D.4, Lemma 3.3)	Imported	[Nag26b, Theorem A.4]; the constant 6 is computed here
Score identity, mixed dependence (Lemmas 3.1 and 3.2)	Adapted	[Nag26b, Lemma A.2 and the proof of Theorem A.4], restated for two circuits with an explicit pointwise bound
Fourier decomposition with selectors (Lemma 3.4)	Adapted	Standard finite Fourier expansion; [Nag26b, Equation (29)]; here also for parity calls
Trace bound for $[-1,1]$ entries (Lemma 4.1)	Adapted	The computation of [Nag26b, Lemma F.1] for 0/1 entries, redone; concentration by [Tro12, Theorem 1.6]
Uniform selector error, potential to error (Lemmas 4.2 and 4.3)	Adapted	[Nag26b, Lemma F.1 and its proof], for matrices with entries in $[-1,1]$ and a subset of columns
Short seed and exact conditional expectations (supporting statement D.9)	Imported	[Nag26b, Lemma F.2 and its proof]
Cycle-sum identity, conditional potentials (Lemma 5.1 and Corollary E.16)	New	
Exact integer mask (Definition 6.1 and Lemma 6.1)	New	Newton forward differences; replaces the $\mathbb{F}_2$ filters of [Nag26b, Appendix D]
Mask guarantees (Lemmas 6.2 to 6.4)	New	Chernoff bounds
Nested rates and telescoping (supporting statement D.12)	Imported	[Nag26c, Lemma 5.1 and Equation (1)]
Walk-sum localization (Lemma 7.1)	Adapted	[Nag26c, Lemma 5.2] for walk sums: summands bounded by $2m$
Moment certificates (Lemma 7.2)	Imported	[Nag26c, Lemma 5.3], any index set
Certified boxes (Lemma 7.3)	Adapted	[Nag26c, Sections 7.3 and 8], for walk sums and the exact mask
One- and two-score reconstruction (supporting statement D.13)	Imported	[Nag26c, Lemmas 4.1–4.4]; Hajdu–Tijdeman [HT01]
Bernstein inequality (supporting statement D.11)	Classical	[Nag26c, Lemma 3.4], from [Ver18, Theorem 2.8.4]
Fermat tests and lifting (supporting statement D.14, Lemma 8.1)	Imported / adapted	[Nag26c, Lemmas 6.6 and 6.7]; Beigel–Tarui [BT94]; here with signed walk weights and the exact mask
Integer normalization (supporting statement D.10)	Classical	[Nag26c, Lemma 3.2] = [CTW26, Lemma 3.2]; [MTT61]; applied on each gate's own inputs
Disjoint exact children, pullback, stacking (supporting statements D.5 to D.7)	Imported	[Nag26b, Lemmas C.2, C.3, C.6]; [HP10, CW19]
Separated printer (supporting statement D.8)	Imported	[Nag26b, Theorem C.5]; Williams [WilACC14], Coppersmith [Cop82], Matoušek [Mat91]
Printing walk polynomials (Lemma 8.2)	New	Per-walk atom pools; load $mk + D \cdot O(\lambda)$
Algorithm, soundness, completeness, time, cap (Sections 9 to 11)	New	Structure follows [Nag26c, Sections 7–9]
Almost-everywhere transfer (Section 12, Appendix A)	Imported / adapted	[Nag26c, Section 11 and Appendices C–D], from [CLW20, CW19, BW24, MW20, GJS76]; wire measure in place of top weight, every class-dependent step re-derived in Appendix A

Table 2: Attribution of the components of the proof.

D.1 Matrix concentration

Supporting statement D.1 (Tropp [Tro12, Theorem 1.6])**.** Let $X_1, \dots, X_Q$ be independent random real $d_1 \times d_2$ matrices with $\mathbb{E}X_j = 0$ and $\|X_j\| \le L$ almost surely, and let $\sigma^2 = \max\{\|\sum_j \mathbb{E}X_j X_j^{\mathsf{T}}\|, \|\sum_j \mathbb{E}X_j^{\mathsf{T}} X_j\|\}$. Then for all $x \ge 0$,

$$\Pr\Big[\Big\|\sum_j X_j\Big\| \ge x\Big] \le (d_1 + d_2)\exp\Big(-\frac{x^2/2}{\sigma^2 + Lx/3}\Big).$$

Here and below $\|\cdot\|$ is the operator norm.

D.2 Product-star geometry and exact printing

Supporting statement D.2 ([Nag26b, Lemma A.1])**.** Let $2 \le r \le n/4$ be even, $s = \lfloor \log_2(n/r+1) \rfloor$, $b = 2^s - 1$ and $k = rs$. There are 2^k explicitly described partitions of $\{0,1\}^n$, each into 2^{n-k} cells of size 2^k. A point of a cell is specified by one choice from $\{0, e_1, \dots, e_b\}$ in each of r disjoint blocks of coordinates. For every partition, a uniform cell and a uniform point give a uniform input. Averaging over the partition and its uniformly chosen center gives a uniform center, independently of the point choices. The center bits can be divided into two halves differing in size by at most one while keeping each block on its assigned side.

Since $2^{s+1} > n/r + 1$, we have $b + 1 > (n/r + 1)/2 > n/(2r)$. We write a cell's address, or *column*, as $z \in \{0,1\}^{n-k}$ and its center as $c(z)$. The $r/2$ blocks on the left side give $N = 2^{k/2}$ left patterns $u \in [N]$, and the $r/2$ blocks on the right side give N right patterns $v \in [N]$; we identify a pattern with its error vector, so the points of the cell are $c(z) \oplus u \oplus v$. Let $F(u)$ be the set of at most $r/2$ coordinates flipped by u; it depends only on u and the partition, not on z.

Supporting statement D.3 ([Nag26b, Lemma A.2])**.** For every threshold function T and all c, u, v as above, $\Delta_T(c,u,v) = T(c \oplus u \oplus v) - T(c \oplus u) - T(c \oplus v) + T(c) \in \{-1, 0, 1\}$.

Supporting statement D.4 ([Nag26b, Theorem A.4])**.** Average over a uniform partition, a uniform center, and uniform patterns u, v. For a threshold function T that depends on f coordinates, $\Pr[\Delta_T \ne 0] \le 2f/(b+1)^{3/2}$.

The next four statements compile threshold atoms into tables printable over all columns.

Supporting statement D.5 ([Nag26b, Lemma C.2])**.** Let $H(x) = \mathbf{1}[\sum_{i=1}^f w_i x_i > T]$ have integer parameters, let $B_0' = \max(1, |T|, |w_1|, \dots, |w_f|)$ and $\ell = \max\{1, \lceil \log_2((f+1)B_0' + 1)\rceil\}$. There is an explicitly constructible list of at most $\max(1, f\ell)$ integer affine equations $E_j(x) = 0$ with $H(x) = \sum_j \mathbf{1}[E_j(x) = 0]$ on the Boolean cube, the sum always lying in $\{0,1\}$. Their coefficients have $O(\ell + \log(f+2))$ bits. Weak thresholds are included by replacing $\sum_i w_i x_i \ge T$ with $\sum_i w_i x_i > T - 1$.

Supporting statement D.6 ([Nag26b, Lemma C.3 and the paragraph after it])**.** In the product-star parametrization, with each block kept on its side of the column split $z = (z_{\mathrm{L}}, z_{\mathrm{R}})$, each original input coordinate of $c(z) \oplus x$, for a fixed point x of the cell, is a Boolean function, computable in polynomial time, of one of the two column halves. Substituting these functions into a disjoint exact child $c_j + \sum_i d_{j,i} x_i = 0$ of supporting statement D.5 yields a *separated* equation $a_j(z_{\mathrm{L}}) + b_j(z_{\mathrm{R}}) = 0$ with polynomial-bit values; the children remain disjoint and represent the pulled-back threshold exactly over the integers.

Supporting statement D.7 ([Nag26b, Lemma C.6])**.** If integer separated equations satisfy $|E_j(z)| \le V$ for all j, z and $H = 2V + 2$, then $\prod_{j=1}^t \mathbf{1}[E_j(z) = 0] = \mathbf{1}[\sum_j H^{j-1} E_j(z) = 0]$, and the stacked equation is separated with $O(t \log(V+2))$-bit coefficients.

Supporting statement D.8 ([Nag26b, Theorem C.5])**.** Fix a polynomial description envelope and $n/2 \le n_{\mathrm{out}} \le n$. Let $E_i(z) = a_i(z_{\mathrm{L}}) + b_i(z_{\mathrm{R}})$, $1 \le i \le s$, be separated equations over $z \in \{0,1\}^{n_{\mathrm{out}}}$ with polynomial-bit half-scores computable in polynomial time, and let v_i be polynomial-bit integers. If $s \le 2^{n_{\mathrm{out}}/100}$, the table $z \mapsto \sum_{i=1}^{s} v_i \mathbf{1}[E_i(z) = 0]$ is printable in $2^{n_{\mathrm{out}}}\,\mathrm{poly}(n)$ bit operations, in addition to the construction of the list of the s equations.

The printer reduces the table to weighted dominance counting, which Matoušek's bucketing [Mat91] reduces to Williams' uniform rectangular matrix multiplication [WilACC14, Corollary 4.4].

Supporting statement D.9 ([Nag26b, Lemma F.2 and its proof])**.** Let $N = 2^{k/2}$, let Q and h be positive integers, take an integer $\ell_f \ge \max\{k, \lceil \log_2(Q+1) \rceil\}$ and the field $\mathbb{F}_{2^{\ell_f}}$, label the Q samples by distinct field elements, evaluate a uniformly random polynomial of degree $2h-1$ at the labels, and project each value to k bits, read as a pair in $[N]^2$. The seed has $S = 2h\ell_f$ bits, and the Q sample positions are $2h$-wise independent and uniform on $[N]^2$. For every seed prefix, the conditional probability of any conjunction of events "sample j lies at position x" is 0 or 2^{-a} with $a \le S$, and it is computable by Gaussian elimination over $\mathbb{F}_2$; repeated sample labels repeat or augment the same linear system. The two one-bit extensions of a prefix average to the prefix: for every function Ψ of the seed, $\mathbb{E}[\Psi \mid \pi] = \frac{1}{2}(\mathbb{E}[\Psi \mid \pi 0] + \mathbb{E}[\Psi \mid \pi 1])$.

D.3 Normalization, sampling rates, and residue reconstruction

Supporting statement D.10 ([CTW26, Lemma 3.2])**.** Every threshold function on $f \ge 1$ Boolean inputs has an equivalent representation whose weights and threshold are integers in $\pm f^f$.

This is [CTW26, Lemma 3.2], attributed there to Muroga, Toda, and Takasu [MTT61].

Supporting statement D.11 ([Ver18, Theorem 2.8.4])**.** If independent centered random variables Y_t satisfy $|Y_t| \le 1$ and $\sum_t \mathbb{E}Y_t^2 \le V$, then for every $T > 0$, $\Pr[|\sum_t Y_t| \ge \sqrt{2TV} + 2T/3] \le 2e^{-T}$.

Supporting statement D.12 ([Nag26c, Lemma 5.1 and Equation (1)])**.** Let $T > 0$, let $h \ge 64T$ be an integer, and let $\ell_0 \ge 0$. Define

$$\bar{\ell} = \max\{\ell_0, h^2/(256T)\}, \quad \pi^* = \frac{h^2}{8T\bar{\ell}}, \quad s(\pi) = \sqrt{2T\pi\bar{\ell}} + \frac{2T}{3}, \quad M_1 = \lceil 4\bar{\ell}/h \rceil, \quad \pi_1 = 1/M_1.$$

While $x_r = \pi_r/\pi^* < 2^{-8}$, set $a_r = \lfloor 1/(4\sqrt{x_r}) \rfloor$ and $\pi_{r+1} = a_r\pi_r$; stop at the first J with $x_J \ge 2^{-8}$. Then:

(i) $a_r \ge 4$, all rates lie in $(0,1]$, and $\pi_r = \prod_{i<r} a_i/M_1$;

(ii) $\pi_1\bar{\ell} + s(\pi_1) \le h/2$, and $s(\pi_{r+1}) + a_r s(\pi_r) \le h/2$ for $r < J$;

(iii) $J \le \lceil \log_2 \log_2(h/T) \rceil + 1$;

(iv) $s(\pi_J)/\pi_J \le 86T\bar{\ell}/h$.

Put $A = \prod_{i=1}^{J-1} a_i$, $A_0 = A$ and $A_r = \prod_{i=r+1}^{J-1} a_i$ for $1 \le r < J$. For arbitrary integer vectors $K_1, \dots, K_J$, with $E_0 = K_1$ and $E_r = K_{r+1} - a_r K_r$, the identities $K_J = \sum_{r=0}^{J-1} A_r E_r$ and $\pi_J = A/M_1$ hold.

The telescoping identity holds for *any* proposed vectors, not only for nested samples; Lemma 7.3(d) uses it for arbitrary witness vectors.

Supporting statement D.13 ([Nag26c, Lemmas 4.1–4.4 and the paragraph after Lemma 4.3])**.** Let $\mathcal{P}$ be the set of primes in $(P/2, P]$ and $D' = 1 + \sum_{p\in\mathcal{P}}(p-1)$. For a finitely supported $F : \mathbb{Z}^d \to \mathbb{Q}$ and a prime p, let $F_p(\rho) = \sum_{y\equiv\rho \pmod p} F(y)$ for $\rho \in (\mathbb{Z}/p)^d$.

(a) A rational function supported on at most D' consecutive integers with $F_p = 0$ for all $p \in \mathcal{P}$ is zero.

(b) Let $S_\varphi(q) = \sum_{t\le q} t\varphi(t)$, with φ Euler's function, and let L_2 be the largest $L \ge 2$ with $S_\varphi(\lfloor (D'-1)/(L-1)\rfloor) \ge L$, or $L_2 = 1$ if there is none. A rational function supported on a given box in $\mathbb{Z}^2$ with both sides at most L_2 is uniquely determined, among such functions, by its push-forwards F_p, $p \in \mathcal{P}$.

(c) As $P \to \infty$, $D' = \Theta(P^2/\ln P)$ and $L_2 = \Theta(P^{3/2}/(\ln P)^{3/4})$.

Exact Gaussian elimination, with one unknown per box point and one equation per residue class, finds the unique solution when the data come from a function on the box.

Part (b) is the quantitative modular form, proved in [Nag26c], of Hajdu and Tijdeman's factorization method for discrete tomography [HT01, Theorem 1]. If $L > D'$ then $\lfloor (D'-1)/(L-1)\rfloor = 0$ and $S_\varphi(0) = 0 < L$, so $L_2 \le D'$.

Supporting statement D.14 ([BT94, Section 2.3])**.** For $e \ge 1$,

$$\mathrm{Amp}_e(t) = 1 - (1-t)^e \sum_{j=0}^{e-1} \binom{e-1+j}{j} t^j$$

is an integer polynomial of degree $2e-1$. Modulo p^e, it maps every integer congruent to 0 modulo p to 0 and every integer congruent to 1 modulo p to 1.

This is the modulus-amplifying polynomial of Beigel and Tarui [BT94, Section 2.3].

E Proofs of statements from earlier work

The following proofs make the supporting interfaces self-contained. Citations identify the origin of an argument, rather than an assumed result. The published foundations are those of Section 1.4; standard field, probability, and arithmetic facts are used in their usual published forms. In the reproduced machine constructions, local exponents named G denote the general-circuit exponent written $G^\star$ in Appendix A; other variables are local to their stated interfaces.

E.1 The product-star partition

This is [Nag26b, Lemma A.1]; the proof is reproduced with notation adapted.

Proof. Use rb coordinates in r blocks, leaving the others unchanged. Let H be the $s \times b$ matrix whose columns are the nonzero vectors of $\mathbb{F}_2^s$. Its rank is s, so its kernel $\mathcal{C}$ has 2^{b-s} elements. Every $x \in \mathbb{F}_2^b$ has exactly one representation $x = c+e$, with $c \in \mathcal{C}$ and $e \in \{0, e_1, \dots, e_b\}$: the syndrome Hx uniquely specifies e. The same assertion holds with centers in any fixed coset of $\mathcal{C}$. Use all 2^s cosets in each block. This gives 2^{rs} partitions, and the displayed cardinalities follow by multiplication, including the unblocked coordinates as center coordinates.

A uniform coset followed by a uniform center is uniform on a block. Choose all errors independently of both. This proves the averaging claim. Assign $r/2$ whole blocks to each side. Each

contributes exactly $b - s$ free center bits, so the two blocked parts have equal dimension. Split the remaining $n - rb$ coordinates as evenly as possible. A fixed linear basis of each code gives polynomial-time encoding and decoding. Finally $b + 1 > (n/r + 1)/2$, hence $b + 1 = \Theta(n/r)$. □

E.2 Range of the four-corner difference

This is [Nag26b, Lemma A.2]; the proof is reproduced with notation adapted.

Proof. The underlying real scores at the corners have the form $a, a + \ell, a + t, a + \ell + t$: the left and right error supports are disjoint. Their opposite diagonal sums agree. A difference of 2 would require the two positively signed threshold values in the defining four-corner difference to be one and the other two to be zero. Their score sums would then be respectively at least and strictly below twice the threshold, a contradiction. The case -2 is identical with the diagonals reversed. □

E.3 The influence estimate

This elementary argument is reproduced from [Nag26b, Lemma A.3].

Lemma E.1 (A threshold's total influence)**.** *If a threshold depends on f input coordinates, then $\sum_i \mathrm{Inf}_i(T) \le \sqrt{f}$ under the uniform cube distribution.*

Proof. Complement coordinates of negative weight to make the function monotone. These complements preserve every influence. Use spin variables X_i and the spin-valued output $g = 2T - 1$. For a monotone function, conditioning on all coordinates except i shows $\mathbb{E}[g(X)X_i] = \mathrm{Inf}_i(T)$: the conditional expectation is one exactly when coordinate i is pivotal and zero otherwise. Therefore

$$\sum_i \mathrm{Inf}_i(T) = \mathbb{E}\left[g(X) \sum_{i \in \mathrm{supp}\, T} X_i\right] \le (\mathbb{E}g^2)^{1/2} \left(\mathbb{E}\left(\sum_{i \in \mathrm{supp}\, T} X_i\right)^2\right)^{1/2} = \sqrt{f}.$$

This elementary consequence of unateness is the only influence bound needed below. □

E.4 The mixed-dependence population bound

This is [Nag26b, Theorem A.4]; the proof is reproduced with notation adapted.

Write $I_L(T) = \sum_{i \text{ on the left}} \mathrm{Inf}_i(T)$ and let f_R be the right support size. The estimate proved below is

$$\Pr[\Delta_T \ne 0] \le \frac{2I_L(T)}{b+1} \min\{1, f_R/(b+1)\} \le \frac{2f}{(b+1)^{3/2}} \le 6\frac{r^{3/2}f}{n^{3/2}}.$$

Proof. If v misses the support, the mixed difference is zero. Otherwise, $\Delta_T \ne 0$ requires at least one of the left paths from c to $c + u$, or from $c + v$ to $c + v + u$, to cross a pivotal edge. Order the selected left coordinates canonically. After fixing both errors, every intermediate vertex is a fixed translate of the uniform ambient center, and so is uniform. The union bound on the two paths is $2\sum_{i \text{ selected on left}} \mathrm{Inf}_i(T)$.

The errors on the two sides are independent. Each left coordinate is selected with probability $1/(b+1)$, and a union bound on the right support gives $\Pr[v \text{ hits support}] \le \min\{1, f_R/(b+1)\}$. Averaging the path bound proves the first inequality. Apply Lemma E.1 and $\min\{1, t\} \le \sqrt{t}$ with $t = f/(b+1)$ for the second. The last follows from supporting statement D.2. Sum over bottom gates; their support sizes sum to at most the wire count. □

E.5 A sharp disjoint exact-child decomposition

This is [Nag26b, Lemma C.2], whose exact-child construction is [Nag26a, Lemma A.4]; the proof is reproduced with notation adapted.

Proof. Put $a_i = |w_i|$, let $z_i = x_i$ when $w_i \geq 0$ and $z_i = 1 - x_i$ when $w_i < 0$, and set

$$A_- = \sum_{i:w_i<0} a_i, \qquad T' = T + A_-.$$

The identity

$$\sum_i w_i x_i = \sum_i a_i z_i - A_-$$

reduces the gate to $\mathbf{1}[\sum_i a_i z_i > T']$. If $T' < 0$, use the single always-true equation $0 = 0$; if $T' \geq \sum_i a_i$, use the single always-false equation $0 = 1$. Hence assume $0 \leq T' < \sum_i a_i$. Let L be the least positive integer for which $T', a_1, \ldots, a_f \leq 2^L - 1$. Then $L \leq \ell$, and write

$$a_i = \sum_{b=1}^{L} 2^{b-1} a_{i,b}, \qquad a_{i,b} \in \{0, 1\}.$$

For $(a, b) \in [f] \times [L]$, define the partial score

$$P_{a,b}(z) = \sum_{i=1}^{f} z_i \left(\sum_{j>b} 2^{j-1} a_{i,j} + \mathbf{1}[i \geq a]\, 2^{b-1} a_{i,b} \right).$$

Prepend the explicit sentinel 0, and then order these scores as

$$P_{f,L}, P_{f-1,L}, \ldots, P_{1,L}, P_{f,L-1}, \ldots, P_{1,1}.$$

The sequence is nondecreasing and ends at $\sum_i a_i z_i$. Let $P^-_{a,b}$ denote the entry immediately before $P_{a,b}$, with the prepended sentinel used at the first position. Directly from the definition,

$$P_{a,b} - P^-_{a,b} = 2^{b-1} a_{a,b} z_a.$$

Consequently, on every accepting input there is a unique first pair (a, b) whose partial score exceeds T'. At that pair $P^-_{a,b} \leq T' < P_{a,b}$, so its step is nonzero and hence $a_{a,b} = z_a = 1$. Write

$$T' = 2^{b-1} Q_b + R_b, \qquad 0 \leq R_b < 2^{b-1}, \qquad Q_b \geq 0.$$

Both $P_{a,b}$ and $P^-_{a,b} = P_{a,b} - 2^{b-1}$ are multiples of 2^{b-1}. Therefore the two inequalities $P^-_{a,b} \leq T' < P_{a,b}$ hold if and only if $P_{a,b} = 2^{b-1}(Q_b + 1)$. Thus (a, b) is the first crossing if and only if

$$a_{a,b} = 1, \qquad z_a = 1, \qquad P_{a,b}(z) = 2^{b-1}(Q_b + 1).$$

For a zero bit $a_{a,b}$ omit the child. For a one bit use the single exact equation

$$2P_{a,b}(z) + z_a = 2^b(Q_b + 1) + 1.$$

Parity in the child equation forces $z_a = 1$, after which the equation is precisely the crossing conditions. Thus each accepting input satisfies its unique first-crossing child and every rejecting input satisfies none: the OR is disjoint.

There are at most fL nonconstant children. A coefficient on the left of the child equation is at most $2(2^L - 1) + 1 = 2^{L+1} - 1$. Since $Q_b \le 2^{L-b+1} - 1$, its right-hand side is at most $2^{L+1} + 1$. Substitution of $z_i = 1 - x_i$ negates the corresponding coefficient and subtracts it from the target, giving coefficients of magnitude at most $(f + 2)2^{\ell+1}$, hence the $O(\ell + \log(f + 2))$ bits of supporting statement D.5. Finally, in the nonconstant case $T' < \sum_{i=1}^{f} a_i \le fB'_0$, so $L \le \ell$; all loops and arithmetic are polynomial in the displayed output size. □

E.6 Separated pullback through the center coordinates

This is [Nag26b, Lemma C.3]; the proof is reproduced with notation adapted.

Proof. Write an original child as $c_j + \sum_i d_{j,i} x_i = 0$. Put its constant and the terms whose coordinate depends on z_{L} into $a_j(z_{\mathrm{L}})$, and the remaining terms into $b_j(z_{\mathrm{R}})$. Evaluating at most n coordinate functions and adding polynomial-bit integers takes polynomial time and increases bit length by at most $O(\log n)$. The identity and disjointness hold for every original Boolean input, so substitution preserves both.

□

In the product-star parametrization, keep each entire code block on one column side. Each coordinate of its center, including the subsequent exclusive-or with a fixed error row, then depends on that column side alone. Supporting statement D.6 applies even though a center bit may be a parity of many message bits. It does *not* assert that a real threshold remains a real threshold under a linear map over $\mathbb{F}_2$.

E.7 Stacking equations

This is [Nag26b, Lemma C.6]; the proof is reproduced with notation adapted.

Proof. The forward implication is immediate. Otherwise choose the largest j with $E_j(z) \neq 0$. Its contribution has magnitude at least H^{j-1}, whereas all lower contributions have total magnitude at most $V(H^{j-1} - 1)/(H - 1) < H^{j-1}$. Cancellation is impossible. Sum the left half-scores and the right half-scores separately with the same powers of H. Since $H = 2V + 2$, the largest power has $O(t \log(V + 2))$ bits, proving the bounds. For the empty conjunction, use the equation $0 = 0$. □

The published multiplication input.

Supporting statement E.2 (Williams)**.** There is one deterministic multitape Turing machine that, for all sufficiently large N, multiplies over the integers any two 0-1 matrices of dimensions $N \times \lceil N^{0.1} \rceil$ and $\lceil N^{0.1} \rceil \times N$ in $N^2 \operatorname{poly}(\log N)$ steps.

This is [WilACC14, Corollary 4.4 and Appendix C].

E.8 Signed rectangular multiplication

This is [Nag26b, Lemma C.4]; the proof is reproduced with notation adapted.

Lemma E.3 (Signed-by-Boolean multiplication)**.** *Fix C. Suppose $d \le \lceil U^{0.1} \rceil$, $A \in \mathbb{Z}^{U \times d}$, and $B \in \{0,1\}^{d \times U}$. If the entries of A have at most $(\log_2(U + 2))^C$ bits, then the exact product AB takes $U^2 \operatorname{poly}(\log U)$ bit operations.*

Proof. Let b bound the magnitude bit length, and write $A = \sum_{t=0}^{b-1} 2^t (A_t^+ - A_t^-)$, using separate Boolean magnitude planes for positive and negative entries. Apply supporting statement E.2 to the $2b$ products $A_t^{\pm} B$, padding the middle dimension with zeros. Their entries lie in $[0, d]$, so signed recombination uses $O(b + \log(d+1))$ bits per answer and the asserted total time. This is the binary-slicing reduction of [Nag26a, Lemma B.2]. □

E.9 The separated exact printer

This is [Nag26b, Theorem C.5]; the proof is reproduced with notation adapted.

Proof. This is the weighted-dominance proof of [Nag26a, Theorem B.3], with arbitrary evaluable half-scores in place of linear half-scores. We give the reduction to verify that this is the only property of linearity it uses. Replace an equality by the two successor cuts

$$\mathbf{1}[E_i = 0] = \mathbf{1}[E_i \geq 0] - \mathbf{1}[E_i \geq 1].$$

For even n_{out}, put $U = 2^{n_{\text{out}}/2}$. Enumerate the half-scores of the resulting $m \leq 2s$ cuts to express the required table as

$$P[a, b] = \sum_{j=1}^{m} \omega_j \mathbf{1}[A[a, j] \leq B[j, b]].$$

This takes $Um\,\text{poly}(n)$ time. For all sufficiently large n_{out}, $m \leq 2U^{0.02} \leq U^{0.049}$. Pad the gate dimension to $\lceil U^{0.049} \rceil$ using zero coefficients.

For each j, sort the $2U$ labeled entries in its A column and B row, putting every A entry before every B entry at an equal numerical value. Thus a weak comparison is true exactly when the A rank is earlier. Partition the ordered list into consecutive buckets of at most $\lceil U^{0.951} \rceil$ entries. For each pair (a, j), inspect the B entries in the same bucket, and add ω_j precisely when their ranks are ordered correctly. There are $O(U^2)$ such inspections.

For the remaining comparisons, form

$$A'[a, (j, h)] = \omega_j \mathbf{1}[A[a, j] \text{ is in bucket } h], \qquad B'[(j, h), b] = \mathbf{1}[B[j, b] \text{ is in a later bucket than } h].$$

The common dimension is $O(U^{0.098})$, hence at most $\lceil U^{0.1} \rceil$ eventually. Lemma E.3 computes $A'B'$ in $U^2\,\text{poly}(n)$ bit operations. Its entries are exactly the cross-bucket contributions. Add the already computed same-bucket contributions. All sorting, table construction, and additions fit the same bound. Coefficient and output lengths remain polynomial because each output is a sum of at most $2s$ polynomial-bit integers.

For odd n_{out}, add one ignored bit to the smaller half, run the even case, and retain its zero slice. This multiplies time by at most two. The finitely many sizes below the onset of supporting statement E.2 can be treated directly. At no point was a half-score required to be linear in its argument. □

E.10 A short seed and exact conditional expectations

This is [Nag26b, Lemma F.2]; the proof is reproduced with notation adapted.

Proof. Use the standard polynomial-evaluation construction of limited independence, in the universal-hashing tradition of Carter–Wegman [CW79]; its needed properties are verified here. Take $\ell \geq \max\{k, \lceil \log_2(Q+1) \rceil\}$ and a field $\mathbb{F}_{2^\ell}$. Label the Q samples by distinct field elements. Evaluate a uniformly random polynomial of degree $2h - 1$ at those labels and project each value to k binary

coordinates. Evaluations at any $2h$ distinct labels are independent uniform field elements, so the projected positions are $2h$-wise independent and uniform on $[N]^2$. The seed consists of $2h\ell$ binary coefficient bits. A defining irreducible polynomial and the arithmetic can be found and implemented in $2^{O(\ell)}$ elementary preprocessing time, which is allowed below; no randomness in field construction is needed.

For a fixed prefix, every assertion that a sample lies at a specified position is a system of binary linear equations in the remaining seed bits. A product of such indicators is their conjunction. Gaussian elimination gives conditional probability zero if it is inconsistent, and 2^{-a} if its remaining rank is a. The denominator divides 2^S. The trace expansion is finite, so these probabilities compute all conditional coefficients exactly. Repeated sample labels simply repeat or augment the same system.

The averaging identity is the ordinary partition of the remaining uniform seed space into its two equally sized children. □

E.11 Recovering distributions from modular counts

This reproduces [Nag26c, Section 4], with notation adapted. The counter of Section 8 will supply exact residue counts of integer scores. Residues must determine a *distribution*, not just a single integer. For example, knowing how often a score is even does not identify which even scores occurred. Our reconstruction lemmas apply to arbitrary rational-valued functions; positivity is not required. This permits a clean uniqueness proof by subtracting two candidate distributions.

Let $\mathcal{P}$ be a nonempty finite set of primes and put

$$D' = 1 + \sum_{p \in \mathcal{P}} (p - 1).$$

For a finitely supported function $F : \mathbb{Z}^d \to \mathbb{Q}$, define its modular push-forward by

$$F_p(\rho) = \sum_{u \equiv \rho \pmod p} F(u), \qquad \rho \in (\mathbb{Z}/p\mathbb{Z})^d.$$

Thus F_p records the total mass of each residue class.

E.12 One score: a roots-of-unity argument

Lemma E.4. *If $F : \mathbb{Z} \to \mathbb{Q}$ is supported on at most D' consecutive integers and $F_p = 0$ for every $p \in \mathcal{P}$, then $F = 0$.*

Proof. Shift the support to $\{0, \dots, D' - 1\}$ and form $f(t) = \sum_u F(u) t^u$. For every p-th root of unity ζ,

$$f(\zeta) = \sum_{\rho=0}^{p-1} F_p(\rho) \zeta^\rho = 0.$$

Distinct primes' root sets intersect only at one. Their union therefore contains D' distinct roots. A polynomial of degree below D' with this many roots is zero. □

E.13 Two scores: short projections and binomial factors

Let φ be Euler's totient function and set $S(q) = \sum_{m=1}^{q} m\varphi(m)$, with $S(0) = 0$. Hajdu and Tijdeman's discrete-tomography method [HT01, Theorem 1 and its preliminary divisibility lemma] turns zero line sums into binomial divisibility. We prove this over $\mathbb{Q}$ and count the usable directions.

Lemma E.5. *Let $L \geq 2$ and $q = \lfloor (D'-1)/(L-1) \rfloor$. If $S(q) \geq L$, then a rational-valued function supported on a given box with both sides at most L is uniquely determined, among functions supported on that box, by its push-forwards F_p, $p \in \mathcal{P}$.*

Idea of proof. Take the difference of two functions with the same data. Each of its projections along a short primitive direction is a one-score function with zero data, hence zero by the one-score lemma. Each vanishing projection makes an irreducible binomial divide the generating polynomial. These binomials have more total degree in one variable than the polynomial has, so the difference is zero.

Proof. Subtract two functions with the same data and translate their common box to $\{0, \ldots, L-1\}^2$. It suffices to show that a function with zero data is zero. Write its generating polynomial as

$$f(u,v) = \sum_{a,b=0}^{L-1} F(a,b) u^a v^b.$$

First, every short primitive projection is zero. A vector $(\alpha, \beta) \neq (0,0)$ is primitive if $\gcd(|\alpha|, |\beta|) = 1$. If $|\alpha| + |\beta| \leq q$, then

$$G(c) = \sum_{\alpha a + \beta b = c} F(a,b)$$

is supported on an interval containing at most $(L-1)q + 1 \leq D'$ integers. Its residue data are zero, since projection commutes with reduction modulo p. By Lemma E.4, $G = 0$. Equivalently, $f(t^\alpha, t^\beta) = 0$ in $\mathbb{Q}[t, t^{-1}]$.

Second, each vanishing projection forces a distinct prime factor. Choose one vector from each pair $\pm(\alpha, \beta)$, normalized so that $\alpha \geq 0$, and $\beta = 1$ if $\alpha = 0$. The kernel of the substitution $u \mapsto t^\alpha, v \mapsto t^\beta$ on $\mathbb{Q}[u,v]$ is generated by the binomial

$$B_{\alpha,\beta} = \begin{cases} u^\beta - v^\alpha, & \beta \geq 0, \\ u^{|\beta|} v^\alpha - 1, & \beta < 0. \end{cases}$$

In the Laurent polynomial ring, choose integers i, j with $\alpha i + \beta j = 1$. The monomials $\omega_1 = u^i v^j$ and $\omega_2 = u^\beta v^{-\alpha}$ form a coordinate system, because their exponent matrix has determinant -1. Substitution sends ω_1 to t and ω_2 to one, so its kernel there is $(\omega_2 - 1)$. Up to a Laurent monomial unit, this is the displayed binomial. That binomial has no factor u or v. It is irreducible in $\mathbb{Q}[u,v]$: a factorization gives a Laurent unit factor, since $\omega_2 - 1$ is irreducible in the Laurent ring, hence a monomial, which must be constant. Clearing Laurent denominators and using unique factorization gives the asserted polynomial kernel.

Nonparallel primitive directions give nonassociate binomials: substitution in the other direction gives two powers of t whose exponents differ by $\pm(\alpha\beta' - \beta\alpha') \neq 0$. By unique factorization, the product of these binomials divides f.

Finally, there are too many factors for a nonzero f. The sum of their degrees in u is $\sum |\beta|$, over primitive directions modulo sign of norm at most q. At norm one, the horizontal and vertical directions contribute one in total. At norm $m \geq 2$, the directions contribute

$$2 \sum_{\substack{1 \leq a < m \\ \gcd(a,m) = 1}} (m - a) = m\varphi(m),$$

using the symmetry $a \leftrightarrow m - a$. Thus the product has u-degree $S(q) \geq L$, whereas $\deg_u f \leq L - 1$. This is impossible unless $f = 0$. □

For a concrete picture, the projection $(\alpha, \beta) = (1, 1)$ adds masses along diagonals $a + b = c$. If all those sums vanish, $u - v$ divides f. The projection $(1, -1)$ instead forces $uv - 1$. The lemma uses many such independent factors, not just the coordinate projections.

Lemma E.6. *For every integer $q \geq 1$,*

$$S(q) \geq \frac{2q^3}{\pi^2} - q^2(4/3 + \ln q).$$

Call $L \geq 2$ admissible if $S(\lfloor (D' - 1)/(L - 1) \rfloor) \geq L$, as in Lemma E.5. Let L_2 be the largest admissible L, or one if none exists. Then $L_2 = \Theta(D'^{3/4})$ as $D' \to \infty$. In particular $L_2 \geq \left\lfloor (3/5)D'^{3/4} \right\rfloor$ for all sufficiently large D'.

The proof is a standard totient-summation calculation, given in Section E.15.

Uniqueness also gives an algorithm: introduce one rational unknown per box point and one equation per residue class. The matrix entries are zero or one. Exact Gaussian elimination finds the unique distribution whenever the data come from a function on the box. An algorithm may also detect inconsistency and report uncertainty. A box with one point is determined by its total mass, so the singleton convention in the definition of L_2 is also sound.

E.14 How many useful primes are available?

For an integer $P \geq 3$, let $\mathcal{P}$ consist of all primes in $(P/2, P]$. Only the order of magnitude of this set is needed.

Lemma E.7. *For this prime set, as $P \to \infty$,*

$$D' = \Theta(P^2/\ln P), \qquad L_2 = \Theta\left(P^{3/2}/(\ln P)^{3/4}\right).$$

Proof. By Chebyshev-type bounds (for instance Erdős' proof of Bertrand's postulate [Erd32]), the interval $(P/2, P]$ contains $\Theta(P/\ln P)$ primes. Each such prime contributes more than $P/2 - 1$ and at most P to D', so $D' = \Theta(P^2/\ln P)$. Apply Lemma E.6. □

Remark E.8 (The limitation of full-box residue reconstruction). For distributions on $\{0, \ldots, L-1\}^d$, the residue map has rank at most $1 + \sum_{p \in \mathcal{P}}(p^d - 1)$: all primes report the same total mass. Injectivity therefore requires $L^d \leq 1 + \sum_p (p^d - 1)$. For two scores this yields $L = O(P^{3/2}/\sqrt{\ln P})$. Our box bound has the same power of P, with an additional factor $(\ln P)^{1/4}$ loss. This is a restriction on recovering *arbitrary* box-supported distributions from these data, not an impossibility theorem for all circuit-analysis methods or for more structured distributions.

E.15 Totient sums and the two-score window

Proof of Lemma E.6. This reproduces [Nag26c, Appendix A], with notation adapted. These are standard totient-summation calculations; see Apostol [Apo76, Chapters 2–3 and 11] for the Möbius identities, arithmetic averages, and Euler products. Write μ for the Möbius function. Since $m\varphi(m) = \sum_{d|m} \mu(d) m^2/d$,

$$S(q) = \sum_{d \leq q} \mu(d) d \sum_{e \leq q/d} e^2.$$

The sum-of-squares formula gives $\left|\sum_{e\le y} e^2 - y^3/3\right| \le y^2$ for $y \ge 1$. Therefore

$$S(q) \ge \frac{q^3}{3}\sum_{d\le q}\frac{\mu(d)}{d^2} - q^2\sum_{d\le q}\frac{1}{d}.$$

Use $\sum_{d\ge 1}\mu(d)/d^2 = 6/\pi^2$, a tail of magnitude at most $1/q$, and $\sum_{d\le q} 1/d \le 1 + \ln q$. For $L = \left\lfloor cD'^{3/4}\right\rfloor$, the associated q is $(1/c + o(1))D'^{1/4}$. Thus $S(q) \ge L$ eventually whenever $c^4 < 2/\pi^2$; this includes $c = 3/5$. Conversely $S(q) \le \sum_{m\le q} m^2 = O(q^3)$, and $q \le (D'-1)/(L-1)$, so every admissible L is $O(D'^{3/4})$. □

E.16 Nested subsampling produces short integer digits

This reproduces [Nag26c, Section 5], with notation adapted. We now turn a potentially large live sum into short forms that can be identified from one residue. Sampling at a single very small rate would make the first sampled sum short, but leave a poor estimate of the original sum. The chain increases the rate while keeping each new correction short. Integer rate ratios preserve exact arithmetic.

Lemma E.9 (Choice of rates)**.** *Let $T > 0$, let h be an integer with $h \ge 64T$, and let $\ell_0 \ge 0$. Define*

$$\bar{\ell} = \max\{\ell_0, h^2/(256T)\}, \quad \pi^* = \frac{h^2}{8T\bar{\ell}}, \quad s(\pi) = \sqrt{2T\pi\bar{\ell}} + \frac{2T}{3},$$

$$M_1 = \left\lceil 4\bar{\ell}/h\right\rceil, \qquad \pi_1 = 1/M_1.$$

While $x_j = \pi_j/\pi^* < 2^{-8}$, set

$$a_j = \left\lfloor 1/(4\sqrt{x_j})\right\rfloor, \qquad \pi_{j+1} = a_j\pi_j.$$

Stop at the first index J with $x_J \ge 2^{-8}$. Then

(i) $a_j \ge 4$, *all rates lie in* $(0,1]$, *and* $\pi_j = \prod_{i<j} a_i/M_1$.

(ii) $\pi_1\bar{\ell} + s(\pi_1) \le h/2$, *and* $s(\pi_{j+1}) + a_j s(\pi_j) \le h/2$ *for* $j < J$.

(iii) $J \le \lceil \log_2\log_2(h/T)\rceil + 1$.

(iv) $s(\pi_J)/\pi_J \le 86T\bar{\ell}/h$.

Proof. Since $h \ge 64T$, the second entry of the maximum gives $\bar{\ell} \ge h^2/(256T) \ge h/4$. Thus $4\bar{\ell}/h \le M_1 \le 8\bar{\ell}/h$, and

$$T/h \le x_1 \le 2T/h \le 1/32.$$

Write $\tau(\pi) = \sqrt{2T\pi\bar{\ell}} = (h/2)\sqrt{\pi/\pi^*}$. For the first digit, $\pi_1\bar{\ell} \le h/4$, $\tau(\pi_1) \le \sqrt{Th/2} \le h/8$, and $2T/3 \le h/96$.

During a step, $a_j \ge 4$, $a_j\sqrt{x_j} \le 1/4$, and $a_j x_j \le \sqrt{x_j}/4$, so $\sqrt{a_j x_j} \le x_j^{1/4}/2 < 1/8$. Consequently

$$\tau(\pi_{j+1}) + a_j\tau(\pi_j) = \frac{h}{2}\left(\sqrt{a_j x_j} + a_j\sqrt{x_j}\right) \le \frac{3h}{16}.$$

The x_j increase, so $a_j \le \sqrt{h/T}/4$, and

$$(1 + a_j)\frac{2T}{3} \le h/96 + \sqrt{hT}/6 \le h/32.$$

These bounds prove (ii). Every newly produced x_{j+1} is below $1/64$, while $x_1 \le 1/32$. Also $\pi^* \le 32$, so all rates are at most one.

Since $1/(4\sqrt{x_j}) > 4$ during a step, flooring loses at most a factor two, and $x_{j+1} \ge \sqrt{x_j}/8$. For $y_j = \log_2(1/x_j)$, this means $y_{j+1} - 6 \le (y_j - 6)/2$. If the chain has not stopped earlier, iteration gives $y_j \le 6 + 2^{-(j-1)} y_1$, with $y_1 \le \log_2(h/T)$. A step at index j requires $y_j > 8$, hence $2^{j-1} < y_1/2$ and $j < \log_2 y_1$. Applied to $j = J - 1$ when $J \ge 2$, this gives $J < \log_2 \log_2(h/T) + 1$, which proves (iii). At the stopping index, $x_J \ge 2^{-8}$, so $\pi_J \ge h^2/(2048 T\bar{\ell})$; with $T/h \le 1/64$ this gives

$$\frac{s(\pi_J)}{\pi_J} \le \frac{64 T\bar{\ell}}{h} + \frac{1366 T^2 \bar{\ell}}{h^2} \le \frac{86 T\bar{\ell}}{h}.$$

This proves (iv) and completes the argument. □

Expand a top sum X into signed unit copies $\sigma_t g_{i(t)}$. Choose independent uniform $u_t \in [0,1]$ and define

$$K_j = \sum_t \mathbf{1}[u_t \le \pi_j] \sigma_t g_{i(t)}, \qquad E_0 = K_1, \quad E_j = K_{j+1} - a_j K_j \ (1 \le j < J).$$

Let

$$A = \prod_{i=1}^{J-1} a_i, \qquad A_0 = A, \qquad A_j = \prod_{i=j+1}^{J-1} a_i \quad (1 \le j < J).$$

The identities

$$K_J = \sum_{j=0}^{J-1} A_j E_j, \qquad \pi_J = A/M_1 \tag{7}$$

follow by repeatedly substituting $K_{j+1} = a_j K_j + E_j$. The first identity holds for *any* proposed coefficient vectors $K_1, \dots, K_J$, whether or not they arose from nested sampling. Proposition 10.1 uses this.

E.17 The Beigel–Tarui lifting polynomial

This reproduces [Nag26c, Lemma 6.6], with notation adapted.

Lemma E.10. *For $e \ge 1$, define*

$$\mathrm{Amp}_e(t) = 1 - (1-t)^e \sum_{j=0}^{e-1} \binom{e-1+j}{j} t^j.$$

This integer polynomial has degree $2e-1$, with $t^e \mid \mathrm{Amp}_e(t)$ and $(1-t)^e \mid 1 - \mathrm{Amp}_e(t)$. Consequently, modulo p^e, it maps an integer congruent to zero modulo p to zero, and one congruent to one to one.

This is the modulus-amplifying polynomial of Beigel and Tarui [BT94, Section 2.3].

Proof. The power-series identity $(1-t)^{-e} = \sum_{j \ge 0} \binom{e-1+j}{j} t^j$ shows that truncating at degree $e-1$ leaves an error divisible by t^e. Multiplication by $(1-t)^e$ proves the first divisibility assertion. The second is immediate from the definition. Both imply the stated congruences, including for negative integer arguments. □

E.18 Pointwise proximity and the five-weight gadget

This reproduces [Nag26c, Sections 11.1–11.3], with notation adapted.

Supporting statement E.11 (Pointwise three-query PCPP)**.** There are a rational constant $0 < \gamma < 1$, a polynomial-time computable linear encoding Enc, and a polynomial-time transformation taking a Boolean circuit D on q inputs to a polynomial-size list of three-literal clauses on variables $Y \cup Z$, such that the following hold for every $x \in \{0,1\}^q$ with $Y = \mathrm{Enc}(x)$:

- If $D(x) = 1$, an auxiliary assignment computable in polynomial time satisfies every clause.
- If $D(x) = 0$, every auxiliary assignment leaves at least a γ fraction of the clauses false.

Each systematic coordinate is a parity of input coordinates. The number of clauses can be taken to be a power of two.

This is the pointwise form of Chen–Williams' perfectly complete three-query PCPP [CW19, Lemma 24 and Appendix A], using their linear code and decoder [CW19, Lemma 27], which Section E.34 derives from [CW19, Lemma 26]. As in the proof of [CW19, Theorem 5], here is the reduction to the stated form. Take a fixed proximity radius smaller than the code's decoding radius, totalize the decoder by returning an arbitrary fixed string on failure, and apply the PCPP to $\widetilde{D}(y) = D(\mathrm{Dec}(y))$. If $D(x) = 0$, every word sufficiently close to $\mathrm{Enc}(x)$ decodes to x, and is therefore rejected by $\widetilde{D}$. Thus $\mathrm{Enc}(x)$ is farther than the proximity radius from every accepting input of $\widetilde{D}$, and proximity soundness applies. If $D(x) = 1$, perfect completeness and the efficient honest-proof procedure of [CW19, Lemma 24] apply. The underlying proximity construction is due to Ben-Sasson, Goldreich, Harsha, Sudan, and Vadhan [BGHSV06]. In the constant-size clause conversion in the proof of [CW19, Lemma 24], pad to a power-of-two number of clauses with tautologies before selecting a clause uniformly. This preserves perfect completeness and changes only the fixed rejection constant. Finally choose a smaller rational γ if necessary.

For one three-literal clause, let x_1, x_2, x_3 be its literal values. The classical Garey–Johnson–Stockmeyer gadget [GJS76], in the form recorded by Chen–Williams [CW19, Lemma 47], introduces a Boolean helper y and uses the ten constraints

$$x_1,\ x_2,\ x_3,\quad \neg x_1 \vee \neg x_2,\ \neg x_2 \vee \neg x_3,\ \neg x_3 \vee \neg x_1,\quad y,\quad x_1 \vee \neg y,\ x_2 \vee \neg y,\ x_3 \vee \neg y.$$

If $k = x_1 + x_2 + x_3$, their total satisfaction score is $\mathrm{sat}(x, y) = 6 + k - \binom{k}{2} + y(k-2)$. We use the penalty

$$\begin{aligned} Q(x_1, x_2, x_3, y) &= 7 - \mathrm{sat}(x, y) \\ &= 1 - \sum_i x_i + \sum_{i<j} x_i x_j + 2y - y \sum_i x_i. \end{aligned} \tag{8}$$

Its values on Boolean arguments are

k	0	1	2	3
$Q(k,0)$	1	0	0	1
$Q(k,1)$	3	1	0	0

Thus $Q \geq 0$, and a false original clause forces $Q \geq 1$ for either helper. For a true original clause, the choice $y^* = x_1 x_2 x_3$ gives penalty zero.

Lemma E.12 (Weighted stability of the gadget)**.** *For every true honest triple $x \neq 000$, with $y^* = x_1 x_2 x_3$, and every Boolean replacement x', y', we have*

$$Q(x', y') \leq \sum_{i=1}^{3} |x_i - x_i'| + 2|y^* - y'|.$$

Proof. If $y' = y^* = 0$, the honest triple has weight one or two. The only nonzero penalties with helper zero occur at weights zero and three, each requiring an input change. If $y' = y^* = 1$, the honest triple is 111, and the table gives $Q(k', 1) \le 3 - k'$, its Hamming distance from that triple. If the helper changes, its weight already pays two. The only penalty exceeding two is $Q(0, 1) = 3$; this also requires at least one input change from a true honest triple. □

The aggregate seven-versus-six score is classical [GJS76], and Chen–Williams use it in this aggregate form to prove [CW19, Lemma 25]; what is added here is its weighted stability under proof-table errors. This estimate uses cancellation within the ten-clause score. Applying a separate union bound to every clause would discard that cancellation. The helper has weight two, and each original coordinate has weight one.

E.19 A certified upper estimate using pair means

Writing the two-literal clauses in the XOR basis, as Chen–Williams do for two-query predicates [CW19, proof of Theorem 5], the identity $ab = (a + b - (a \oplus b))/2$ gives

$$Q = 1 + \frac{1}{2}\left(y - \sum_i x_i - \sum_{i<j}(x_i \oplus x_j) + \sum_i (x_i \oplus y)\right). \tag{9}$$

There are ten probability terms, all with coefficient of magnitude $1/2$. A literal is a guessed Boolean proof coordinate or its complement, or a prescribed systematic parity or its complement. Hence every nonexact term is a single-circuit, pair-XOR, or circuit-XOR-raw-parity call. Pure parity terms, constants, and repeated-coordinate identities are evaluated exactly. No closure of the target class under XOR is used.

For each nonexact term let $(\widehat{p}, \widehat{\kappa})$ be the counter's estimate and certified radius, and let $\sigma \in \{-1, 1\}$ be its sign in Equation (9). Substitute $\sigma\widehat{p} + \widehat{\kappa}$ for the signed probability and average the resulting expressions over all gadgets. Call the result U, and write $\mathbb{E}Q$ for the exact average of the penalty over gadgets and over $x \in \{0, 1\}^q$, evaluated on the guessed coordinates. Soundness of the intervals gives

$$U \ge \mathbb{E}Q \quad \text{for every witness.} \tag{10}$$

For a useful witness (Proposition 10.2) with all radii at most ω, it also gives

$$U \le \mathbb{E}Q + 10\omega. \tag{11}$$

Indeed a signed probability endpoint loses at most 2ω, and its coefficient has magnitude $1/2$. Arbitrary witnesses may return broader intervals; only the existence of a useful witness asserts the small-radius bound.

E.20 The succinct source construction

This is [Nag26c, Section 11.4]; the construction is reproduced with notation adapted. The general-circuit exponent is $G^\star$ as in Section A.7; $h_{\mathrm{sim}} = 6$ by Lemma A.5. The derived short-PCP interface is supporting statement A.1.

For a general circuit B with $a \ge \nu$ inputs, let $\mathrm{tt}_N(B) \in \{0, 1\}^N$ be the word whose ith bit is $B(\mathrm{bin}_\nu(i)0^{a-\nu})$, where $\mathrm{bin}_\nu(i)$ is the ν-bit binary expansion of $i < N$. Fix a self-delimiting binary description of general circuits that lists the gates, so that a description of n bits has at most n gates, and whose length is polynomial in the numbers of gates and inputs. Fix an integer $G' \ge 2$, depending only on $G^\star$ and on this description format, such that for all large ν every description

of a circuit with at most $q^{G^\star}$ gates on q inputs has fewer than $\nu^{G'}$ bits. Such a G' exists: these descriptions have length at most a fixed polynomial in $q = O(\nu)$ whose degree depends only on $G^\star$ and the format, and any G' above that degree works. Put

$$L_\nu = \nu^{G'},$$

and, when the description of B has at most L_ν bits, let $\langle B\rangle_\nu \in \{0,1\}^{L_\nu}$ be that description followed by zeros.

Lemma E.15 in Section E.21 gives one deterministic multitape verifier $\mathcal{V}_{G^\star}(u, y)$, a nondecreasing time bound $T_{G^\star}(n) \geq n$ within which $\mathcal{V}_{G^\star}$ halts on every input of length n and every witness, and a constant $c_{G^\star}$. They satisfy $T_{G^\star}(L_\nu) = c_{G^\star}\mathcal{T}(N)$, and, for every such B, some witness of length $T_{G^\star}(L_\nu)$ makes $\mathcal{V}_{G^\star}$ accept $\langle B\rangle_\nu$ exactly when $\mathrm{tt}_N(B) \in L_{\mathcal{T}}$. The only size bound that $\mathcal{V}_{G^\star}$ places on B is the length L_ν, which depends on N and the fixed G' alone; the weak machine enforces the bound $q^{G^\star}$ separately. Only three properties of L_ν are used:

- it is a fixed polynomial in ν;
- it has degree at least two;
- it eventually exceeds the description length of every circuit with at most $q^{G^\star}$ gates on q inputs.

L_ν is fixed before $\mathcal{V}_{G^\star}$, and these properties hold whatever constants are chosen after it.

We apply supporting statement A.1 once, to $\mathcal{V}_{G^\star}$ with the time bound $T_{G^\star}$, which is computable from the input length in polynomial time, as that input requires (Lemma E.15(i)). On an input $\langle B\rangle_\nu$ it constructs, in time $\mathrm{poly}(L_\nu, \log T_{G^\star}(L_\nu)) = \nu^{O_{G^\star,C_{\mathrm{clk}}}(1)}$, an oracle circuit $C^{O}_{\langle B\rangle_\nu}$ of size $\nu^{O_{G^\star,C_{\mathrm{clk}}}(1)}$ with $\ell_{\mathrm{out}} = \log_2 \mathcal{T}(N) + O(\log\log \mathcal{T}(N))$ inputs. Pad it to the exact native width

$$q(N) = \lceil \log_2 \mathcal{T}(N)\rceil + A_{\mathrm{pad}}\lceil \log_2(\log_2 \mathcal{T}(N) + 2)\rceil + A_{\mathrm{pad}},$$

by adding inputs that it ignores, for a sufficiently large fixed integer A_{pad}, chosen after C_{clk} as at the end of Section E.21 so that $\ell_{\mathrm{out}} \leq q(N)$. Then $q = \Theta(\log N)$ and $2^q/\mathcal{T}(N) = q^{O(1)}$.

The outer verifier. The source-circuit test is

$$B_{\mathrm{src}}(\mathrm{bin}_\nu(i)0^{q-\nu}) = \zeta_i \qquad \text{for every } i < N, \tag{12}$$

that is, $\mathrm{tt}_N(B_{\mathrm{src}}) = \zeta$, and rejects otherwise. It constructs $C^{O}_{\langle B_{\mathrm{src}}\rangle_\nu}$, whose number ℓ_{out} of inputs may depend on B_{src}, and guesses a general circuit B_{pf} with q inputs and at most $q^{G^\star}$ gates, the outer proof oracle, again rejecting malformed descriptions. It answers an oracle query $a \in \{0,1\}^{\ell_{\mathrm{out}}}$ by

$$O_{B_{\mathrm{pf}}}(a) = B_{\mathrm{pf}}(a0^{q-\ell_{\mathrm{out}}}),$$

the same zero padding as in (12). Write $B = (B_{\mathrm{src}}, B_{\mathrm{pf}})$. Substituting $O_{B_{\mathrm{pf}}}$ for every oracle gate and negating the output gives the outer verifier

$$D_B(x) = \neg\, C^{O_{B_{\mathrm{pf}}}}_{\langle B_{\mathrm{src}}\rangle_\nu}(x_1, \ldots, x_{\ell_{\mathrm{out}}}), \qquad x \in \{0,1\}^q,$$

which ignores its last $q - \ell_{\mathrm{out}}$ inputs. The negation turns the convention of supporting statement A.1 into the usual one: $D_B(x) = 1$ means that the outer verifier accepts the random string x. Since $L_\nu \leq q^{G'}$, D_B is a circuit of size $q^{O_{G^\star,C_{\mathrm{clk}}}(1)}$ on q inputs. The exponent depends on C_{clk} as well, through $\mathcal{V}_{G^\star}$; the constants that depend on it, beginning with the exponent $d_{G^\star}$ below, are chosen

after C_{clk} (Section E.21). Apply supporting statement E.11. Choose a fixed $d_{G^\star}$ bounding all resulting clause counts and repeat the whole clause list to exactly

$$M = 2^{\lceil d_{G^\star} \log_2(q+2) \rceil}$$

clauses. If the list is longer than M, which happens at only finitely many lengths, the machine rejects. The original length divides this number. This fixes the eventual table arity independently of the guessed pair.

E.21 Multitape readings of the refuter and the succinct verifier

This is [Nag26c, Appendix C]; its proof is reproduced here, with notation adapted. The Lean companion derives the refuter separately, through its source adapters; the written argument is not a transcription of that formal proof. This appendix supplies the machine-model facts used by the lower-bound transfer (Appendix A). The transfer applies supporting statement A.2 and supporting statement A.1 to multitape machines, the model of Section 2. The refuter's proof in [CLW20, Section 4], however, assumes random-access machines for simplicity. The appendix has three parts.

- Lemma E.13 derives the multitape reading of the refuter from the proof in [CLW20, Section 4]. It imports the accepting-time universal simulation of [SFM78, BGW70], used with genuine input and witness prefixes, and it gives the hierarchy language $L_{\mathcal{T}}$ a multitape verifier $\mathcal{D}$. Corollary E.14 then proves the exact framed convention of the Lean interface, including at linear clocks.
- Lemma E.15 constructs the one fixed machine to which the short PCP is applied, the succinct hierarchy verifier $\mathcal{V}_G$. It reads a source input through the short description of a circuit and runs $\mathcal{D}$ on the described word. Section E.25 proves the short PCP for it.
- The paragraph after Lemma E.15 fixes the order in which the constants of the transfer are chosen.

Lemma E.13 (Multitape refuter)**.** *Supporting statement A.2 holds with every machine in it, the class* NTIME *and the procedure $R_{\mathcal{T}}$ read for multitape Turing machines, with the same quantifier order. The language $L_{\mathcal{T}}$ has a nondeterministic multitape verifier that halts within $O(\mathcal{T}(N))$ steps on every input and every guess.*

Idea of proof. The diagonal language of Chen, Lyu, and Williams runs each enumerated machine through a universal simulation. We use the accepting-time simulation of Seiferas, Fischer, and Meyer for multitape machines, under an explicit clock of $O(\mathcal{T}(N))$ steps (Step 1). For each admitted machine, every verdict required by the diagonal recurrence has a faithful accepting simulation that survives this clock at large lengths; other branches may time out. Thus [CLW20, Lemma 4.3] holds for it (Step 2). The rest of their proof uses the machine model only through that lemma and through SAT-oracle queries, which the Cook–Levin tableau handles for multitape machines (Step 3). So their refuter applies verbatim (Step 4), and the guess budgets are unchanged (Step 5).

Proof. Conventions. A nondeterministic multitape machine reads its guessed bits from a witness tape and is deterministic once the witness is fixed. It guesses at most $g(N)$ bits when every run on an input of length N reads at most $g(N)$ witness cells. Fix $\mathcal{T}$, which is computable from 1^N in $O(\mathcal{T}(N))$ steps, and an enumeration of these machines. As in [CLW20, Section 4.1], $V_M(x, T, z)$ is the outcome of running M on x with witness z for at most T steps. It is $\perp$ when the run does not halt within T steps or reads more than $|z|$ witness bits. A machine that guesses at most

$g(N) \le |z|$ bits behaves the same with the witness z and with its prefix of length $g(N)$. So it may equally be given exactly $g(N)$ witness bits. This proof uses the ordinary witness-tape convention. Corollary E.14 treats the framed convention directly, without adding preprocessing to the weak machine.

The universal simulation. We use the accepted-input time definition of [SFM78, p. 147], its tape-reduction Lemma 3 (pp. 149–150), and its universal-simulation Condition 2 (p. 152). Composing the tape reduction with that simulator gives a fixed universal nondeterministic acceptor U that preserves the language of each simulated machine P: if P has an accepting computation of length t on u, then U on its code and u has one of length at most $c_P(t+1)$. Here c_P depends only on P; the additive one covers initialization and zero-step computations. No halting or time-bound hypothesis is imposed on other inputs or branches. The tape reduction is due to Book, Greibach, and Wegbreit [BGW70]; [SFM78] sketches its construction and states Condition 2 as a property of the program coding. Fortnow and Santhanam use the linear-time simulation in their hierarchy argument [FS16, proof of Theorem 7.1].

Shared prefix simulation. First prepare the actual input and supplied witness tapes, with their prescribed boundaries and heads at zero, in a uniform $c_0 N$ setup independent of M. For each simulation the hierarchy verifier guesses a positive length r in unary and copies cells $0, \dots, r-1$ of both prepared tapes, including genuine blank cells when necessary. It packs these two prefixes, their boundary flags, and cutoff markers at cell r into one binary word of length $O(r+1)$, using a fixed encoding independent of M. The contents and flags are copied, not guessed. Two fixed acceptors P_M^+ and P_M^- unpack this word into separate tapes, copying the second prefix to a work tape and rewinding in $O_M(r+1)$ time. They then execute the deterministic computation of M with this witness fixed, accepting only upon, respectively, an accepting or a rejecting halt of M. Reaching a cutoff aborts without either verdict, before executing an original transition there. The cutoff and boundary flags are kept on protected tracks, so writes by M cannot erase them. In the present ordinary-witness convention, an attempt to read beyond the supplied witness has exactly the V_M failure behavior $\perp$, and neither acceptor accepts it. In the framed convention of Corollary E.14, the copied tapes instead include the prescribed frame terminators and blank zeros beyond them, with no artificial failure at a frame terminator. These are separate acceptors for the two verdicts; we never complement the universal acceptor.

For every guessed r, the initial simulated cells agree with the original tapes. Induction on the simulated steps, including their writes, preserves the state, heads, and all visited cells until a cutoff or a genuine halt. Hence every verified verdict is genuine. Conversely, if the original computation halts in t steps, the choice $r = t+2$ keeps both heads strictly before the cutoffs: a head starts at zero and moves by at most one cell per step. Prefix copying, unpacking, and this computation cost $O_M(t+1)$, and the universal simulation therefore has a corresponding accepting branch within $a_M + b_M(t+1)$ steps, after increasing fixed constants. The choice of r is existential in this argument; the algorithm does not compute t or a runtime bound for M. The original work tapes remain unbounded. A computation on some other prefix can run arbitrarily long without reaching a cutoff, which is why we use an accepting-time simulation rather than assert a time bound for every input of P_M^+ or P_M^-.

Step 1: the hierarchy verifier. Let $\mathcal{D}$ be the following clocked variant of the algorithm $A_{\mathrm{FS}}^{\mathcal{T}}$ of [CLW20, Definition 4.2]. First $\mathcal{D}$ computes $\mathcal{T}(N)$ and prepares its clock in $O(\mathcal{T}(N)+1)$ steps. It then parses its input as $\langle M, N, z\rangle$ with $|z| = \lfloor N/10 \rfloor$, rejecting otherwise. This takes at most $c_0 N$ steps for a constant c_0 that does not depend on M, enlarged to include preparation of the actual tapes described below. It then uses the shared prefix construction for two universal simulations:

(a) the rejecting-verdict acceptor P_M^-, for M on input $\langle M, N, 0^{\lfloor N/10 \rfloor}\rangle$ with witness z;

(b) when $z \neq 1^{\lfloor N/10 \rfloor}$, the accepting-verdict acceptor P_M^+, for M on input $\langle M, N, z+1 \rangle$ with a guessed witness of $\lfloor N/10 \rfloor$ bits.

It accepts exactly when the first simulation and, when required, the second simulation accept. The original run in (a) has a fixed witness and is deterministic; an accepted first simulation certifies its rejecting halt, never a cutoff or a failed simulation. The clock rejects every branch whose steps after clock preparation exceed $(c_0+2)(\mathcal{T}(N)+1)$, including overlong prefix guesses, machine-description preparation, and universal-simulation branches. Hence every branch halts within $O(\mathcal{T}(N)+1)$ steps on every input, and $L_{\mathcal{T}} = L(\mathcal{D})$ has the stated verifier.

The language $L(\mathcal{D})$ may differ from $L(A_{\mathrm{FS}}^{\mathcal{T}})$ on inputs where a simulation exceeds the clock or the encoded machine breaks its promise. Step 2 shows that it obeys the formula of [CLW20, Lemma 4.3] on every input that the proofs use.

Step 2: Lemma 4.3 of [CLW20] holds for every admitted machine. Fix M that runs in time $t_M(N) = o(\mathcal{T}(N))$ on every input and witness and guesses at most $\lfloor N/10 \rfloor$ bits. Consider an input $x_i^{(N)} = \langle M, N, w_i^{(N)} \rangle$ of [CLW20, Section 4.2]. Whenever the original computations give the requested verdicts, the shared prefix argument supplies a branch verifying them with $r = t_M(N)+2$. After computing $\mathcal{T}(N)$ and preparing the clock, its total cost is at most

$$c_0 N + a_M + 2b_M(t_M(N)+1) < (c_0+2)(\mathcal{T}(N)+1)$$

for all sufficiently large N, since $N \leq \mathcal{T}(N)$ and $t_M(N) = o(\mathcal{T}(N))$. The constants a_M, b_M cover the fixed machine descriptions and prefix simulations; only c_0 is uniform in M. Thus every required verdict has an accepting simulation that survives the cap, whereas faithfulness rules out any branch certifying a false verdict. Other branches can time out. Every run of M halts within $t_M(N) \leq \mathcal{T}(N)$ steps and reads at most $\lfloor N/10 \rfloor$ witness bits. Hence $\mathcal{D}$ accepts $x_i^{(N)}$ exactly when $V_M(x_1^{(N)}, \mathcal{T}(N), w_i^{(N)}) = 0$ and, for $i < 2^{\lfloor N/10 \rfloor}$, $M(x_{i+1}^{(N)}) = 1$. This is the formula of [CLW20, Lemma 4.3], which therefore holds for $\mathcal{D}$ and M at every sufficiently large N. The overhead is the constant factor b_M, so the little-o promise needs no logarithmic slack.

Step 3: the oracle queries. The proofs of [CLW20, Theorems 1.12 and 4.6] use the machine model only through Lemma 4.3 and through SAT-oracle queries of three kinds:

- whether $\mathcal{D}$ accepts a given input;
- whether M accepts a given input within $\mathcal{T}(N)$ steps with some witness in a given lexicographic interval;
- whether $V_M(x, \mathcal{T}(N), w) = 1$ for given x and w.

Each query concerns runs of $O(\mathcal{T}(N))$ steps of one nondeterministic multitape machine. The Cook–Levin tableau turns it into a CNF formula of size $O(\mathcal{T}(N)^2)$. The formula is constructed in $\mathrm{poly}(\mathcal{T}(N))$ time by a polynomial whose degree does not depend on M; for queries about M, only the constant factor depends on M. For the companion's exact SAT oracle, apply the standard equisatisfiable Tseitin conversion and repeat literals in unit and binary clauses so that every clause has exactly three literals; an empty clause is represented by a fixed contradictory pair of padded unit clauses. Encode each such clause using the companion's native literal-list coding, and encode the formula using its balanced clause-list coding. Because the clause length is fixed and the outer list is balanced, these encodings and the query framing have polynomial length and construction time, with a degree independent of M. Thus the oracle answers and the stated polynomial bounds are preserved.

Step 4: the refuter. With Steps 2 and 3, the proof of [CLW20, Theorem 4.6] applies verbatim. $R_{\mathcal{T}}$ first asks whether M accepts $x_1^{(N)}$. If not, it runs the binary search of [CLW20, Corollary 4.5]

on the lists $\mathcal{D}(x_i^{(N)})$ and $M(x_i^{(N)})$, $1 \le i \le 2^{\lfloor N/10 \rfloor}$. If so, it finds by binary search the least j with $V_M(x_1^{(N)}, \mathcal{T}(N), w_j^{(N)}) = 1$ and runs the same search on the first j entries. Either way it makes $O(N)$ adaptive queries and runs in $\mathrm{poly}(\mathcal{T}(N))$ time, with a degree that does not depend on M. At every sufficiently large N it returns a word $x_p^{(N)}$ with $\mathcal{D}(x_p^{(N)}) \neq M(x_p^{(N)})$.

Step 5: the guess budgets. The guesses made by the simulations in Step 1 are guesses of $\mathcal{D}$, not of M, so the promise on M is unchanged. The weak machine of Appendix A guesses at most $\lfloor N/16 \rfloor \le \lfloor N/10 \rfloor$ bits and runs in time $o(\mathcal{T}(N))$, so the lemma applies to it. □

Corollary E.14 (Refuter for framed machines)**.** *The conclusion of supporting statement A.2 also holds with the following convention, for every time-constructible $N \le \mathcal{T}(N) \le 2^{\mathrm{poly}(N)}$. Set $g(N) = \lfloor N/16 \rfloor$ and*

$$\mathrm{fr}(b_1 \cdots b_m) = 1b_1\, 1b_2 \cdots 1b_m\, 0, \qquad \mathrm{fr}(\epsilon) = 0.$$

A weak machine is a deterministic multitape verifier whose initial tapes contain $\mathrm{fr}(x)$ and $\mathrm{fr}(w)$, with $|x| = N$ and $|w| = g(N)$; all other tapes are blank, all heads start at the leftmost cell, and blank cells contain zero. It halts within $t_M(N) = o(\mathcal{T}(N))$ steps for every such x, w, and $M(x) = 1$ means that some such w is accepted. The hierarchy language has a verifier with the same framing convention, time bound $c(\mathcal{T}(N) + 1)$, and exactly $c(\mathcal{T}(N) + 1)$ witness bits, for a fixed positive integer c. The hierarchy language and the refuter are fixed before M; only the eventual onset and the coefficient in the refuter's polynomial time bound may depend on M.

Proof. We apply the diagonal construction in the proof of Lemma E.13 directly to an enumeration of these framed verifiers. Thus M retains its original runtime and its original initial tapes. The two conventions may give different hierarchy languages, as supporting statement A.2 permits.

For fixed M, N, enumerate all $g(N)$-bit strings as $w_1, \ldots, w_s$, where $s = 2^{g(N)}$, and use the length-N records $x_i = \langle M, N, w_i \rangle$ of [CLW20, Section 4.1]. For every fixed M these records exist at all sufficiently large N. Write $V_M^{\mathrm{fr}}(x, t, w)$ for the outcome of the original verifier within t steps on exactly the two framed tapes above, with value $\perp$ if it has not halted. In particular the terminator is placed immediately after the $g(N)$th witness bit; no longer witness is substituted for w.

Uniform setup cost. On the record x_i, the hierarchy verifier prepares $\mathrm{fr}(x_1)$ and $\mathrm{fr}(w_i)$ for the first simulation. For the second simulation, when $i < s$, it prepares $\mathrm{fr}(x_{i+1})$ and $\mathrm{fr}(v)$, where it guesses exactly $g(N)$ bits v. Parsing the outer record, incrementing its witness field, writing these frames, and returning their heads to zero cost at most $c_0 N$ steps, for an absolute constant c_0. The frames have lengths $2N + 1$ and $2g(N) + 1$. Their bit representation is independent of M; it does not encode every data bit by a machine-dependent block.

Apply the shared prefix simulation from the proof of Lemma E.13 to these prepared framed tapes. The copied data include the actual terminator immediately after the prescribed witness, if it lies within the prefix, and actual blank zeros beyond it. Thus the deterministic computations being verified have exactly the initial tapes in the statement, even if M reads or writes past a frame terminator. Only the separate protected prefix cutoff aborts without a verdict. In particular an aborted prefix does not supply the rejecting verdict required in the first simulation.

For each promised computation choose $r = t_M(N) + 2$ in that argument. The fixed-machine decoding cost is a_M, and prefix processing and a useful accepting simulation cost at most $b_M(t_M(N)+1)$ per run. Whenever the original computations give the requested verdicts, there is therefore a branch verifying both with total cost at most

$$c_0 N + a_M + 2b_M(t_M(N) + 1).$$

Here c_0 is independent of the enumerated machine, while a_M, b_M may depend on it. In particular there is no machine-dependent coefficient multiplying the initial data length.

After computing $\mathcal{T}(N)$ and preparing a clock, impose the uniform step cap $(c_0+2)(\mathcal{T}(N)+1)$, as in Step 1 of Lemma E.13. For each fixed promised M,

$$c_0 N + a_M + 2b_M(t_M(N)+1) < (c_0+2)(\mathcal{T}(N)+1)$$

eventually, since $N \le \mathcal{T}(N)$ and $t_M(N) = o(\mathcal{T}(N))$. The cap rejects overlong branches; an invalid simulation cannot certify a verdict. Every required verdict has an accepting simulation within the cap at these lengths, and faithfulness prevents spurious accepting branches. The resulting diagonal language, say $L^{\mathrm{fr}}_{\mathcal{T}}$, thus satisfies the exact recurrence

$$L^{\mathrm{fr}}_{\mathcal{T}}(x_i) = \mathbf{1}[V^{\mathrm{fr}}_M(x_1, \mathcal{T}(N), w_i) = 0] \cdot \begin{cases} M(x_{i+1}), & i < s, \\ 1, & i = s. \end{cases}$$

This is the identity used by [CLW20, Lemma 4.3 and Theorem 4.6], with $g(N)$ in place of $\lfloor N/10 \rfloor$. Its proof and the binary-search refuter use only the complete enumeration of witnesses, so Steps 3–4 of Lemma E.13 apply. The SAT queries now encode the specified framed tapes and their fixed terminators. Their initial descriptions have length $O(N)$, and their time bounds are $O(\mathcal{T}(N))$; the same polynomial tableau bounds and $O(N)$ query count follow. Encoding the refuter's request and output with the companion's nested frame encoding changes these polynomial bounds only by constant factors. The polynomial degree is fixed with the refuter, before M.

Finally the clocked diagonal verifier uses $O(\mathcal{T}(N)+1)$ steps and nondeterministic bits on every branch. A deterministic verifier reads those choices from a framed witness, decoding each requested bit in constant time. Taking a sufficiently large fixed integer c gives both the step bound and the witness length $c(\mathcal{T}(N)+1)$; unused witness bits are ignored. This is the hierarchy convention in the statement. All setup and simulation guesses belong to this hierarchy verifier. The weak machine still has exactly $g(N)$ witness bits and its original runtime. The argument uses only $N \le \mathcal{T}(N)$, so it includes linear clocks. □

Lemma E.15 (Succinct hierarchy verifier)**.** *Fix integers $G \ge 1$, $G' \ge 2$ and $C_{\mathrm{clk}} \ge 2$, let $\mathcal{T}(N) = N^{C_{\mathrm{clk}}}$, let $\mathcal{D}$ be the verifier of $L_{\mathcal{T}}$ from Lemma E.13, and let $L_\nu = \nu^{G'}$. Use the self-delimiting binary description of general circuits fixed in Appendix A, and let tt_N and $\langle B \rangle_\nu$ be as there. There are a deterministic multitape machine $\mathcal{V}_G(u, y)$, with input u and witness y, and a constant $c_G \ge 1$ such that, with*

$$T_G(n) = c_G\, 2^{C_{\mathrm{clk}} \left\lfloor n^{1/G'} \right\rfloor},$$

the following hold.

(i) T_G is nondecreasing, $T_G(n) \ge n$ for every $n \ge 1$, $T_G(n)$ is computable from n in $\mathrm{poly}(n)$ steps, and $T_G(L_\nu) = c_G \mathcal{T}(2^\nu)$ for every $\nu \ge 1$.

(ii) For every $n \ge 1$, every $u \in \{0,1\}^n$ and every y, $\mathcal{V}_G$ halts within $T_G(n)$ steps.

(iii) Let $\nu \ge 1$ and $N = 2^\nu$, and let B be a general circuit with at least ν inputs whose description has at most L_ν bits. Some y of length $T_G(L_\nu)$ makes $\mathcal{V}_G(\langle B \rangle_\nu, y)$ accept if and only if $\mathrm{tt}_N(B) \in L_{\mathcal{T}}$.

Idea of proof. The verifier $\mathcal{V}_G$ checks that its input has the length L_ν of a padded description. It then expands the described circuit B into its truth table $\mathrm{tt}_N(B)$ and runs $\mathcal{D}$ on that word with its own witness. Expanding takes $O(NL_\nu^3)$ steps and running $\mathcal{D}$ takes $O(\mathcal{T}(N))$ steps, so at the lengths L_ν the time bound T_G is a constant multiple of $\mathcal{T}(N)$. At every other length $\mathcal{V}_G$ rejects within $O(n)$ steps.

Proof. On (u, y) with $|u| = n$, the machine $\mathcal{V}_G$ works in four stages.

(1) It writes n in binary, computes $\nu = \left\lfloor n^{1/G'} \right\rfloor$, and rejects unless $\nu \geq 1$ and $n = \nu^{G'}$. This takes $O(n)$ steps.

(2) It parses u as the description of a general circuit B followed by zeros. It rejects if the description is malformed or B has fewer than ν inputs. This takes $O(n)$ steps.

(3) For $i = 0, 1, \ldots, N-1$, where $N = 2^\nu$, it evaluates B at $\mathrm{bin}_\nu(i)0^{a-\nu}$, where a is the number of inputs of B. It writes the value into cell i of a work tape, which then holds $\mathrm{tt}_N(B)$. The circuit has at most n gates. Keeping the gate values on a tape and fetching the two arguments of each gate by a scan evaluates it in $O(n^3)$ steps. So this stage takes $O(Nn^3)$ steps.

(4) It runs $\mathcal{D}$ on $\mathrm{tt}_N(B)$ and accepts exactly when $\mathcal{D}$ accepts. The work tape of stage (3) serves as the input tape of $\mathcal{D}$, its own witness tape as the witness tape of $\mathcal{D}$, and further work tapes as those of $\mathcal{D}$. By Lemma E.13 this takes at most $c_{\mathcal{D}}\mathcal{T}(N)$ steps for a constant $c_{\mathcal{D}}$.

If $n = L_\nu$, the total is

$$O(n) + O(NL_\nu^3) + c_{\mathcal{D}}\mathcal{T}(N) \leq c' 2^{C_{\mathrm{clk}}\nu}$$

for a constant c', because $L_\nu^3 = \nu^{3G'}$ and $C_{\mathrm{clk}} \geq 2$. Otherwise the run stops in stage (1) within $c''n$ steps, and $n < (\nu+1)^{G'}$ with $\nu = \left\lfloor n^{1/G'} \right\rfloor$. Choose $c_G \geq c'$ so large that $c_G 2^{C_{\mathrm{clk}}\nu} \geq (c''+1)(\nu+1)^{G'}$ for every $\nu \geq 1$. This is possible because $(\nu+1)^{G'} 2^{-C_{\mathrm{clk}}\nu}$ is bounded. Then (ii) holds, and so does $T_G(n) \geq n$. The other assertions of (i) are immediate from the definition.

For (iii), if $\mathcal{V}_G(\langle B \rangle_\nu, y)$ accepts, then $\mathcal{D}$ accepts $\mathrm{tt}_N(B)$ with a witness, so $\mathrm{tt}_N(B) \in L(\mathcal{D}) = L_{\mathcal{T}}$. Conversely, suppose that $\mathrm{tt}_N(B) \in L_{\mathcal{T}}$, and take an accepting run of $\mathcal{D}$ on it. The run has at most $c_{\mathcal{D}}\mathcal{T}(N) \leq T_G(L_\nu)$ steps, so it reads only witness cells below $T_G(L_\nu)$. Writing the witness bits it reads into a word y of length $T_G(L_\nu)$, with zeros elsewhere, gives the same run, and $\mathcal{V}_G(\langle B \rangle_\nu, y)$ accepts. □

Use in the transfer and the order of constants. In Appendix A, $\mathcal{D}$ is the multitape verifier of $L_{\mathcal{T}}$ from Lemma E.13, with $\mathcal{T}(N) = N^{C_{\mathrm{clk}}}$, and supporting statement A.1 is applied to $\mathcal{V}_G$ with the time bound T_G. The constants are chosen in this order.

1. $d_0 = 2$. It bounds the exponent of N in the check (12) and depends on no PCP construction.
2. The general-circuit exponent G, from h_{sim} and b_{amp}; it does not depend on C_{clk}.
3. The exponent $G' \geq 2$, from G and the description format of general circuits (Appendix A). It fixes $L_\nu = \nu^{G'}$, and it does not depend on C_{clk} or A_{pad}.
4. An integer $C_{\mathrm{clk}} > d_0 + 2$. This fixes $\mathcal{D}$, and Lemma E.15 then fixes $\mathcal{V}_G$, c_G and T_G.
5. Supporting statement A.1, applied to $\mathcal{V}_G$ and T_G, fixes a constant $c_\ell \geq 1$ with

   $$\ell \leq \log_2 T_G(n) + c_\ell \log_2(\log_2 T_G(n) + 2)$$

   for every n, and the polynomials bounding the size of C_u^O and the running time of its construction. (Theorem E.33 derives them from the constants of [BCGT13] and of Lemmas E.31 and E.32.)
6. An integer $A_{\mathrm{pad}} \geq c_\ell$ with $A_{\mathrm{pad}} \geq b + c_\ell \log_2(b+1)$, where $b = \log_2 c_G$. For $N = 2^\nu$ write $\tau = \log_2 \mathcal{T}(N) = C_{\mathrm{clk}}\nu$. Since $T_G(L_\nu) = c_G 2^\tau$ and $\tau + b + 2 \leq (\tau+2)(b+1)$,

   $$\ell_{\mathrm{out}} \leq \tau + b + c_\ell \log_2(\tau + b + 2) \leq \tau + c_\ell \log_2(\tau+2) + b + c_\ell \log_2(b+1) \leq q(N).$$

So the outer verifier fits the native width. Its oracle queries $a \in \{0,1\}^{\ell_{\mathrm{out}}}$ can be zero-padded to the q inputs of B_{pf}, and the tables X_ζ and $\widehat{O}^*$ of Section E.22 have q inputs. Moreover

$$2^{q(N)} \le 2 \cdot 4^{A_{\mathrm{pad}}} \mathcal{T}(N) (\log_2 \mathcal{T}(N) + 2)^{A_{\mathrm{pad}}},$$

so $2^q/\mathcal{T}(N) = q^{O(1)}$.

7. Then the remaining constants, in this order:

   - the clause-count exponent d_G;
   - finally the padding and cap constants: K_{amp}, B_{len} and K_d in Appendix A.

   The constant η_0 of Appendix A depends only on Theorem 2.1.

No constant depends on one chosen after it. In particular, $\mathcal{V}_G$ bounds the size of a source circuit only through the length L_ν. That length depends on N and G' alone, not on q or A_{pad}. The weak machine enforces the bound q^G itself.

E.22 Extracting the weighted hard table

This reproduces [Nag26c, Section 11.5], with notation adapted.

Fix such a rejected yes-instance $\zeta \in \{0,1\}^N$, with $N = 2^\nu$. The two cases below, the SAT-checked canonical choices, and the clause-indexed proof table follow Chen–Lyu–Williams [CLW20, Sections 6.2.4–6.2.5]. Their proof of [CLW20, Lemma 6.5] averages cofactors into real-valued proofs; we instead select one Boolean cofactor per name and weight the list positions. Call a general circuit with q inputs and at most $q^{G^\star}$ gates a *source circuit for* ζ if it satisfies (12). Descriptions are ordered lexicographically, and "first" refers to this order. The canonical choices are made in three steps, which yield the three cases named in Section A.5.

(i) (NoSource) If ζ has no source circuit, let $X_\zeta : \{0,1\}^q \to \{0,1\}$ map u to ζ_i, where $\mathrm{bin}_\nu(i)$ is the first ν coordinates of u. A general circuit of at most $q^{G^\star}$ gates computing X_ζ would be a source circuit for ζ. Hence X_ζ has general circuit complexity greater than $q^{G^\star}$.

(ii) Otherwise let B^*_{src} be the first source circuit for ζ. Then $\mathrm{tt}_N(B^*_{\mathrm{src}}) = \zeta \in L_{\mathcal{T}}$, so some witness of length $T_{G^\star}(L_\nu)$ makes $\mathcal{V}_{G^\star}$ accept $\langle B^*_{\mathrm{src}} \rangle_\nu$, and supporting statement A.1 provides an oracle on ℓ_{out} inputs that makes $C^O_{\langle B^*_{\mathrm{src}} \rangle_\nu}$ unsatisfiable, that is, accepted on every random string. Let O^* be the first such oracle, viewed as a truth table, and let $\widehat{O}^*(av) = O^*(a)$ for $a \in \{0,1\}^{\ell_{\mathrm{out}}}$ and $v \in \{0,1\}^{q-\ell_{\mathrm{out}}}$.

(iii) (NoProof) If no general circuit B_{pf} with q inputs and at most $q^{G^\star}$ gates satisfies $O_{B_{\mathrm{pf}}} = O^*$, then $\widehat{O}^*$ has general circuit complexity greater than $q^{G^\star}$, since a circuit computing it would satisfy that equation.

Cases (NoSource) and (NoProof) are treated together. In each of them we obtain a function on q inputs with general circuit complexity greater than $q^{G^\star}$. By supporting statement A.3 and the choice of $G^\star$, the amplified function, on at most $a_{G^\star} q$ inputs, has agreement $1/2 + o(1)$ against every target circuit considered here. This is eventually below $1 - \gamma/5 + \epsilon$.

Otherwise, in case (Tautology), let B^*_{pf} be the first general circuit with q inputs and at most $q^{G^\star}$ gates such that $O_{B^*_{\mathrm{pf}}} = O^*$, and put $B = (B^*_{\mathrm{src}}, B^*_{\mathrm{pf}})$. Then D_B is a tautology, because $C^{O^*}_{\langle B^*_{\mathrm{src}} \rangle_\nu}$ is unsatisfiable, and its source circuit passes the check (12), so the weak machine can guess B. This canonical choice suffices: the argument below uses only that B is a pair whose source circuit passes

the check, that D_B is a tautology, and that the weak machine rejected ζ on every branch, including those that guess B. The efficient honest proof from supporting statement E.11 defines Boolean coordinate functions $\Pi_X(x)$ that make every three-literal clause true. For each fresh helper, define its honest function to be the AND of that clause's three honest literal values. For gadget a, form the five-entry list of underlying coordinate names

$$(X_{a,1}, X_{a,2}, X_{a,3}, Y_a, Y_a).$$

The first three entries are unsigned names: the clause's literal signs remain in the verifier, not in the table definition. Fill P_q address positions by cycling through this list, and define the target table to return the corresponding honest coordinate. Its exact arity is

$$q_0(q) = q + \log_2 M + \log_2 P_q = q + O(\log q). \tag{13}$$

Weighted-cofactor incidence. Let $M_{\mathrm{cl}} = M$ be the number of gadgets, let $P_q \geq 5$ be the positions per gadget, and put $t = \lfloor P_q/5 \rfloor$. Suppose a Boolean candidate table $C(x, \alpha)$ has uniform error δ against the honest table above. For each auxiliary unsigned name X, choose an address cofactor minimizing its error e_X against Π_X; systematic names use their prescribed parities and have $e_X = 0$. Let I_X be its total stability weight across the gadgets: one for each literal position and two for each helper. The five-entry lists repeat every name in at least tI_X table positions, so

$$t \sum_X I_X e_X \leq M_{\mathrm{cl}} P_q \delta.$$

Weighted gadget stability (Lemma E.12), averaged over gadgets and inputs, therefore gives

$$\mathbb{E}Q \leq \frac{1}{M_{\mathrm{cl}}} \sum_X I_X e_X \leq \frac{P_q}{t}\delta \leq (5+\epsilon)\delta.$$

For the paper's choice $P_q = 128/\epsilon$, with $0 < \epsilon < 1$, we have $t \geq 4/\epsilon$, so $P_q/t \leq 5 + 4/t \leq 5 + \epsilon$. Repeated names are counted with their multiplicities, complements preserve error, and systematic positions only strengthen the table-error bound. This is the incidence argument of [Nag26c, Equation (18)].

E.23 Uniform recovery of the canonical choices

This is [Nag26c, Section 11.6]; the proof is reproduced with notation adapted.

All canonical choices are recoverable in E^{NP}. The fixed-machine refuter runs in $\mathrm{poly}(\mathcal{T}(N_\nu)) = 2^{O(q)}$ time with SAT access. The choices of Appendix A are recovered in three stages, by prefix self-reduction on SAT formulas of size $2^{O(q)}$. First, a formula over the description of B_{src} with the N constraints (12), of size $Nq^{O(G^\star)}$, finds the first source circuit B^*_{src}, or shows that there is none, in which case the table is X_ζ. Second, the algorithm of supporting statement A.1 is run deterministically on $\langle B^*_{\mathrm{src}} \rangle_\nu$, in $q^{O_{G^\star, C_{\mathrm{clk}}}(1)}$ time; a formula with one variable per oracle address and the constraints $C^{O}_{\langle B^*_{\mathrm{src}} \rangle_\nu}(R) = 0$ for all $2^{\ell_{\mathrm{out}}}$ inputs R, of size $2^{2\ell_{\mathrm{out}}} q^{O_{G^\star, C_{\mathrm{clk}}}(1)} = 2^{O(q)}$ (each oracle gate becomes a selector over the $2^{\ell_{\mathrm{out}}}$ oracle variables), finds the first oracle O^*. Third, a formula over the description of B_{pf} with the $2^{\ell_{\mathrm{out}}}$ constraints $B_{\mathrm{pf}}(a0^{q-\ell_{\mathrm{out}}}) = O^*(a)$, of size $2^{\ell_{\mathrm{out}}} q^{O(G^\star)}$, finds the first B^*_{pf}, or shows that there is none, in which case the table is $\widehat{O}^*$. No formula contains the algorithm of supporting statement A.1. Run the general amplifier on X_ζ or $\widehat{O}^*$. If no oracle O^* exists at an exceptional initial source length, the second stage detects that case and the slice is set to zero. In the Tautology case, the honest PCPP proof and each AND helper are computable in polynomial time once the pair is known. The fixed-error schedule of Section A.2 is computable in polynomial time in q. Since $q = O(n)$, the final functions belong to E^{NP}.

E.24 Exact conditional cycle potentials

This section gives the conditional-expectation calculation used by the counter.

Corollary E.16 (Conditional potentials)**.** *Use the generator of supporting statement D.9 with $h = m$ and seed length S. For a seed prefix π put $c_\pi(w) = \mathbb{E}[\Gamma_w \mid \pi]$. Then $2^S c_\pi(w) \in \mathbb{Z}$, $|c_\pi(w)| \le (N^2Q)^{2m}$, $c_\pi(w)$ does not depend on z, it is computable exactly in $(Q+1)^{2m}\,\mathrm{poly}(n,S)$ time, and*

$$\mathbb{E}[\Phi_z \mid \pi] = \frac{2|\Theta|}{4^m} \sum_w c_\pi(w)\,\mathrm{mask}_w(z) \sum_\sigma \kappa_\sigma \mathbf{1}[Y_{1,\sigma}(z,w) = 0]\,\mathbf{1}[Y_{2,\sigma}(z,w) = 0].$$

For the empty prefix, $\mathbb{E}\Phi_z$ equals its value under independent uniform samples.

Proof. Expand $\Gamma_w = \prod_{l=1}^{2m}(Q - N^2 \sum_{j=1}^{Q} \mathbf{1}[(U_j, V_j) = e_l])$ into $(Q+1)^{2m}$ products, each an integer times a conjunction of at most $2m$ events "sample j lies at e_l". By supporting statement D.9 each conjunction has conditional probability 0 or 2^{-a} with $a \le S$, computed by Gaussian elimination; repeated sample labels or repeated walk points only repeat or augment the linear system, and a label required to lie at two different points gives probability 0. Since $\mathrm{mask}_w(z)$ and the indicators do not depend on the seed, linearity gives the display. The bound $|\Gamma(x)| \le N^2Q$ gives the bound on $|c_\pi|$. Each conjunction involves at most $2m$ distinct labels, and the positions are $2m$-wise independent and uniform, so the unconditional expectation of every term equals its value under independent samples. □

E.25 The short PCP for the verifiers used here

This is [Nag26c, Appendix D]; the complete proof of [Nag26c] is reproduced with notation adapted. Its two published premises are the BCGT verifier and Shoup's irreducible-polynomial algorithm from Section 1.4. This appendix proves supporting statement A.1, the short PCP in the form of Murray and Williams, for deterministic multitape verifiers with computable time bounds (Theorem E.33). This case covers its only use. The transfer applies supporting statement A.1 to one verifier: the succinct hierarchy verifier $\mathcal{V}_G$ of Lemma E.15, in Appendix A. We use the PCP construction of Ben-Sasson, Chiesa, Genkin, and Tromer for algebraic constraint problems [BCGT13] and import two results: their strong Reed–Solomon proximity verifiers, with the soundness bounds of [BCGT13, Theorem 11.1], and Shoup's deterministic algorithm for irreducible polynomials [Sho88, Corollary 3.2], as used in [BCGT13, Algorithm 17]. The reduction from the verifier to an algebraic constraint problem is proved here. References to numbered statements of [BCGT13] are to its full version.

Plan of the proof. The verifier is first simulated by a random-access machine whose run has an exact length. That run is then written as an assignment to the points of an affine subspace of a binary field, so that local polynomial equations hold everywhere on the subspace exactly when the run accepts. The PCP of [BCGT13] checks such an assignment. It amplifies the soundness of its proximity sub-verifiers by naive sequential repetition, which multiplies the randomness by a polylogarithmic factor [BCGT13, Remark 11.2 and Lemmas C.6–C.8], whereas supporting statement A.1 has randomness $\log_2 T + O(\log\log T)$. We therefore run each sub-verifier once and amplify the whole verifier with a small-bias set, as in [BV14, Section 4.4]. The appendix has the following parts.

- Lemmas E.17 and E.18: small-bias sets and the amplification that uses them; Lemma E.19: the randomness and proof length of one run of a Reed–Solomon proximity verifier.

- Definition E.1 and Lemma E.21: random-access machines, and the simulation of a multitape verifier by one whose run has an exact length.
- Definition E.2: affine constraint instances, the problem the PCP checks.
- Theorem E.27: an accepting run as an event trace, checked by local equations in time order and in address order.
- Theorem E.30: the event trace as an affine constraint instance, with a routing network joining the two orders; Lemmas E.31 and E.32: its degree, its field, and its uniform construction.
- Theorem E.33: assembly in four steps.
- The routine proof of Lemma E.21 comes last.

Notation. As in supporting statement A.1, n and $T(n)$ are the input length and the time bound of a general verifier. All other symbols introduced in this appendix are local to it. Logarithms are base two. Write $\alpha \cdot \sigma$ for the inner product over $\mathbb{F}_2$.

E.26 Small-bias sets and one Reed–Solomon test

Lemma E.17 (Small-bias sets)**.** *Let $r \geq 1$ and $0 < \varepsilon < 1$. A multiset $S \subseteq \{0,1\}^r$ is* ε-biased *if $|\mathbb{E}_{\sigma \in S}(-1)^{\alpha \cdot \sigma}| \leq \varepsilon$ for every nonzero $\alpha \in \{0,1\}^r$.*

(i) An ε-biased multiset of size less than $(2r/\varepsilon)^2$ can be computed deterministically in $\mathrm{poly}(r/\varepsilon)$ steps.

(ii) If S is ε-biased and $\mathcal{B} \subseteq \{0,1\}^r$ has density $\mu > 0$, then the set of $R \in \{0,1\}^r$ with $R \oplus \sigma \notin \mathcal{B}$ for every $\sigma \in S$ has density at most ε^2/μ.

Proof. (i) This is the powering construction of Alon, Goldreich, Håstad, and Peralta [AGHP92]; see also Naor and Naor [NN93]. Let m be the least integer with $2^m \geq r/\varepsilon$, so that $2^m < 2r/\varepsilon$. Find an irreducible polynomial of degree m over $\mathbb{F}_2$ by exhaustive search, in $2^m\,\mathrm{poly}(m)$ steps, and identify $\mathbb{F}_{2^m}$ with $\mathbb{F}_2^m$. For $a, b \in \mathbb{F}_{2^m}$ let $\sigma_{a,b} \in \{0,1\}^r$ have coordinates $a^j \cdot b$, $0 \leq j < r$, and let S consist of all $2^{2m} < (2r/\varepsilon)^2$ vectors $\sigma_{a,b}$. For nonzero α, $\alpha \cdot \sigma_{a,b} = p_\alpha(a) \cdot b$, where $p_\alpha(X) = \sum_j \alpha_j X^j$ is a nonzero polynomial of degree less than r. If $p_\alpha(a) \neq 0$, the bit $p_\alpha(a) \cdot b$ is unbiased over b, and $p_\alpha(a) = 0$ for fewer than r values of a. Hence the bias is less than $r/2^m \leq \varepsilon$.

(ii) Let $g(R) = \mathbb{E}_{\sigma \in S} \mathbf{1}_{\mathcal{B}}(R \oplus \sigma)$, and write $\widehat{h}(\alpha) = \mathbb{E}_R h(R)(-1)^{\alpha \cdot R}$. Then $\mathbb{E}_R g(R) = \mu$ and $\widehat{g}(\alpha) = \widehat{\mathbf{1}_{\mathcal{B}}}(\alpha)\mathbb{E}_{\sigma \in S}(-1)^{\alpha \cdot \sigma}$. By Parseval,

$$\mathbb{E}_R\big(g(R) - \mu\big)^2 = \sum_{\alpha \neq 0} \widehat{g}(\alpha)^2 \leq \varepsilon^2 \sum_{\alpha \neq 0} \widehat{\mathbf{1}_{\mathcal{B}}}(\alpha)^2 = \varepsilon^2(\mu - \mu^2).$$

On the set in question $g = 0$, so $(g - \mu)^2 = \mu^2$ there, and its density is at most $\varepsilon^2(\mu - \mu^2)/\mu^2 \leq \varepsilon^2/\mu$. □

Part (ii) is the hitting property [BV14, Lemma 4.7]. Equivalently, it follows from the expander mixing lemma of Alon and Chung [AC88] for the Cayley graph of $\mathbb{F}_2^r$ generated by S, whose nontrivial eigenvalues are the biases $\mathbb{E}_{\sigma \in S}(-1)^{\alpha \cdot \sigma}$.

Lemma E.18 (Randomness-efficient amplification)**.** *Let V_0 be a PCP verifier that uses r_0 random bits on an instance x, and let $0 < s \leq 1$ and $0 < \sigma_0 \leq 1$. Let $S \subseteq \{0,1\}^{r_0}$ be ε-biased with $\varepsilon^2 \leq s\sigma_0$. Let V accept x with proof π and random string $R \in \{0,1\}^{r_0}$ exactly when V_0 accepts x with π and $R \oplus \sigma$ for every $\sigma \in S$.*

(i) V uses r_0 random bits and $|S|$ times as many queries as V_0; if V_0 is non-adaptive, so is V.

(ii) A proof that V_0 accepts with probability one is accepted by V with probability one.

(iii) If V_0 rejects π with probability at least σ_0, then V accepts π with probability at most s.

Proof. Parts (i) and (ii) are immediate, because $R \oplus \sigma$ is again a random string of V_0. For (iii), let $\mathcal{B}$ be the set of random strings on which V_0 rejects π; its density is $\mu \geq \sigma_0$. The verifier V accepts exactly on the set of Lemma E.17(ii), whose density is at most $\varepsilon^2/\mu \leq s\sigma_0/\sigma_0 = s$. □

The proximity sub-verifiers of [BCGT13] test whether a function on an affine subspace of the field is close to the evaluation table of a polynomial of bounded degree, a Reed–Solomon codeword. Their basic case is the verifier $V_{\mathrm{RS},=}$ with its prover $P_{\mathrm{RS},=}$ [BCGT13, Algorithms 16 and 7], which recurse on subspaces of about half the dimension. On a subspace of dimension κ, the number of random bits of $V_{\mathrm{RS},=}$ and the number of field elements in the proof of $P_{\mathrm{RS},=}$ are given by the recursions (27) and (28) of [BCGT13, Section C.8]. These recursions depend on the parameter tuple $(\eta, \kappa_0, \gamma_{\mathrm{rep}}, \mu_{\mathrm{rep}})$ of [BCGT13, Theorem 11.1]. We use the tuple $(\eta, \kappa_0, \gamma_{\mathrm{rep}}, \mu_{\mathrm{rep}}) = (2, 4, 0, 1)$, which satisfies the conditions listed in that theorem (Equation (6) of [BCGT13, Theorem 11.3]). At this tuple the two recursions read as follows.

Lemma E.19 (One Reed–Solomon test at the tuple $(2, 4, 0, 1)$)**.** *Define* $\mathrm{rand}_=(\kappa) = \mathrm{len}_=(\kappa) = 0$ *for* $\kappa \leq 4$ *and, for* $\kappa \geq 5$,

$$\mathrm{rand}_=(\kappa) = 1 + \max\{\lceil \kappa/2 \rceil + \mathrm{rand}_=(\lfloor \kappa/2 \rfloor + 2),\ \lfloor \kappa/2 \rfloor + 1 + \mathrm{rand}_=(\lceil \kappa/2 \rceil)\},$$
$$\mathrm{len}_=(\kappa) = 2^{\lceil \kappa/2 \rceil} \cdot 2^{\lfloor \kappa/2 \rfloor + 2} + 2^{\lceil \kappa/2 \rceil}\, \mathrm{len}_=(\lfloor \kappa/2 \rfloor + 2) + 2^{\lfloor \kappa/2 \rfloor + 1}\, \mathrm{len}_=(\lceil \kappa/2 \rceil).$$

Then for every integer $\kappa \geq 1$,

$$\mathrm{rand}_=(\kappa) \leq \kappa + 3\log_2(2\kappa), \qquad \mathrm{len}_=(\kappa) \leq 2^\kappa \kappa^3.$$

Proof. For $\kappa \leq 4$ both functions vanish, so both bounds hold. We first show, by strong induction on $\kappa \geq 5$, that for every integer j with $\kappa - 4 < 2^j$,

$$\mathrm{rand}_=(\kappa) + 3 \leq \kappa + 3j, \qquad \mathrm{len}_=(\kappa) + 2^\kappa \leq 2^\kappa 6^j. \tag{14}$$

Here $j \geq 1$, because $\kappa - 4 \geq 1$. The two arguments $a = \lfloor \kappa/2 \rfloor + 2$ and $b = \lceil \kappa/2 \rceil$ of the recursion are smaller than κ. Moreover $a - 4 = \lfloor (\kappa - 4)/2 \rfloor$ and $b - 4 < a - 4$, so $a - 4$ and $b - 4$ are less than 2^{j-1}. Hence each $c \in \{a, b\}$ satisfies (14) with $j - 1$ in place of j when $c \geq 5$, by induction. When $c \leq 4$, it has $\mathrm{rand}_=(c) = \mathrm{len}_=(c) = 0$.

Randomness. Since $\lceil \kappa/2 \rceil + a = \kappa + 2$ and $\lfloor \kappa/2 \rfloor + 1 + b = \kappa + 1$, a branch of the maximum whose argument c is at least 5 is at most $\kappa + 2 + 3(j-1) - 3$. A branch with $c \leq 4$ is at most $\lfloor \kappa/2 \rfloor + 1 \leq \kappa - 1$. Both are at most $\kappa + 3j - 4$, so $\mathrm{rand}_=(\kappa) \leq \kappa + 3j - 3$.

Length. By the cases above, $\mathrm{len}_=(c) + 2^c \leq 2^c 6^{j-1}$ for $c \in \{a, b\}$. Multiply the case $c = a$ by $2^{\lceil \kappa/2 \rceil}$ and the case $c = b$ by $2^{\lfloor \kappa/2 \rfloor + 1}$. Since $2^{\lceil \kappa/2 \rceil} \cdot 2^a = 4 \cdot 2^\kappa$ and $2^{\lfloor \kappa/2 \rfloor + 1} \cdot 2^b = 2 \cdot 2^\kappa$, adding the two inequalities gives

$$\mathrm{len}_=(\kappa) + 2 \cdot 2^\kappa \leq 6 \cdot 2^\kappa 6^{j-1} = 2^\kappa 6^j,$$

which implies (14).

Now let $\kappa \geq 5$, let $m = \lfloor \log_2(\kappa - 4) \rfloor$, so that $2^m \leq \kappa - 4 < 2^{m+1}$, and apply (14) with $j = m + 1$. Since $2^{m+1} \leq 2(\kappa - 4) < 2\kappa$, we have $m + 1 \leq \log_2(2\kappa)$, and therefore $\mathrm{rand}_=(\kappa) \leq \kappa + 3(m+1) - 3 \leq \kappa + 3\log_2(2\kappa)$. For the length, $\mathrm{len}_=(\kappa) \leq 2^\kappa 6^{m+1}$, and $6^{m+1} \leq (2^m + 4)^3 \leq \kappa^3$. The first of these inequalities is checked directly for $m \leq 6$, with equality at $m = 5$. For $m \geq 7$, $6^{m+1} \leq 8^m = (2^m)^3$, since it holds at $m = 7$ and, as m increases by one, the two sides grow by the factors 6 and 8. □

Remark E.20 (The general bound in [BCGT13, Lemma C.1]). We use Lemma E.19 instead of [BCGT13, Lemma C.1]; at the tuple $(2, 4, 0, 1)$ it gives the same bounds, proved directly from recursions (27)–(28).

E.27 Random-access machines

The verifier is first simulated by a random-access machine, because memory accesses of such a machine can be checked by sorting (Theorem E.27).

Definition E.1 (Random-access machines)**.** A *random-access machine* M has a word width $w \geq 1$, a number K of registers, and a code $(I_0, \ldots, I_{L-1})$ with $1 \leq L \leq 2^w$. A *word* is an integer in $[0, 2^w)$, written with w bits. A *configuration* consists of a program counter pc, which is a word, and registers $r_0, \ldots, r_{K-1}$, each a word or the end-of-tape symbol #. The memory has 2^w cells, each holding a word. Two read-only tapes, tape A for the input and tape B for the witness, are read in consecutive words of w bits; bit j of a word is the j-th bit of its block, least significant first. An operand is a register, pc or a constant word; a register holding # reads as 0. The instructions are:

- `readA` i and `readB` i: r_i receives a word or #;
- `load` $i\,j$: $r_j \leftarrow \mathrm{mem}[r_i]$, and `store` $j\,x$: $\mathrm{mem}[r_j] \leftarrow x$;
- `iseof` $i\,j$: $r_i \leftarrow 1$ if $r_j = \#$, and $r_i \leftarrow 0$ otherwise;
- `arith` $o\,i\,x\,y$: $r_i \leftarrow o(x, y)$ for $o \in \{\mathrm{add}, \mathrm{sub}, \mathrm{and}, \mathrm{cmp}\}$, where addition and subtraction are modulo 2^w, and is bitwise, and $\mathrm{cmp}(x, y)$ is 1, 0 or $2^w - 1$ according as $x < y$, $x = y$ or $x > y$;
- `nop`;
- `je` $i\,c$, `jne` $i\,c$ and `jmp` c: $pc \leftarrow c \bmod 2^w$ if $r_i = 0$, if $r_i \neq 0$, or always;
- `out` accept and `out` reject: halt.

Every instruction except a taken jump and `out` also sets $pc \leftarrow pc + 1 \bmod 2^w$. A configuration whose pc is at least L or points to `out` has no successor. The initial configuration has $pc = 0$ and all registers 0; a configuration is accepting if $I_{pc} =$ `out` accept.

Let $T \geq 1$. An *accepting run of length* T on binary strings $(\bar{x}, y)$ consists of configurations $S^0, \ldots, S^{T-1}$ and memories $\mathrm{mem}_0, \ldots, \mathrm{mem}_{T-1}$ such that S^0 is initial, $\mathrm{mem}_0 = 0$, S^{T-1} is accepting, and $(S^{i+1}, \mathrm{mem}_{i+1})$ results from (S^i, mem_i) by executing I_{pc} for every $i < T - 1$; moreover the words other than # that the run reads from tape A, concatenated in time order, form a prefix of $\bar{x}$, and those read from tape B form a prefix of y. The machine *accepts* $\bar{x}$ *in time* T if it has such a run for some y with $|y| \leq T$.

A read may return # at any step; the prefix conditions make every word that is returned the next word of its tape. This frees the trace from tracking where a tape ends. The machines M_n never rely on it: a read of # sends them into a loop that never halts (Lemma E.21).

Lemma E.21 (Multitape machines as random-access machines)**.** *Let $V(x, y)$ be a deterministic multitape machine that halts within $T(n) \geq n$ steps on every $x \in \{0,1\}^n$ and every y. There are constants $c_V, K, L \geq 1$ with the following properties. For each $n \geq 2$ let the word width $w_n = 2^{\lceil \log_2 (\lceil \log_2 T(n) \rceil + c_V) \rceil}$ be the least power of two with $w_n \geq \lceil \log_2 T(n) \rceil + c_V$, for $x \in \{0,1\}^n$ let $\bar{x} = x10^j$ be its padding to length $2^{\lceil \log_2 \max(n+1, w_n) \rceil}$, a multiple of w_n, and let T_n be the least power of two with $T_n \geq c_V(T(n) + w_n)$. There is a random-access machine M_n with word width w_n, K registers and a code of length L such that, for every $x \in \{0,1\}^n$, $V(x, y)$ accepts for some*

$y \in \{0,1\}^{T(n)}$ if and only if M_n accepts $\bar{x}$ in time T_n. Moreover $|\bar{x}| \le T_n < 2^{w_n}$, and the code of M_n is a fixed code, depending only on V, into which the integers n, $T(n)$ and $T_n - 8$ are written.

The proof of Lemma E.21 is a direct simulation followed by an idle phase that pads the run to exactly T_n configurations. It is routine, and we give it at the end of this appendix.

E.28 Affine constraint instances

The target of the reduction is the following algebraic problem. Let $\mathrm{irr} \in \mathbb{F}_2[X]$ be irreducible of degree f, and let $F = \mathbb{F}_2[X]/(\mathrm{irr})$, the field with 2^f elements. An affine subspace of F over $\mathbb{F}_2$ is given by a basis $b_1, \dots, b_m$ and an offset o; its i-th element, $0 \le i < 2^m$, is $o + \sum_j i_j b_j$, where $i_1, i_2, \dots$ are the binary digits of i, least significant first.

Definition E.2 (Affine constraint instances)**.** An *affine constraint instance* consists of an irreducible irr of degree f, an affine subspace $H \subseteq F$ of dimension m_H, neighbors $N_\nu(X) = a_\nu X + b_\nu$ with $a_\nu \neq 0$, for $1 \le \nu \le c_N$ and some $c_N \ge 1$, a constraint polynomial $P(X, Y_1, \dots, Y_{c_N})$ over F given by an arithmetic circuit, and an affine subspace $I \subseteq H$ of dimension m. Give X weight 1 and each Y_ν weight $2^{m_H} - 1$. The instance has *degree bound* d_B if every monomial of P has weight at most d_B; we require $4d_B \le 2^f$. The instance *accepts* a binary string $\bar{x}$ of length 2^m if some $A \in F[X]$ of degree less than 2^{m_H} satisfies

$$P(\alpha, A(N_1(\alpha)), \dots, A(N_{c_N}(\alpha))) = 0 \qquad \text{for every } \alpha \in H,$$

and $A(\alpha_i) = \bar{x}_i$ for every $i < 2^m$, where α_i is the i-th element of I and the bits $0, 1$ are read in F.

If $\deg A < 2^{m_H}$, then $B_A(X) = P(X, A(N_1(X)), \dots, A(N_{c_N}(X)))$ has degree at most d_B: substituting $A \circ N_\nu$, of degree at most $2^{m_H} - 1$, raises no monomial above its weight. This is the succinct algebraic constraint satisfaction problem of [BCGT13, Definition 7.1] at a single instance, with affine neighbors. The weighted bound $4d_B \le 2^f$ implies its degree condition (3) and, as just shown, bounds every composition B_A; we use this stronger sufficient condition.

E.29 The run as an event trace

Idea. A run is checked locally in two orders. In time order, each step relates two consecutive configurations, which is a local check, except that a `load` must return the value last stored at its address, possibly many steps earlier. In address order, with ties broken by time, every `load` directly follows the last access to its cell, so the memory check is local again. The trace therefore lists each memory access, and each word read from the input tape, as a *record*: once in time order, at the step that makes it, and once in sorted order. The two lists must be equal as multisets; in the affine instance a routing network enforces this (Lemma E.29). Checking random-access memory by sorting is a standard technique; the argument below is self-contained.

Fix a machine M with word width $w = 2^\omega$, K registers and code length L; an integer $\tau \ge \max(1, \omega)$ with $T = 2^\tau < 2^w$ (for M_n, $T = T_n$, not $T(n)$); and a string $\bar{x}$ whose length is a multiple of w and at most T. Put $\beta = \tau - \omega$, $N_A = |\bar{x}|/w$, $N = 2T$ and $\kappa_{\mathrm{key}} = w + \tau + 2$. Every slot below holds one bit; a w-bit word v occupies w slots $v[0], \dots, v[w-1]$, least significant first.

Definition E.3 (Event trace)**.** A *record* consists of an activity bit *act*, a region bit *rgn* (0 for memory, 1 for tape A), an address or tape index $ad < 2^w$, a phase bit *ph*, a time stamp $ts < 2^\tau$, a value $va < 2^w$, and a store bit *st*. Its *key* is $ts + 2^\tau ph + 2^{\tau+1} ad + 2^{\tau+1+w} rgn < 2^{\kappa_{\mathrm{key}}}$. The active records are memory accesses $(1, 0, a, 1, i, v, st)$ at address a and step i with value v, tape-A reads

$(1,1,c,1,i,v,0)$ of the c-th word read, at step i, and input words $(1,1,c,0,0,v,0)$, word c of $\bar{x}$. The *sentinel* is $(1,0,0,0,0,0,0)$.

The *event trace* $\mathrm{ET}(M,\tau,\bar{x})$ is the following system of bit slots and equations over $\mathbb{F}_2$.

- *T time rows* $\rho_0,\dots,\rho_{T-1}$. Row ρ_i holds the current and the next configuration (pc, and for each register r a flag fl_r, set for #, and a word rg_r; next values are primed), a one-hot instruction vector $e_0,\dots,e_{L-1}$ with its prefix-or $o_0,\dots,o_{L-1}$, operand words X,Y and a flag X_F, the results AD, SB, AN of addition, subtraction and conjunction with carries ca_j, borrows bo_j, a prefix-or oz_j and a bit gt, a result word Z with flag Z_F, a tape word R_V with flag R_F, a memory word M_V, a record REC whose time stamp ts is the row's time stamp, its successor ts', the counts c_A and c_B of words read so far from tapes A and B (τ and $\beta+1$ bits) with successors c'_A, c'_B, increments a_A, a_B, and the carries tc_j, ac_j, bc_j of the three counters.
- *N sorted vertices* $\sigma_0,\dots,\sigma_{N-1}$, each with a record REC, a copy PREV of the first six fields of the previous record, written *pact*, *prgn*, *pad*, *pph*, *pts* and *pva*, and auxiliary bits *both*, ob_l, so_l, u, g, EV, g_{A1} and g_A.
- *N_A input records* $\iota_0,\dots,\iota_{N_A-1}$ and *$N-T-N_A$ dummy records*, each a record only.

The *source list* is $(\mathrm{REC}(\rho_0),\dots,\mathrm{REC}(\rho_{T-1}),\iota_0,\dots,\iota_{N_A-1})$ followed by the dummies; it has length N. The equations are listed below: (T1)–(T10) at every time row, (S1)–(S5) at every sorted vertex, the boundary and copy equations (B1)–(B5), and

(G) every slot is a bit, and

(P) the source list and $(\mathrm{REC}(\sigma_0),\dots,\mathrm{REC}(\sigma_{N-1}))$ are equal as multisets.

The system is *satisfiable* if some assignment of bits to all slots satisfies every equation.

Code tables. Let $e(Q)=\sum_{p\in Q} e_p$ for a set $Q\subseteq[0,L)$ of code positions, a linear form in the slots. The sets used are Ld, St, RdA, RdB, Eof, Jmp, Je, Jne, Out and Ar_o (the positions of `load`, `store`, `readA`, `readB`, `iseof`, the three jumps, `out` and `arith` o), $\mathrm{Rd}=\mathrm{RdA}\cup\mathrm{RdB}$, Acc (the positions of `out` accept), Seq (the positions of all instructions other than jumps and `out`), and Wr_r (the positions of the instructions that write register r). For $p<L$ let $p^+=p+1 \bmod 2^w$, let $c_p=c \bmod 2^w$ for a jump to c, and let $s_p^X[j]$, $s_p^Y[j]$ be bit j of the operands that I_p places on the buses: (x,y) for `arith` $o\,i\,x\,y$; $(r_i,0)$ for `je` and `jne` on r_i; r_i on X for `load` $i\,j$; (r_j,x) for `store` $j\,x$; and 0 otherwise. Each is a current slot or a constant bit. All sums are in $\mathbb{F}_2$, and j ranges over the bit positions of the word concerned.

Time rows.

(T1) $fl_r\, rg_r[j]=0$.

(T2) $o_0=e_0$; $o_p=o_{p-1}+e_p$ and $e_p o_{p-1}=0$ for $1\le p<L$; $o_{L-1}=1$; $pc[j]=\sum_p \mathrm{bit}_j(p)e_p$.

(T3) $X[j]=\sum_p e_p s_p^X[j]$, $Y[j]=\sum_p e_p s_p^Y[j]$, and $X_F=\sum e_p fl_{j'}$ over the positions p with $I_p=$ `iseof` $i\,j'$.

(T4) With $ca_0=bo_0=0$: $AD[j]=X[j]+Y[j]+ca_j$ and $ca_{j+1}=X[j]Y[j]+X[j]ca_j+Y[j]ca_j$ $(j+1<w)$; $SB[j]=X[j]+Y[j]+bo_j$ and $bo_{j+1}=(1+X[j])(Y[j]+bo_j)+Y[j]bo_j$; $AN[j]=X[j]Y[j]$; $oz_0=SB[0]$ and $oz_j=oz_{j-1}+SB[j]+oz_{j-1}SB[j]$; $gt=(1+bo_w)oz_{w-1}$.

(T5) $R_F R_V[j]=0$ and $e(\mathrm{St})(M_V[j]+Y[j])=0$.

(T6) $Z[j]=e(\mathrm{Ar_{add}})AD[j]+e(\mathrm{Ar_{sub}})SB[j]+e(\mathrm{Ar_{and}})AN[j]+e(\mathrm{Ar_{cmp}})CMP_j+e(\mathrm{Ld})M_V[j]+e(\mathrm{Rd})R_V[j]+[j=0]\,e(\mathrm{Eof})X_F$, where $CMP_0=bo_w+gt$ and $CMP_j=gt$ for $j\ge 1$; and

$Z_F = e(\mathrm{Rd})R_F$.

(T7) $rg'_r[j] = e(\mathrm{Wr}_r)Z[j] + (1 + e(\mathrm{Wr}_r))rg_r[j]$ and $fl'_r = e(\mathrm{Wr}_r)Z_F + (1 + e(\mathrm{Wr}_r))fl_r$.

(T8) With $z = oz_{w-1}$,

$$pc'[j] = \sum_{\mathrm{Seq}} e_p \mathrm{bit}_j(p^+) + \sum_{\mathrm{Out}} e_p \mathrm{bit}_j(p) + \sum_{\mathrm{Jmp}} e_p \mathrm{bit}_j(c_p) + \sum_{\mathrm{Je}} (e_p z\,\mathrm{bit}_j(p^+) + (e_p + e_p z)\mathrm{bit}_j(c_p)) + \sum_{\mathrm{Jne}} (e_p z\,\mathrm{bit}_j(c_p) + (e_p + e_p z)\mathrm{bit}_j(p^+)).$$

(T9) $act = e(\mathrm{Ld}) + e(\mathrm{St}) + a_A$, $rgn = a_A$, $ad[j] = (e(\mathrm{Ld}) + e(\mathrm{St}))X[j] + a_A c_A[j]$ (with $c_A[j] = 0$ for $j \geq \tau$), $ph = 1$, $va[j] = (e(\mathrm{Ld}) + e(\mathrm{St}))M_V[j] + a_A R_V[j]$, $st = e(\mathrm{St})$.

(T10) With $tc_0 = 1$: $ts'[j] = ts[j] + tc_j$ and $tc_{j+1} = ts[j]tc_j$. $a_A = e(\mathrm{RdA})(1 + R_F)$ and, with $ac_0 = a_A$, $c'_A[j] = c_A[j] + ac_j$ and $ac_{j+1} = c_A[j]ac_j$. Likewise $a_B = e(\mathrm{RdB})(1 + R_F)$ and $c'_B = c_B + a_B$ with carries bc_j. Finally $a_B c_B[\beta] = 0$.

The last carries of ts', c'_A and c'_B are not kept, so these counters work modulo 2^τ, 2^τ and $2^{\beta+1}$. In words: (T1) a register holding # has word 0; (T2) pc is decoded one-hot; (T3) the operand buses; (T4) sum, difference, conjunction and comparison; (T5) the read and store channels; (T6) the result; (T7) the register write; (T8) the next pc; (T9) the row's record; (T10) the time stamp and the read counters.

Sorted vertices. Let K_l and P_l, $l < \kappa_{\mathrm{key}}$, be the key bits of REC and of PREV, least significant first.

(S1) $(1 + pact)act = 0$ and $both = act \cdot pact$.

(S2) With $ob_0 = 0$: $ob_{l+1} = (1 + P_l)(K_l + ob_l) + K_l ob_l$ for $l < \kappa_{\mathrm{key}}$, and $both(1 + ob_{\kappa_{\mathrm{key}}}) = 0$.

(S3) With $d_0 = prgn + rgn$ and $d_{1+j} = pad[j] + ad[j]$: $so_0 = d_0$ and $so_l = so_{l-1} + d_l + so_{l-1}d_l$ for $1 \leq l \leq w$.

(S4) $u = act(1 + rgn)$, $g = u(1 + st)$, $EV[j] = (1 + so_w)pva[j]$ and $g(va[j] + EV[j]) = 0$.

(S5) $g_{A1} = act \cdot rgn$, $g_A = g_{A1}ph$, $g_A so_w = 0$, $g_A pph = 0$ and $g_A(va[j] + pva[j]) = 0$.

In words: (S1) active records come first; (S2) keys increase; (S3) the record names the same region and address as its predecessor exactly when $so_w = 0$; (S4) the memory rule; (S5) the input rule.

Boundaries and copies.

(B1) At ρ_0: $pc = 0$, $fl_r = 0$ and $rg_r = 0$ for all r, $ts = 0$, $c_A = 0$, $c_B = 0$.

(B2) For $i < T - 1$: $e(\mathrm{Out}) = 0$ at ρ_i, and ρ_{i+1} has current configuration, ts, c_A and c_B equal to the next configuration, ts', c'_A and c'_B of ρ_i.

(B3) At ρ_{T-1}: $e(\mathrm{Acc}) = 1$.

(B4) $\mathrm{PREV}(\sigma_0)$ is the sentinel's first six fields, and $\mathrm{PREV}(\sigma_{j+1})$ is the first six fields of $\mathrm{REC}(\sigma_j)$.

(B5) $\iota_c = (1, 1, c, 0, 0, v_c, 0)$ with v_c word c of $\bar{x}$; every dummy record is zero.

Every equation has degree at most two in the slots; the auxiliary slots Z_F, a_A, a_B, u, g, g_{A1}, g_A, EV, $both$, oz_j and gt exist to keep it so.

Lemma E.22 (Gadgets)**.** *Assume (G). For all words X, Y, the equations (T4) have exactly one solution in the auxiliary slots: the carries, borrows and prefix-ors of the school methods. In it $AD = X + Y \bmod 2^w$, $SB = X - Y \bmod 2^w$, $bo_w = [X < Y]$, $AN = X \wedge Y$, $oz_{w-1} = [X \neq Y]$, $gt = [X > Y]$ and $CMP = \mathrm{cmp}(X, Y)$. Likewise (T10) forces $ts' = ts + 1 \bmod 2^\tau$, $c'_A = c_A + a_A \bmod 2^\tau$ and $c'_B = c_B + a_B \bmod 2^{\beta+1}$, with unique carries.*

Proof. Each auxiliary slot is the left side of one equation whose right side uses inputs and earlier slots only, so the solution exists and is unique. By induction on j, $ca_j = \lfloor (X \bmod 2^j + Y \bmod 2^j)/2^j \rfloor$, because over $\mathbb{F}_2$ the carry equation is the majority of three bits; likewise $bo_j = [X \bmod 2^j < Y \bmod 2^j]$, the majority of $1 + X[j]$, $Y[j]$ and bo_j, and $oz_j = [SB \bmod 2^{j+1} \neq 0]$. Since $SB = 0$ exactly when $X = Y$, $gt = [X \geq Y \text{ and } X \neq Y]$. The counters are the case $Y \in \{0, 1\}$ of the adder. □

Lemma E.23 (One step)**.** *Assume (G).*

(i) (T1) holds for a configuration block exactly when it encodes a configuration: a register is # if its flag is set, and its word is then 0.

(ii) (T2) holds exactly when $pc < L$, e is the indicator vector of pc, and $o_p = [pc \leq p]$.

(iii) Let a time row satisfy (T1)–(T10), let S and S' be its current and next configurations, and let $I = I_{pc}$. If I is not `out`*, then S' results from S by executing I with the row's access: a* `load` *receives M_V from address X, a* `store` *writes $M_V = Y$ at address X, and a read receives # if $R_F = 1$ and R_V otherwise. If I is* `out`*, then $S' = S$. Conversely, every such step, with the auxiliary slots of Lemma E.22, unused channels zero, and $c_B[\beta] = 0$ whenever I is a* `readB` *that receives a word, satisfies (T1)–(T10).*

Proof. (i) restates the encoding. (ii) The equations make o_p the number of ones among $e_0, \ldots, e_p$, keep it at most one, and make it one at $p = L - 1$; then $pc = \sum_p p\, e_p$. In characteristic two, $\sum_p e_p = 1$ alone would allow three ones; the prefix-or chain excludes this. (iii) By (ii) every $e(Q)$ equals $[pc \in Q]$. (T3) puts the operands of I on the buses, Lemma E.22 computes the four operations and the zero test $z = [X \neq 0]$ used by the conditional jumps, (T6) selects the value written, (T7) writes it into the register of I and copies the others, and (T8) computes the next pc. A read of # writes the flag 1 and the word 0 by (T5) and (T6). The converse is the same computation. □

Lemma E.24 (Memory by sorting)**.** *Let a time-ordered list of memory accesses be given, at distinct steps. The following are equivalent.*

(a) There are memories $\mathrm{mem}_0 = 0, \mathrm{mem}_1, \ldots$ such that each store (a, v) at step i gives $\mathrm{mem}_{i+1} = \mathrm{mem}_i[a \mapsto v]$, every other step leaves the memory unchanged, and each load (a, v) at step i has $v = \mathrm{mem}_i(a)$.

(b) Each load (a, v) at step i has v equal to the value of the latest access to a before step i, or $v = 0$ if there is none.

(c) In the list of the memory records sorted by key and preceded by the sentinel, a load record whose predecessor has the same address has the predecessor's value, and every other load record has value 0.

Proof. (a)⇔(b): by induction on i, $\mathrm{mem}_i(a)$ is the value of the latest access to a before step i, or 0. (b)⇔(c): the memory record of an access to a at step i has key $2^{\tau+1}a + 2^\tau + i$. In key order the records of one address are therefore consecutive and in time order. The predecessor of a load at (a, i) is the latest access to a before i if there is one, and otherwise a record of a smaller address or the sentinel, whose address 0 and value 0 give the value 0 in either case. □

Lemma E.25 (Time stamps and counters)**.** *Assume (G), (T1)–(T10) at every time row, (B1) and (B2). Then $ts(\rho_i) = i$, and $c_A(\rho_i)$ and $c_B(\rho_i)$ are the numbers of steps before i that read a word other than # from tape A and from tape B. Hence at most 2^β words are read from tape B, of total length at most $2^\beta w = T$. Conversely, if at most 2^β such words are read, the honest counters satisfy $a_B c_B[\beta] = 0$.*

Proof. Induction on i with Lemma E.22. The time stamp does not wrap, since $i \le T - 1 < 2^\tau$; nor does c_A, since at most $T - 1$ steps precede ρ_{T-1}. A read at $c_B = 2^\beta$ is excluded by $a_B c_B[\beta] = 0$, so $c_B \le 2^\beta < 2^{\beta+1}$. Conversely the r-th read happens at $c_B = r - 1 < 2^\beta$. □

Lemma E.26 (The sorted list)**.** *Assume (G), (P), (S1)–(S5), (B4), (B5), the time-row equations, and $ts(\rho_i) = i$ for all i. Then the active sorted vertices form a prefix of $\sigma_0, \dots, \sigma_{N-1}$, which lists the active records of the source list in increasing key order. The memory records satisfy Lemma E.24(c), and every tape-A read record $(1, 1, c, 1, i, v, 0)$ has $c < N_A$ and v equal to word c of $\bar{x}$. Conversely, let a source list have distinct active keys and pairwise distinct tape-A read indices c, and let its memory and tape-A records have these two properties. Then sorting its active records by key, appending the inactive ones, and filling* PREV *by (B4) satisfies (S1)–(S5) and (P).*

Proof. By (S1) activity does not increase along σ, and the sentinel is active, so the active vertices form a prefix. By (S2) their keys increase strictly; by (P) they are the active source records in key order, and their keys are distinct. A memory load has $u = g = 1$, and (S4) is Lemma E.24(c), because $so_w = 0$ exactly when the two records name the same region and address. A tape-A read $(1, 1, c, 1, i, v, 0)$ has $g_A = 1$, and (S5) makes its predecessor a record in region 1 with index c, phase 0 and value v. Time rows have phase 1 by (T9), so the predecessor is ι_c, which exists only for $c < N_A$ and has the value prescribed by (B5). For the converse, each predecessor in the sorted arrangement is as described, and the same computation verifies every equation. Distinct read indices are needed here: a second read of index c would have a read record of phase 1 as its predecessor and violate (S5). □

Theorem E.27 (Event traces)**.** $\mathrm{ET}(M, \tau, \bar{x})$ *is satisfiable if and only if M accepts $\bar{x}$ in time $T = 2^\tau$.*

Proof. ($\Rightarrow$) Let S^i be the current configuration of ρ_i (Lemma E.23(i)). By (B2) the next configuration of ρ_i is S^{i+1}; by (B1) S^0 is initial; by (B2) and (B3) no configuration before the last executes `out`, and S^{T-1} is accepting. Lemma E.23(iii) gives every step relative to its access, and Lemma E.25 gives the time stamps, so the access records have distinct keys. By Lemmas E.24 and E.26, there are memories with $\mathrm{mem}_0 = 0$ under which every step is a step of M. The words read from tape A have indices $c = 0, 1, 2, \dots$ in time order (Lemma E.25), and each equals word c of $\bar{x}$; so they form a prefix of $\bar{x}$. Let y be the concatenation of the words read from tape B; then $|y| \le T$ by Lemma E.25, and the run is accepting on $(\bar{x}, y)$.

($\Leftarrow$) Take an accepting run on $(\bar{x}, y)$ with $|y| \le T$. Fill ρ_i from S^i, S^{i+1} and the access of step i (the loaded value $\mathrm{mem}_i(a)$, the stored word, or the word read), with the auxiliary slots of Lemma E.22, unused channels zero, the next configuration of ρ_{T-1} equal to its current one, and honest time stamps and counters. At most $|\bar{x}|/w = N_A$ words are read from tape A, and the c-th is word c of $\bar{x}$; at most $|y|/w \le 2^\beta$ words are read from tape B, so the witness bound holds (Lemma E.25). Add the input and dummy records. The active keys are distinct: time-row records differ in their time stamps, input records in their indices, and the two kinds in their phase. The tape-A read indices are the counter values $0, 1, \dots$. Lemma E.26 now provides the sorted vertices, and every equation holds by Lemmas E.22 to E.25. □

Corollary E.28 (Event traces of verifiers)**.** *Let V, $T(n)$ and M_n be as in Lemma E.21, let $n \geq 2$, and let $T_n = 2^{\tau_n}$. For every $x \in \{0,1\}^n$, $V(x,y)$ accepts for some $y \in \{0,1\}^{T(n)}$ if and only if* $\mathrm{ET}(M_n, \tau_n, \bar{x})$ *is satisfiable.*

Proof. Lemma E.21 and Theorem E.27. The hypotheses of the latter hold: $\tau_n \geq \max(1, \omega)$ and $|\bar{x}| \leq T_n < 2^{w_n}$ by Lemma E.21, and w_n divides $|\bar{x}|$. □

E.30 The event trace as an affine constraint instance

Throughout this subsection and the next, assume in addition that $|\bar{x}| = 2^m$ for an integer m with $\omega \leq m \leq \tau$. This holds for the padded inputs of Lemma E.21; the event-trace theorem itself does not need it.

Idea. Each slot of the event trace becomes a point of an affine subspace H of a binary field, and the assignment polynomial A holds its value there. Each equation, at each vertex, has its own host point in H; a *selector* polynomial switches the equation on at its host points only, and the equation reads the slots it needs at affine shifts $\alpha + b$ of the host point α. Three kinds of reads occur: within a vertex, from a time row to its predecessor, and from a sorted vertex to its predecessor. The multiset equality (P) is not local. A Beneš network [Ben65] enforces it: between the source list and the sorted list there are $2k_N - 1$ layers of switches, each of which passes a pair of records straight or swaps them.

Coordinates. Let $k_N = \tau + 1$, so that $N = 2^{k_N}$; let $\ell_S = \lceil \log_2(2k_N) \rceil$; let $F_{\mathrm{tags}} = 32 + 3K + 2L$ (the number of field tags the placement below needs), $d = \lceil \log_2 F_{\mathrm{tags}} \rceil$, $q = d + \omega$ and $C = 2^q$; and let $m_H = k_N + \ell_S + q + 4$. Given an irreducible irr of degree $f \geq m_H$, the residues $e_b = X^b$, $b < m_H$, are linearly independent over $\mathbb{F}_2$. Let H be their span, with offset 0, so that $|H| = 2^{m_H} = 16NC2^{\ell_S}$. A point of H is labeled by a lane $x < N$, a stage $s < 2^{\ell_S}$, a column $c < C$, a role $r < 8$ and an area $a < 2$:

$$E(x,s,c,r,a) = \sum_{b<k_N} x_b e_b + \sum_{b<\ell_S} s_b e_{k_N+b} + \sum_{b<q} c_b e_{k_N+\ell_S+b} + \sum_{b<3} r_b e_{k_N+\ell_S+q+b} + a\, e_{m_H-1},$$

where x_b is bit b of x, and so on. A column $c = 2^\omega t + j$ names a field tag $t < 2^d$ and a bit index $j < w$. Labels add by the bitwise exclusive-or of their coordinates.

Placement. A fixed catalog, which depends only on K and L, assigns every slot and every equation its own point.

- *Area 0, the network.* At stage $s < 2k_N$ and lane x, role 0 holds a network record of C bits, one per column: the seven record fields occupy their own field tags, the value field *va* at tag 0 (the input subspace below relies on this), at bit index j for bit j, and all other columns are padding. Stage 0 holds the source list ($\mathrm{REC}(\rho_i)$ at lane $i < T$, ι_c at lane $T + c$, dummies at the other lanes), and stage $2k_N - 1$ holds $\mathrm{REC}(\sigma_j)$ at lane j. Role 1 at a stage $s < 2k_N - 1$ holds that stage's switch at the lane, copied in every column. The routing equation of a record bit is hosted at that bit's own point, the column equation of a switch at the switch's point, the bit check of a record bit at role 2, the bit check of a switch at role 3, and the pair equation at role 4.
- *Area 1, the machine.* The other slots of ρ_i are at stage 0, lane i, and those of σ_j at stage $2k_N - 1$, lane j. Each word-sized block of a vertex has its own field tag; the blocks use at most $31 + 2K + 2L$ tags.
  - Roles 0 and 1 hold data: current values in role 0; next values, and PREV, in role 1, in the column of the value they are copied to.

– Role 2 hosts the equations, one field tag per equation *template* and bit index j for the instance at j. A template is one displayed equation of (T1)–(T10) or (S1)–(S5) for a fixed register r or code position p; the κ_{key}-bit borrow chain of (S2) is split into three templates of at most w instances each, since $\kappa_{\mathrm{key}} \le 2w+1$. There are exactly $32+3K+2L$ time templates and 17 sorted-vertex templates, so F_{tags} tags suffice. The initial carries ca_0, bo_0, tc_0, ac_0 and bc_0 are shorthand for their displayed values, requiring no additional slots or templates. The respective template counts for (T1)–(T10) are K, $2L+1$, 3, 8, 2, 2, $2K$, 1, 6, 9, whose sum is $32+3K+2L$.

– Roles 3 and 4 host the bit checks (G) of roles 0 and 1, in the same column.

– Role 5 hosts (B1) and the time copies of (B2), role 6 hosts (B4), and role 7 hosts (B3) and (B5).

Neighbors. Every neighbor is a translation $X \mapsto X+b$ with $b \in H$, so it maps H onto H. An equation hosted at a point α reads its slots at $\alpha+b$ for the following b.

- *Within a vertex:* the $16C$ sums $E(0,0,c,r,a)$ over all columns, roles and areas, which change no lane or stage bit; the exclusive-or of the host's and the target's column, role and area selects the target.
- *Previous time row and previous sorted vertex:* for each $t < k_N$, the translation by $\Gamma_t = e_0 + \cdots + e_t$ plus a fixed role and area mask. If t is the least set bit of the lane x, then $x \oplus (2^{t+1}-1) = x-1$, so this translation reads lane $x-1$ at the same column. Two such families of k_N maps are used.
- *Network edges:* the identity; for each stage $s < 2k_N - 1$, the translation δ_s that changes the stage label s to $s+1$ and $\delta_s + e_{d_s}$, which in addition flips lane bit $d_s = \min(s, 2k_N - 2 - s)$; the role masks used by the routing equations; the pair-partner translations $E(0,0,0,5,0)+e_b$, $b < k_N$, by which the pair equation at role 4 reads the switch at role 1 of lane $x \oplus 2^b$; the column successors $\zeta_t = e_{k_N+\ell_S} + \cdots + e_{k_N+\ell_S+t}$, $t < q$; and the lane flips e_b, $b < k_N$ (counted among the neighbors; no equation of this construction reads them).

Hence $c_N \le 8k_N + q + 16C + 3$. The identity is a neighbor, so $c_N \ge 1$.

Selectors. For each coordinate $a < m_H$ there is a *linearized* polynomial $\lambda_a(X) = \sum_{r<m_H} c_{a,r} X^{2^r}$, of degree at most 2^{m_H-1}, with $\lambda_a(e_b) = [a=b]$; since λ_a is $\mathbb{F}_2$-linear, $\lambda_a(\sum_b u_b e_b) = u_a$ for all bits u_b. Indeed, the coefficients solve an $m_H \times m_H$ linear system with matrix $(e_b^{2^r})_{b,r}$. A nonzero vector in its kernel would give a nonzero linearized polynomial of degree at most 2^{m_H-1} vanishing on every e_b, hence on all 2^{m_H} points of H, which is impossible; so the system is invertible. A *pattern* is an assignment of bits to a set of distinct coordinates. It is emitted as

$$\chi_p(X) = \prod_{a:\, p(a)=1} \lambda_a(X) \prod_{a:\, p(a)=0} (1+\lambda_a(X)),$$

which on H is the indicator of the points whose coordinates agree with p, and has degree at most $m_H 2^{m_H-1}$. The selector of an equation is the sum of a list of disjoint patterns: one pattern fixes the host's stage, column, role and area, and it is merged with the patterns of a lane condition. The lane conditions used are the time lanes, the input lanes, the dummy lanes, the first lane, the last time lane, the other time lanes, and, for each t, the lanes whose least set bit is t; each is a list of at most k_N patterns. Two patterns are merged by uniting their assignments before emission, and a conflict emits nothing. Every emitted pattern thus fixes each coordinate at most once.

Routing equations. Write $D_s(x,c)$ for the network record bit at stage s, lane x, column c, and $W_s(x,c)$ for the switch value. At every stage $s < 2k_N - 1$, every lane and every column, the host

points of area 0 carry the equations

$$D_s(x,c) + D_{s+1}(x,c) + W_s(x,c)(D_{s+1}(x,c) + D_{s+1}(x \oplus 2^{d_s}, c)) = 0,$$
$$W_s(x,c)^2 + W_s(x,c) = 0,$$
$$W_s(x,c) + W_s(x, c+1 \bmod C) = 0, \qquad W_s(x,c) + W_s(x \oplus 2^{d_s}, c) = 0,$$

and every network record bit satisfies $D_s(x,c)^2 + D_s(x,c) = 0$. The column successor $c \mapsto c+1 \bmod C$ is the translation by ζ_t at the columns whose lowest zero bit is t, and by ζ_{q-1} at the column of all ones; each case is selected by a column pattern.

Lemma E.29 (Routing)**.** *Let $U, V : [0,N) \times [0,C) \to \{0,1\}$. Switch values satisfying the routing equations with $D_0 = U$ and $D_{2k_N-1} = V$ exist if and only if some permutation π of $[0,N)$ satisfies $V(\pi(x), c) = U(x,c)$ for all x and c.*

Proof. Soundness. The switch equations make each $W_s(x,c)$ a bit, constant in c, and equal at the two lanes x and $x \oplus 2^{d_s}$. So the map p_s sending x to x or to $x \oplus 2^{d_s}$, according to the switch, is an involution, and the routing equation says $D_{s+1}(p_s(x), c) = D_s(x,c)$ for every column. Composing the $2k_N - 1$ involutions gives π; whole records move together, so no record is lost or duplicated, even when values repeat.

Completeness, by induction on k_N, following Beneš [Ben65]. For $k_N = 1$ one switch realizes either permutation of two lanes. For $k_N \geq 2$ form a bipartite multigraph whose left vertices are the source pairs $\{x, x \oplus 1\}$, whose right vertices are the destination pairs $\{y, y \oplus 1\}$, and which has one edge from the pair of x to the pair of $\pi(x)$ for each x. Every vertex has degree two, so the multigraph is a union of even cycles, and its edges can be colored 0 and 1 so that each vertex meets both colors. The first switch of each source pair sends its record of color b to the lane whose lowest bit is b, and the last switch of each destination pair receives its color-b record from that lane. For each color the records induce a permutation of the remaining $k_N - 1$ lane bits, which is routed by induction inside the two subnetworks, whose dimension sequence is $1, \ldots, k_N - 1, \ldots, 1$. Together with the first and last stages this is the sequence $d_s = \min(s, 2k_N - 2 - s)$. Copy every switch value to all columns and propagate the records through the network; then all equations hold. □

Input subspace. Let I have offset $E(T,0,0,0,0)$ and the basis $e_{k_N+\ell_S}, \ldots, e_{k_N+\ell_S+\omega-1}$, $e_0, \ldots, e_{m-\omega-1}$, in this order. Since $c < N_A \leq T$ and T is a single lane bit, $T + c = T \oplus c$, and the i-th element of I is the point of bit $i \bmod w$ of the value of $\iota_{\lfloor i/w \rfloor}$, because *va* has field tag 0. Thus the input clause of Definition E.2 writes $\bar{x}$ directly into the input records. When (B5) is compiled into P, only the equations prescribing these records' value bits are omitted; the input clause enforces them instead. All other equations of (B5), including the zero values of the dummy records, are retained.

Theorem E.30 (The affine instance)**.** *Let $\mathcal{I}(M, \tau, \bar{x})$ be the affine constraint instance with an irreducible polynomial of the degree f of Lemma E.31, the subspace H, the neighbors above, the constraint polynomial P given by the sum, over the equations of the event trace other than the input-value equations of (B5), the routing equations and the bit checks, of the equation's selector times the equation, and the input subspace I. It accepts $\bar{x}$ if and only if $\mathrm{ET}(M, \tau, \bar{x})$ is satisfiable.*

Proof. At every point of H at most one selector is nonzero, because equations have distinct host points or disjoint lane conditions. So $P(\alpha, A(\alpha + b_1), \ldots)$ vanishes on H exactly when every equation holds for the table of values of A on H; all points read lie in H, since every $b_\nu \in H$. ($\Rightarrow$) Given A, read the slots of the trace from its values on H. Every compiled equation of the event trace holds, the bits are bits, and by Lemma E.29 the network records at the last stage are a permutation of

those at stage 0; hence so are their record fields, which is (P). The input clause gives the values of the input records, the remaining equations of (B5). ($\Leftarrow$) Given a satisfying assignment of the event trace, place its values, set unused network padding to 0, choose switches by Lemma E.29 for the permutation of (P), set all other points to 0, and let A be the interpolation polynomial of this table, of degree less than $|H|$. Every equation and the input clause hold. □

E.31 Degree, field, and generator

Lemma E.31 (Degree and field)**.** *The constraint polynomial P of Theorem E.30, as emitted, has weighted degree at most $d_B = (m_H + 4)2^{m_H}$. Let $f = m_H + \lceil \log_2(4(m_H + 4)) \rceil$. Then $4d_B \le 2^f$, $m_H + 2 \le f$ and $2^f < 8(m_H + 4)2^{m_H}$. For the machine M_n of Lemma E.21 with $T_n = 2^{\tau_n}$,*

$$2^f = O(T(n) \log^3 T(n)), \qquad f = \log T(n) + O(\log \log T(n)), \qquad c_N = O(\log T(n)),$$

with constants depending only on V.

Proof. Every term is an emitted pattern, of degree at most $m_H 2^{m_H - 1}$ in X, times an equation of degree at most two in the Y_ν; the position constants of (B5) are single coordinates $\lambda_a(X)$. A term therefore has weight at most $m_H 2^{m_H - 1} + 2(2^{m_H} - 1) \le d_B$, and a sum of terms has no monomial outside theirs. The bound holds for the polynomial as emitted, not only on H; this is why conditions are merged into single patterns before emission. The choice of f gives $2^f \ge 4(m_H + 4)2^{m_H} = 4d_B$ and $2^f < 8(m_H + 4)2^{m_H}$. For M_n, K, L and F_{tags} depend only on V, while w_n and $k_N = \tau_n + 1$ are $O(\log T(n))$ and $T_n = O(T(n))$. Hence $|H| = 16N2^{\ell_S}C = O(T(n) \log^2 T(n))$, and the bounds follow. □

Lemma E.32 (Generator)**.** *Fix V. There is an algorithm that, given $x \in \{0,1\}^n$ with $n \ge 2$ and an integer $t \ge n$ in binary, outputs $\bar{x}$ and the instance $\mathcal{I}_x = \mathcal{I}(M_n, \tau_n, \bar{x})$ of Theorem E.30, built with t in place of $T(n)$ in Lemma E.21, in $\text{poly}(n, \log t)$ steps. The description of $\mathcal{I}_x$ has $\text{poly}(\log t)$ bits, and its constraint circuit has $O(\log^3 t)$ gates, with constants depending only on V.*

Proof. The algorithm computes the numbers w_n, τ_n, m, k_N, ℓ_S, d, q, m_H and f, all of $O(\log t)$ bits, and writes $\bar{x}$. It obtains irr from Shoup's deterministic algorithm [Sho88, Corollary 3.2], which runs in $\text{poly}(f)$ time [BCGT13, Algorithm 17], and solves the $m_H \times m_H$ system for the coefficients of the λ_a with $O(m_H^3)$ field operations. It emits the neighbor offsets, which are sums of basis elements; the guard patterns, at most $4C(k_N + 3) + 4k_N(q + 3) + 4 = O(\log^2 t)$ of them, each with at most m_H coordinates; and, for each equation family and bit index, the selector times the equation, whose reads are the offsets of its catalog entries. The code-dependent constants come from the fixed code of M_n with n, t and $T_n - 8$ inserted. Each pattern costs $O(m_H)$ gates and each equation $O(L)$ gates, which gives $O((m_H + K + L) \cdot \log^2 t + m_H^2 + c_N) = O(\log^3 t)$ gates. Every loop runs over bit indices, code positions, registers, coordinates, neighbors or patterns, never over the points of H, the steps of the run, or a witness; each field operation costs $\text{poly}(f)$ steps. □

E.32 The short PCP

Theorem E.33 (Supporting statement A.1 for multitape verifiers)**.** *Let $V(x, y)$ be a deterministic multitape machine and $T : \mathbb{N} \to \mathbb{N}$ a function with $T(n) \ge n$, such that V halts within $T(n)$ steps on every $x \in \{0,1\}^n$ and every y, and $T(n)$ is computable from n in $\text{poly}(n, \log T(n))$ steps. Then, at every input length $n \ge 1$, the conclusion of supporting statement A.1 holds for V and T. In particular it holds for $\mathcal{V}_G$ and T_G of Lemma E.15.*

Idea of proof. Lemma E.21 turns V into a random-access machine whose run has exactly T_n configurations, and Theorems E.27 and E.30 turn the question whether that machine accepts into an affine constraint instance (Step 1). The PCP verifier of [BCGT13] for such instances repeats its proximity sub-verifiers, which costs randomness; we run each of them once (Step 2). The resulting verifier V_0 is sound only in a weak sense: it rejects every proof of a false instance with probability at least $1/\operatorname{poly}(\log T)$. A small-bias set amplifies this to soundness $1/n$ without new random bits (Step 3). Finally, the verifier is written as a circuit whose proof reads are oracle gates (Step 4). The governing bounds are

$$r_0 \le \log T + O(\log\log T), \qquad \sigma_0 \ge \frac{1}{\operatorname{poly}(\log T)},$$
$$|S| = \operatorname{poly}(n, \log T), \qquad \ell = \log T + O(\log\log T).$$

for the randomness, the base rejection probability, the size of the small-bias set, and the number of inputs of C_x^O. Every step runs in time $\operatorname{poly}(n, \log T)$.

Proof. For $n = 1$, use the zero-input constant-false circuit, as checked below in "The exact Lean interface"; the fixed resource constants can cover this one length. Now fix $n \ge 2$ and $x \in \{0,1\}^n$. Write $T = T(n)$, and take w_n, $\bar{x}$, T_n and M_n from Lemma E.21. The construction needs T explicitly, because w_n, T_n and the code of M_n depend on it. By hypothesis T, and hence w_n and T_n, are computed in $\operatorname{poly}(n, \log T)$ steps. All logarithms below are base two, and $\operatorname{poly}(n, \log T)$ bounds have polynomials depending only on V and T.

Step 1: reduction to an affine constraint instance. Let $T_n = 2^{\tau_n}$. By Corollary E.28 and Theorem E.30, $V(x, y)$ accepts for some y of length T if and only if the affine constraint instance $\mathcal{I}_x = \mathcal{I}(M_n, \tau_n, \bar{x})$ accepts $\bar{x}$; here $|\bar{x}| = 2^m$ with $\omega_n \le m \le \tau_n$ for $w_n = 2^{\omega_n}$, by Lemma E.21. Lemma E.32 computes $\bar{x}$ and $\mathcal{I}_x$ from x and T in $\operatorname{poly}(n, \log T)$ steps. Its neighbors are translations, $m_H + 2 \le f$, its input subspace has dimension $\log_2 |\bar{x}|$, with $1 \le \log_2 |\bar{x}| \le \tau_n < m_H$, and, by Lemma E.31,

$$f = \log T + O(\log\log T), \qquad 1 \le c_N = O(\log T).$$

Step 2: the verifier with single sub-verifier runs. Let V_0 be the verifier of [BCGT13, Construction 8.5] for $\mathcal{I}_x$, with two changes. Each of the amplified proximity verifiers V_{aRS} and V_{aVRS} in its Steps 2, 3 and 5 is replaced by a single run of the corresponding strong proximity verifier V_{RS} or V_{VRS}; and the test of p_1 uses the degree bound d_H below in place of the degree d of [BCGT13, Construction 8.5]. All steps read prefixes of one random string, as in [BCGT13, Algorithm 9]. The proof format is that of [BCGT13, Definition 8.4]: five tables of elements of F, namely an assignment table p_0 and a composition table p_1, both indexed by F, and three proximity proofs. We encode them in binary with f bits per element. The test of p_1 uses the degree bound $d_H = 2^{f-2} = |F|/4$, which V_0 computes from f alone; it bounds the degree d_B of Lemma E.31, since $4d_B \le 2^f$, and it is less than 2^f. We use the following properties.

(a) *Randomness and length.* The sub-verifiers use the parameter tuple $(\eta, \kappa_0, \gamma_{\mathrm{rep}}, \mu_{\mathrm{rep}}) = (2, 4, 0, 1)$ fixed before Lemma E.19. By [BCGT13, Lemmas C.2–C.5], one run of V_{RS} or V_{VRS} on a domain of dimension f uses at most $\max\{\mathrm{rand}_=(f), f\}$ random bits, and its proof has at most $2^f + (2^\eta + 1)(2^f + 2\,\mathrm{len}_=(f))$ field elements. For the second bound, the degree bounds used by V_0 are less than 2^f, so the number of blocks in [BCGT13, Lemma C.3] is at most 2^η. Since $f \ge 1$ and $\eta = 2$, Lemma E.19 bounds these quantities by $f + 3\log(2f)$ random bits and $16 \cdot 2^f f^3$ field elements. The consistency test uses f bits. As the steps share their random string, V_0 uses

$$r_0 \le f + 3\log(2f) = \log T + O(\log\log T)$$

random bits; compare [BCGT13, Lemma C.8], where each term is multiplied by the number of repetitions. The proof consists of two tables of 2^f field elements and three proximity proofs. So the binary proof has L_π bits, where

$$L_\pi \le f(2 \cdot 2^f + 3 \cdot 16 \cdot 2^f f^3) \le 64 \cdot 2^f f^4, \qquad \log L_\pi = \log T + O(\log \log T).$$

(b) *Time and queries.* V_0 runs in time

$$\mathrm{poly}(n, \log T).$$

We bound the parts of its work in turn.

- *Field operations.* The field F is represented by an irreducible polynomial that Shoup's deterministic algorithm finds in $\mathrm{poly}(f)$ steps [BCGT13, Algorithm 17]. So each field operation takes $\mathrm{poly}(f)$ steps.
- *Parameters.* The instance $\mathcal{I}_x$ is generated in $\mathrm{poly}(n, \log T)$ steps (Lemma E.32). Its constraint circuit, which the consistency test evaluates once, has $O(\log^3 T)$ gates, and its c_N neighbors are translations.
- *Sub-verifier runs.* A single run of V_{RS} or V_{VRS} takes $\mathrm{poly}(f)$ field operations. Its recursion roughly halves the dimension of the current affine subspace at each level and does $\mathrm{poly}(f)$ field operations per level [BCGT13, Algorithms 12–16].
- *The instance.* The only step whose cost depends on $|\bar{x}|$ is the evaluation of the low-degree extension A_x of the instance at the queried points (Step 5 of [BCGT13, Construction 8.5]). By Lagrange interpolation over the input subspace I of size $|\bar{x}|$, each evaluation takes $O(|\bar{x}|^2)$ field operations, and $|\bar{x}| = O(n + \log T)$.

The queries of V_0 are computed from the random string alone, by the non-adaptive decompositions of the single-run verifiers [BCGT13, Lemmas 12.16–12.20], combined as in [BCGT13, Lemma 12.23].

(c) *Completeness.* Suppose that $\mathcal{I}_x$ accepts $\bar{x}$ with an assignment polynomial A, and let A_x be the polynomial of degree less than 2^m that interpolates $\bar{x}$ on I. Let π consist of the tables of A and of B_A on F together with the proximity proofs of [BCGT13, Construction 8.3], with π_1 computed for the degree bound d_H. Each table tested by V_0 is a codeword of the code tested there: A has degree less than 2^{m_H}, B_A has degree at most $d_B \le d_H$ and vanishes on H, and $A - A_x$ has degree less than 2^{m_H} and vanishes on I. By the perfect completeness of the strong proximity verifiers [BCGT13, Definition 4.15 and Theorem 11.1], every single run accepts on every random string. The consistency test accepts, since $p_1(\alpha) = B_A(\alpha) = P(\alpha, A(N_1(\alpha)), \ldots)$. So V_0 accepts π with probability one.

(d) *Base soundness.* Suppose that $\mathcal{I}_x$ does not accept $\bar{x}$, and let π be any proof. Put $\delta = 1/(8c_N)$; here $c_N \ge 1$, because the identity is a neighbor. We distinguish four cases. In Cases 1–3, one of the following holds: p_0 has relative distance at least δ from every polynomial of degree less than 2^{m_H} (Case 1); p_1 has relative distance at least $1/8$ from every polynomial of degree at most d_H that vanishes on H (Case 2); or $p_0 - A_x$ has relative distance at least δ from every polynomial of degree less than 2^{m_H} that vanishes on I (Case 3). By strong soundness [BCGT13, Definition 4.15] and [BCGT13, Theorem 11.1], a single run of V_{RS} or V_{VRS} on a table at relative distance at least δ' from its code rejects with probability at least $\delta'/(2 + K_f)$. Here $K_f = 16 f^{\log c}$, with c the constant of [BCGT13, Theorem 11.3] at the tuple fixed in (a);

this is a weakening of the bound printed in [BCGT13, Theorem 11.1], and the companion's source mapping explains the choice.

In Case 4 none of these holds, so there are polynomials A and A' of degree less than 2^{m_H}, with A' vanishing on I, and B' of degree at most d_H, vanishing on H, such that p_0 and A, and $p_0 - A_x$ and A', each differ on at most a δ fraction of F, and p_1 and B' on at most a $1/8$ fraction. The polynomials $A - A_x$ and A' agree wherever both of the first two relations hold, so on at least $(1-2\delta)\,|F| \geq \frac{3}{4}\,|F| > 2^{m_H}$ points, because $m_H + 2 \leq f$. Hence $A - A_x = A'$, and A agrees with $\bar{x}$ on I. Since $\mathcal{I}_x$ does not accept $\bar{x}$, B_A does not vanish on all of H, so $B_A \neq B'$, and these polynomials of degree at most $d_H = |F|/4$ agree on at most $|F|/4$ points. The consistency test of Step 4 draws $\alpha \in F$ uniformly and accepts only if $p_1(\alpha) = P(\alpha, p_0(N_1(\alpha)), \ldots, p_0(N_{c_N}(\alpha)))$. Every N_ν is a bijection of F, so with probability at least $1 - 1/8 - c_N\delta = 3/4$ we have $p_1(\alpha) = B'(\alpha)$ and $p_0(N_\nu(\alpha)) = A(N_\nu(\alpha))$ for every ν, and then the test accepts only if $B'(\alpha) = B_A(\alpha)$. So it rejects with probability at least $3/4 - 1/4 = 1/2$.

Since V_0 rejects whenever one of its steps rejects, and $\delta/(2 + K_f)$ is the smallest of the four bounds, it rejects π with probability at least

$$\sigma_0 = \frac{1}{8c_N(2 + K_f)} \geq \frac{1}{\mathrm{poly}(\log T)}.$$

Step 3: amplification. Choose fixed positive integers K_1, b_1 such that $1/(K_1 f^{b_1}) \leq \sigma_0$. The consistency test uses $f \geq 1$ random bits, so $r_0 \geq 1$. Choose the least nonnegative integer m such that $2^{2m} \geq r_0^2 n K_1 f^{b_1}$, and use the powering construction in the proof of Lemma E.17 over $\mathbb{F}_{2^m}$. Its bias is at most $\varepsilon = r_0/2^m$, with $\varepsilon^2 \leq 1/(nK_1 f^{b_1}) \leq \sigma_0/n$. Thus Lemma E.18 applies with $s = 1/n$. Since $n \geq 2$, minimality of m gives

$$|S| = 2^{2m} < 4r_0^2 n K_1 f^{b_1} = \mathrm{poly}(n, \log T).$$

Choosing m uses only integer comparisons; constructing the field and listing the multiset also take $\mathrm{poly}(n, \log T)$ time. The resulting verifier V_1 uses r_0 random bits. It accepts the proof of (c) with probability one when $\mathcal{I}_x$ accepts $\bar{x}$, and it accepts every proof with probability at most $1/n$ otherwise. It runs in $\mathrm{poly}(n, \log T)$ time.

Step 4: the oracle circuit. Let

$$\ell = \max\{r_0, \lceil \log L_\pi \rceil\}, \qquad \text{so that} \qquad \ell = \log T + O(\log\log T).$$

For $R \in \{0,1\}^\ell$, consider the computation of V_1 on $\bar{x}$ with the random string formed by the first r_0 bits of R. It runs in $\mathrm{poly}(n, \log T)$ steps, and its proof bits are read at addresses below L_π, written with ℓ bits. Convert this computation into a circuit in the standard way, and replace each read of a proof bit by an oracle gate on the ℓ address bits. This gives an oracle circuit of size $\mathrm{poly}(n, \log T)$ that outputs the verdict of V_1 on the proof π_O whose bit at address $a < L_\pi$ is $O(a)$. Let C_x^O be its negation. The algorithm A computes T, $\mathcal{I}_x$ (Lemma E.32), S and C_x^O in $\mathrm{poly}(n, \log T)$ time.

The properties of C_x^O. If $V(x, y)$ accepts for some y of length T, Step 1 and property (c) give a proof π that V_1 accepts on every random string. The oracle O with $O(a) = \pi_a$ for $a < L_\pi$ and $O(a) = 0$ otherwise makes C_x^O unsatisfiable. If $V(x, y)$ rejects for every y of length T, $\mathcal{I}_x$ does not accept $\bar{x}$ (Step 1). So for every oracle O, V_1 accepts π_O on at most a $1/n$ fraction of the strings in $\{0,1\}^{r_0}$. Hence C_x^O has at least

$$2^\ell(1 - 1/n)$$

satisfying assignments. Each oracle gate has ℓ inputs, and C_x^O has ℓ inputs and size $\mathrm{poly}(n, \log T)$.

Finally, $\mathcal{V}_G$ and T_G satisfy the hypotheses by Lemma E.15(i)–(ii). □

Corollary E.34 (Clock supplied as input)**.** *Fix a deterministic multitape verifier V. There is a uniform algorithm that, given $x \in \{0,1\}^n$, $n \geq 2$, and a binary integer $t \geq n$, constructs the oracle circuit of supporting statement A.1 with $T(n)$ replaced by t, provided t bounds the running time of $V(x,y)$ for every witness y. Its running time and circuit size are $\mathrm{poly}(n, \log t)$, and its randomness and oracle-address lengths are $\log_2 t + O(\log\log t)$. Completeness and soundness have the same quantifiers as in supporting statement A.1, with witnesses of length t. In particular, for any bound $T(n) \geq n$, the compiler may be given $(x, T(n))$; it need not compute the function T.*

Proof. In the proof of Theorem E.33, computing $T(n)$ serves only to obtain the integer used to choose the word width, padded clock, reduction parameters, and verifier circuit. Supply that integer as t and perform the same construction, clocking the simulation after t steps. All these choices are effective in $\mathrm{poly}(n, \log t)$ time, with constants depending on the fixed verifier. Under the stated promise, the clock changes no computation on x. The completeness and soundness arguments of Steps 1–4 are pointwise in x and this integer, so apply unchanged. The construction does not test the promise. □

The exact Lean interface. The companion states supporting statement A.1 as `OrdinaryOracleCircuitPCP`, with an ordinary multitape machine as the constructor. We complete it as follows. Its verifier starts with $\mathrm{fr}(x)$ and $\mathrm{fr}(y)$ on its two input tapes, where $\mathrm{fr}(b_1 \cdots b_j) = 1b_1 \cdots 1b_j 0$, followed by blank zeros. The predicate `RunsInTime` bounds this actual computation by $t = T(n)$ on every witness list; the correctness clauses quantify over witnesses of exactly t bits. For $n \geq 2$, use the construction of Corollary E.34 with these initial tapes. The machine M_n of Lemma E.21 writes $\mathrm{fr}(x)$ and $\mathrm{fr}(y)$ onto the two simulated tapes, where y consists of the first t witness bits. This writes the frame terminators at their prescribed positions and leaves zeros beyond them; subsequent writes by the simulated verifier are preserved. Loading takes $O(n + t + w_n)$ steps, so, since $n \leq t$, $T_n = O(t + \log t)$, and M_n accepts exactly for the required t-bit witnesses. The remaining PCP construction is unchanged.

For $n = 1$, output a zero-input, one-node constant-false circuit, with $\ell = 0$ and oracle arity zero. It is unsatisfiable for every oracle, so completeness holds; the soundness lower bound is $2^0(1 - 1/1) = 0$. Since $T(1) \geq 1$, both resource inequalities hold after enlarging the fixed constants. For $n = 0$, use the same fixed output: the two resource and two correctness clauses apply only for $n \geq 1$. Thus one constructor covers every input length.

The constructor receives $\mathrm{fr}(x)\,\mathrm{fr}(\mathrm{bits}(t))$, of length $2n + 2\,|\mathrm{bits}(t)| + 2$, and never computes the function T. For $n \geq 2$, its bounds give, for fixed integers K, e,

$$2^\ell \leq Kt\,\lceil \log_2(t+2) \rceil^e, \qquad \left| C_x^O \right| \leq K\,(n + \mathrm{natBitLength}(t) + 1)^e.$$

Here $\mathrm{natBitLength}(t) = \lfloor \log_2 t \rfloor + 1$ for $t \geq 1$, and $\lceil \log_2(t+2) \rceil$ is the companion's `logScale`. Inputs, constants and oracle gates count as nodes; each oracle gate has ℓ arguments, and the circuit outputs true precisely on rejection, as above. A circuit with s nodes has a word in the companion's self-delimiting format of length $O((s(\ell+1)+1)\log(s+\ell+2))$: write its arity, node count, node tags and addresses, and output address in order. Hence both computing and writing this word take $\mathrm{poly}(n, \log(t+2))$ time on an ordinary multitape machine, which is the constructor the interface asks for.

Remark E.35 (Non-adaptivity). The circuit C_x^O of the proof is non-adaptive: the address bits of every oracle gate are computed from the inputs R alone ([BCGT13, Lemmas 12.16–12.20], combined as in [BCGT13, Lemma 12.23], and Lemma E.18(i)); the oracle answers enter only the decision.

E.33 Proof of Lemma E.21

The construction has two parts. A random-access machine (Definition E.1) simulates V step by step. An idle phase then makes the number of configurations exactly T_n, as acceptance in time T_n requires.

Idea of proof. The machine M_n copies the input and the witness onto the simulated tapes of V, which it keeps in its memory, interleaved. It then simulates V step by step. Each block of its code pads its own length to a multiple of 8 and adds that length to a step counter. Once V accepts, a counted idle loop pads the run, so that the `out` instruction comes at exactly step $T_n - 1$.

Proof of Lemma E.21. Layout. Let V have k_V tapes, counting its input and witness tapes. Memory cell $2k_V j + i$ of M_n holds cell j of tape i of V; with this stride, a head at position 0 is recognized by one comparison. The tape symbols are coded as small integers, and the blank as zero. The $K = 9 + 2k_V$ registers hold a step counter, the verdict, the state of V, the head positions, the scanned symbols, and scratch values.

Loading the tapes. First M_n reads from tape A the words that contain the first n bits of $\bar{x}$, which are x, and writes x bit by bit onto the input tape of V. It then reads from tape B the words that contain the first $T(n)$ witness bits, and writes these bits, y, onto the witness tape of V. (For the companion's verifiers the two tapes receive the framed words $\mathrm{fr}(x)$ and $\mathrm{fr}(y)$; see "The exact Lean interface" above.) The integers n and $T(n)$ are written into the code. If a read returns #, M_n enters a one-instruction loop that never reaches `out`. Loading takes $O(n + T(n) + w_n)$ steps.

Simulating V. One step of V is then simulated by a block that loads the scanned symbols, forms the index of the pair (state, scanned symbols), and branches through a chain of comparisons to one leaf per pair. The leaf stores the written symbols, moves the heads and sets the new state. A halted rejecting state, or a pair without a transition, leads to the loop that never halts; the accepting state records the verdict and starts the idle phase. A step costs $O(1)$ instructions for fixed V, so the simulation ends within $O(n + w_n + T(n)) = O(T(n) + w_n)$ steps.

Counting. The code of M_n consists of blocks: the start, the two loading loops, the simulation of one step of V with its leaves, and the recording of the verdict. Every path through a block is padded with `nop` instructions so that the number of steps it takes is a multiple of 8. On that path the block adds this number to a counter register, which is zero in the initial configuration. Hence, whenever a block ends, the counter equals the number of steps made so far, and this number is a multiple of 8. This at most multiplies the running time by a constant.

Idling. Once the verdict is recorded in a register, M_n idles. A fixed code cannot hold a number of `nop`s that depends on the run, so the padding is a counted loop: an idle block of exactly 8 instructions. In order, they are:

- `add` 8 to the counter;
- `cmp` the counter with the constant $T_n - 8$;
- five `nop`s;
- `jne` back to the start of the idle block if the comparison found the counter different from $T_n - 8$; as the counter never exceeds $T_n - 8$, this means smaller.

The counter takes only multiples of 8. Since T_n is a power of two and at least 16, $T_n - 8$ is also a multiple of 8. So if the counter is at most $T_n - 16$ on entry, the loop ends after the pass at whose end exactly $T_n - 8$ steps have been made.

The verdict. Then M_n executes six `nop`s and one `je` on the verdict register. The `je` jumps to an instruction (`out`, *reject*) if the verdict is rejection, and otherwise falls through to an instruction

(`out`, *accept*). The instruction `out` takes a constant argument, so this branch is needed. After these seven steps exactly $T_n - 1$ steps have been made, and the `out` instruction with the verdict is executed at step $T_n - 1$.

Constants. A suitable constant $c_V \geq 16$ ensures three things.

- The simulation and the recording of the verdict end within $T_n - 16$ steps.
- $|\bar{x}| \leq 2(n+1) + 2w_n \leq T_n$.
- $2^{w_n} \geq 2^{c_V} T(n)$ is larger than T_n and every address.

The code of M_n is a fixed code with the integers n, $T(n)$ and $T_n - 8$ written into it; its length L and the number K of registers depend only on V.

Runs. If $V(x, y)$ accepts for some $y \in \{0,1\}^{T(n)}$, let tape B hold y padded with zeros to a multiple of w_n, of length at most $T(n) + w_n \leq T_n$. The computation of M_n on $\bar{x}$ and this tape, in which every read returns the next word, is an accepting run of length T_n. Conversely, no read of an accepting run returns #, since that leads to the loop that never halts. So the run reads the first words of $\bar{x}$ and of a witness string, simulates V on x and the first $T(n)$ bits y of that string, and reaches the accepting state; hence $V(x, y)$ accepts. □

E.34 Amplification and uniform proximity interfaces

These are the arguments of [Nag26c, Appendix B]; they are reproduced with notation adapted. Their published premises are the STV reconstruction theorem and the two Chen–Williams results in Section 1.4.

Full-range amplification interface. We justify the formulation of [CLW20, Lemma 3.9] used by the companion, for every time-constructible $S : \mathbb{N} \to \mathbb{N}$. Here circuit size counts all DAG nodes, including input and constant nodes. Following the construction of [STV01, Lemmas 25 and 28] and the circuit argument of its Theorem 24, the polynomial reconstruction theorem [STV01, Theorem 29] gives absolute constants $A \geq 1$ and an integer $a \geq 1$ with the following consequence. An s'-node circuit agreeing with the encoded function on more than a $1/2 + \delta$ fraction yields an exact circuit for the original n-input function with at most $A(n/\delta)^a s' + n + 2$ nodes. The polynomial reconstruction overhead includes fixing the decoder's randomness and conversion to our fan-in-two basis; the last term allows input and constant nodes. Choose an integer $c > 2a + 2$, once and before S.

Every n-input function has a circuit with fewer than $U(n) = 2^{n+3}$ nodes. Indeed Shannon expansion combines two $(n-1)$-input circuits as $(x \wedge f_1) \vee (\neg x \wedge f_0)$, adding at most five nodes. Thus $B(0) = 1$ and $B(n) \leq 2B(n-1) + 5$ give $B(n) \leq 6 \cdot 2^n - 5 < U(n)$. For all sufficiently large n, construct the output as follows. If $S(n) = 0$, use a constant function: the target size is zero and no circuit has zero nodes. If $1 \leq S(n) \leq n^c$, use parity on $n+1$ bits. A circuit with at most $S(n)^{1/c} \leq n$ nodes omits an input, so pairing assignments that differ only in that input gives agreement exactly $1/2 < 1/2 + S(n)^{-1/c}$. If $S(n) \geq U(n)$, use a constant function, since no input function satisfies the hardness premise.

In the remaining regime $n^c < S = S(n) < U(n)$, put $\epsilon = S^{-1/c}$. Compute the least integer k such that $2^{ck} \geq S$, and use the dyadic accuracy $\delta = 2^{-k-2}$; then $\epsilon/8 < \delta \leq \epsilon/4$. All parameter choices use integer arithmetic. Apply the binary encoder of [STV01, Lemma 28] to the truth table. A circuit of size at most $S^{1/c}$ with agreement at least $1/2 + \epsilon$ would have agreement strictly greater than $1/2 + \delta$. After increasing the absolute constant A to absorb 8^a, reconstruction gives size at most

$$An^a S^{(a+1)/c} + n + 2 < S.$$

The strict inequality holds uniformly in this regime from a fixed onset: dividing by $S > n^c$ bounds the left side by

$$\frac{A}{n^{c-2a-1}} + \frac{n+2}{n^c} < 1.$$

This contradicts the premise and proves the required strict agreement bound.

On the truth-table layout: in the construction of [STV01, Lemmas 25 and 28] choose a binary extension field, rounding its size up to the required scale, and round the number of polynomial variables up. The evaluation domain is then $\mathbb{F}^m$; concatenation with a Hadamard block indexes the binary codeword by a product of binary cubes. Its length is a power of two, with no padding of the codeword, and remains $\mathrm{poly}(2^n, 1/\delta)$. These roundings preserve the polynomial reconstruction bound above. Since $S < U(n)$, we have $1/\delta = 2^{O(n)}$, so the output has $O(n)$ input bits and its entire truth table is computable in $2^{O(n)}$ time.

Choose one onset $n_0 \geq 2$ for these estimates, also large enough that $n^c < U(n)$ for $n \geq n_0$. At each of the finitely many lengths $n < n_0$ with $S(n) > 0$, output parity on $S(n)+1$ bits instead. Since $S(n)^{1/c} \leq S(n)$, the same missing-input argument gives hardness at every such length, for every supplied function. The case $S(n) = 0$ is as above. These finitely many choices can be hardcoded after S is fixed; the arity and runtime bounds in the companion have onsets, while its hardness clause holds at every length.

Finally the eventual runtime must hold even on easy input functions and for very large $S(n)$. Fix a time constructor computing $S(n)$ from unary n within $C(S(n)+1)$ steps. For $n \geq n_0$, run it for only $C(U(n)+1)$ simulated steps. A timeout implies $S(n) > U(n)$; a returned value at least $U(n)$ also selects the constant output. Otherwise the returned value determines the cases above. Initialization, the capped simulation, integer operations, and reading and writing the framed arities and truth tables all take $2^{O(n)}$ time. The usual polynomial-overhead simulation into `TM2`, the multi-stack machine model of the Lean mathematical library in which the companion states this input, preserves this bound. After increasing its exponent and onset, the bound is 2^{en} as required by the companion. The machine never tests whether the input function is hard. Its constants and onsets may depend on S; the exponent c in the hardness conclusion does not.

Uniform three-query PCPP and code interfaces. We spell out the uniformity and boundary conventions of the companion's two inputs underlying supporting statement A.4. For a fixed real $0 < \delta < 1$, choose a positive rational $\eta < \delta$ and apply [CW19, Lemma 22] at proximity η and soundness $1/2$, followed by the constant-size clause conversion of [CW19, Appendix A] that proves [CW19, Lemma 24]. Distance at least δ from every accepting word implies the source's strict distance condition at η. In that clause conversion, round the common clause count up to a power of two and pad each block with tautological three-literal clauses; use a dummy proof coordinate if necessary and repeat literals in shorter clauses. Uniform binary coins then select exactly three literals, and padding decreases the rejection probability by at most a fixed factor. The verifier is nonadaptive ([CW19, Definition 20]): for a fixed circuit C and coin string, its three queried addresses and their signs are determined without oracle answers. Enumerate all $2^r = \mathrm{poly}(|C|)$ coin strings, list the proof addresses they use, and sort and relabel the distinct addresses. There are at most $3 \cdot 2^r$ of them. Restricting an original proof to these addresses preserves every verifier outcome; conversely every trimmed proof extends arbitrarily at unused addresses and has the same outcomes. Thus trimming preserves both perfect completeness and soundness.

The verifier's uniform algorithms compute each address in polynomial time, so the addresses have polynomial bit length. Enumeration, comparison, relabeling, and emission of the three-literal clauses are polynomial-time operations. Apply the same restriction to the honest proof: the proof produced by [CW19, Lemma 22], extended to the auxiliary bits of the clause conversion by evaluating its

constant-size circuits ([CW19, Appendix A]). Its uniform polynomial construction is preserved. If m is the companion's circuit node count, then $n \le m$, and a topological node list has ordinary description length $O((m+1)\log(m+2))$. The companion's balanced binary encodings also have polynomial length in $m+1$: each pairing multiplies bit length by a constant and the list tree has logarithmic depth. Encoding and decoding these descriptions take polynomial time. Consequently the proof length and number of coin strings are bounded by $K(m+1)^e$, and the constructor and honest-proof algorithm have polynomial-time TM2 realizations on the stated encodings.

For the code, the cited statement is [CW19, Lemma 26]: a constant-rate binary linear code with positive minimum relative distance, an efficient encoder, and an efficient decoder that corrects a constant fraction of errors. The main result uses only the encoder facts: linearity, so that each output coordinate is a parity of a fixed support of message coordinates; constant rate; positive minimum relative distance; and a polynomial-time encoder. These are what the checked circuit and the distance argument after supporting statement A.4 use. The finitely many short lengths can use identity codes, with the common distance constant decreased if needed. At $n = 0, 1$ choose arbitrary fixed interfaces, since the companion's code clauses concern $n \ge 2$. The encoder has a polynomial-time TM2 realization.

The companion's code input is exactly these encoder facts, for $n \ge 2$, derived from [CW19, Lemma 26] as just described. The companion also proves the stronger full code statement of [CW19, Lemma 27], which is not needed here. Its three-block construction $\mathrm{Enc}(x) = \mathrm{Enc}_0(x_1) \circ \mathrm{Enc}_0(x_2) \circ \mathrm{Enc}_0(x_3)$, on the thirds x_1, x_2, x_3 of the message and with the code Enc_0 of [CW19, Lemma 26], bounds every support by $\lceil n/3 \rceil \le \lfloor n/2 \rfloor$ for $n \ge 2$; the stronger statement also supplies a decoder. For it, the decoding constant is decreased together with the distance at short lengths, and $0 < \rho < d < 1$ are chosen below, respectively, the guaranteed decoding radius and minimum relative distance. Choosing ρ strictly inside the decoding guarantee includes the closed ball of radius ρ. Make the decoder total, returning a fixed n-bit word on failure; this does not change its value on a promised input. The total decoder also has a polynomial-time TM2 realization. The support bound and the decoder are retained as proved results; neither is required by the main encoder interface or by the proof of supporting statement A.4.

AI methodology statement

Generative AI tools from OpenAI, Anthropic, and Google (the models Astra, Opus, Fable, Sol, and Argon) were used substantively throughout this work: exploratory proof search and parameter selection; drafting and revising all sections; constructing the Lean formalization; literature navigation and comparison of cited statements with their sources; and adversarial checks of candidate arguments. These tools materially affected the mathematical exposition and proof development; their review passes are not independent human validation. The author selected the research targets and retained arguments and takes responsibility for the mathematical claims, proofs, originality, citations, and final submission. No model is an author.

The correctness of both theorems, as formally stated, is certified by a kernel-checked Lean proof from the six published results listed in Section 1.5; the faithfulness of the formal statements to this paper and to the cited sources is documented statement by statement in the companion's source mapping. AI review is not relied on for correctness.

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